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STEM Education
National Schools of Excellence
Physics Student's Resource Book
Grade 12
Term 3

<u>DOMAIN</u>	<u>PHYLA</u>
Science	Physics
Technology	
Engineering	
Mathematics	

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THEMA: ARTIFICIAL INTELLIGENCE (AI)

Introduction

Artificial Intelligence (AI) refers to the intelligence demonstrated by machines rather than natural intelligence expressed by humans or animals. This mimicked intelligence by machines refers to the cognitive functions, problem-solving methodologies and behavior associated particularly with humans and this is the common perception that is expressed by most people. In this sense, AI gleaned its inspiration from the human brain.

In 1956, the academic discipline of Artificial Intelligence was founded and established by a group of experts from different disciplines (including Mathematics, Engineering, Psychology and Political Science) that sought to discover the possibilities of creating an artificial brain. As such, AI research had explored and rejected a variety of methodologies, including brain simulation and problem-solving modeling. In the first two decades of the twenty-first century, highly mathematical and statistical modelling dominated the subject and created useful applications such as search engines, recommendation systems, and self-driving cars. As technologies improved with research and resources to develop more capable machines became readily available, actions that were previously called "intelligent" were gradually being excluded from AI definitions.

AI technologies are currently used for a variety of applications including robotics for space and underwater exploration, an automated assembly line of robots in large manufacturing industries, and auto correct and face recognition on smartphones. As AI research and technology improves, it is becoming more accessible and integral to the common every-day human being and could become a predominant tool in resolving almost all of humanities problems. This would also suggest that there will be a lesser dependency on humans themselves or human intervention to solve our own problems. This has planted doubts into AI skeptics, developers, and companies on the uncertainties of AI technologies with concerns of it being a possible threat to human survival which is raising many controversial debates today. However so, it is an undisputable fact that AI has improved and aided in the design of the modern world as we know it today.

It is important that students are to be introduced to the concept of AI in preparation of its vast applications in the future. With this introduction, students should be thinking about how they can become creators and innovators in the field of AI and AI development, and not only end-users.

MODULE 12.1: DATA AND ARTIFICIAL INTELLIGENCE

4 weeks

The field of Artificial Intelligence is deeply rooted in using data and information. Based on how its model is created, the data is manipulated to develop "reasoning" that allows the AI to perform its intended function. This is the general field of AI that then diverges into a variety of sub-fields that are categorized according to certain distinct objectives and methods. In this Module, we will explore and discuss some of these sub-fields and focus on some of the current practices used in AI.

Topic 1: Introduction to Artificial Intelligence

From a coined term as a component of science fiction to one of the mainstream technologies in our present-day, AI is an integrated part of the common every-day human activities, especially in urban cities. In this topic, we will discuss the fundamentals of AI and some examples of how it is used in our everyday activities.

Note

This topic is similar to the introductory topics for Engineering and Technology for this THEMA: ARTIFICIAL INTELLIGENCE. It is recommended to identify key concepts and briefly explain what AI is and its influence to Physics.

1.1 Introduction

Artificial Intelligence (AI) is the science that involves the theory and development of computer systems that are able to perform tasks that normally require human intelligence. This can include visual perception, decision-making and speech recognition. Essentially AI is a branch of computer engineering, designed to create machines that behave like humans.

1.2. Branches of Artificial Intelligence

AI is composed of six main branches: Machine Learning, Deep Learning, Robotics, Expert Systems, Fuzzy Logic, and Natural Language Processing. They all play a significant role in how an AI functions and behaves towards tasks. Below is a brief description of each branch respectively.

1.2.1. Robotics

Robotics is the branch of AI that designs, builds, and operates intelligent robots using scientific and engineering approaches. As we know, Robots are mechanical beings that are frequently used to carry out jobs that are difficult for humans to complete consistently.

Robotics has a wide range of applications. One of its first and common practical use is in the manufacturing industry. Automated production is the best technology for producing more products in less time and with higher quality because the manufacturing process is generally continuous and large-scale. Thus, manufacturers use robots in the production lines to enhance output and deal with poor supplier deliveries to avoid a supply chain crisis as much as possible.

Other uses include submarine and space explorations, as well as agricultural procedures such as sowing, weed and insect management, and harvesting, to name a few. Drones, for example, can be used in military applications such as strike reconnaissance or the delivery of medical supplies to ally soldiers.

Note

We will look at Robotics in the next THEMA: ROBOTICS.

1.2.2 Machine Learning

Machine Learning is a technique that allows computers and machine systems to translate, process, analyze, and interpret data in order to find answers to real-world issues. This branch of AI enables computers to work without the need for programming, allowing them to learn from previous events/experiences as well as a large amount of stored and displayed data. They utilize algorithms that forecast and increase the probability of outcomes in the performance of tasks by learning from their experiences. As a result, Machine Learning is the most difficult branch of AI to master.

Machine Learning has led to technological breakthroughs such as web search, face recognition, and virtual personal assistants like Amazon's Alexa and Apple's Siri.

1.2.3 Deep Learning

Deep Learning is a highly sophisticated method for overcoming Machine Learning's "machine-like behavior" by developing an artificially complex and multi-layered "neural network." Much like the human brain that transfers information through electrical signals in neurons, data can move across nodes in this complicated network. Deep Learning is commonly referred to as Artificial Neural Networks since it takes its inspiration from the human brain and mimics the function of the neurons inside.

In the past, it was very difficult to develop and analyze deep networks. Naturally, with the development of technology, it is now possible to analyze many layers of neurons in Deep Learning. Neural Networks are used today by Internet companies such as Amazon to provide product recommendations, and by Facebook to identify faces in photos.

Deep Learning is also a key technology behind some autonomous technology such as driverless cars and robotics such as in the manufacturing industry. By using Deep Learning to automate the management of the production site and production line of almost all the operations, it will be possible to manufacture products completely unattended. This can be used to control the factory temperature control, product inspection, and robot adjustment for quality improvement. In addition, a function that automatically detects when a person or object intrudes around a heavy machine in operation and stops the operation safely has also been put into practical use.¹

Note

- *The concepts of Deep Learning and Machine Learning will be covered in greater detail in the subsequent topics in this Thema.*

1.2.4. Expert Systems

This branch of AI relates to computer systems that replicate the decision-making intellect of a human expert, as the name implies. Expert systems are interactive decision-making systems that have access to a vast database of knowledge. This database is dedicated to a specific topic and is used to answer complicated decision-making problems (in that domain) that would otherwise require the assistance of a human expert.

¹ (Editors of Maga AI, 2022)

The system gets more effective and dependable as the amount of expert information acquired or included inside the knowledge base grows. Expert systems are known for being exceptionally responsive, dependable, intelligible, and capable of excellent execution.

Expert Systems can be used in the healthcare industry where it can identify various bacteria that could cause acute infections. From diagnosis of the patient, the expert system can recommend. It can also predict the degree and type of lung cancer.

Mineral, oil, and gas prospecting, genetic engineering, and even corporate planning, scheduling, and financial management are some of the other applications.

1.2.5 Fuzzy Logic

Fuzzy logic is a branch of AI that is used to recognize and identify the certainty of an outcome or situation from a set of conditions. This can include We often face situations where it is difficult to recognize whether a condition or a set of conditions are true or not. Such as boiling the rice to the right texture and moisture over the fire and how long that would take, believing if someone is telling the absolute truth about something, or even a color you cannot clearly identify (e.g., purple and indigo). This can give rise to inaccurate judgments and decision making and increase the uncertainties of the conditions. The vagueness and attempt to clear the uncertainties is known as fuzzy logic.

Fuzzy logic is a branch of AI that employs a technique that represents and modifies uncertain information by measuring the degree to which the hypothesis is true. Essentially, it uses standard logic to identify whether a condition or concept exhibits a degree of truth: If is completely true, standard logic is 1.0, if it is completely false, the standard logic is 0.0. However, there are cases where a concept can either be partially true or partially false.

Fuzzy logic is convenient and flexible to implement Machine Learning techniques and assist in imitating human thought logically. In essence, an intelligent agent or computer system is given the ability to resolve the fuzzy logics that humans face in their day-to-day activities with the aim of finding a solution.

In these cases, fuzzy logic can have application in automobile industry particularly in automatic gearboxes, healthcare for medicine decision making and even washing machines where they can predict how heavy the wash load is and determine the correct amount of detergent and water, speed of agitation, and duration of wash cycles.

1.2.6 Natural Language Processing

A computer communicates information using binary digits. When it wants to communicate and interact with a human, it uses Natural Language Processing (NLP). This has already been explained as one of the Goals for AI (*see 12.2. Goals for Artificial Intelligence*).

NLP basically conducts searches, analysis, understands and derives information from the text form of data. In order to teach computers how to extract meaningful information from the text data, NLP libraries are used by programmers.

A common example of NLP is spam detection where computer algorithms can check whether an email is a scam or not just by looking at the subject of a line, or text of an email.

NLP is also useful in companies that handle with texts everyday such as Facebook or Twitter. Twitter uses NLP technique to filter terroristic language from various tweets. Amazon implements NLP for interpreting customer reviews and enhancing their e-shopping experience. Other applications involve text translation, sentiment analysis, and speech recognition.

1.3 Applications of Artificial Intelligence

There are many applications of AI that is utilized today. Some of these are listed below:

1.3.1 Navigation – Autonomous Vehicles

AI can be used to build self-driving vehicles which will revolutionize the modern modes of transport systems either for land, sea, or air.

For land vehicles, AI can be used along with the vehicle's camera, radar, cloud services, GPS, and other control signals to operate the vehicle. Together, they can improve the in-vehicle experience and provide additional systems like emergency braking, blind-spot monitoring, and driver-assist steering.

The Waymo LLC, an American company that develops autonomous driving technology, has developed an AI that commercializes “self-driving taxi's” from pickup to destination. Their services operate in most parts of Phoenix, Arizona of which Chandler city is fully mapped (which is also one of the suburbs of Phoenix).

Other transport-related industries such as Railways and Shipping companies have also dabbled in integrating AI technology for increasing safety and economic benefits. France's railway operator (SNCF) wants to start having autonomous trains on its railway network by 2023.

1.3.2 Health Care

AI is also used in the healthcare sector for many diverse applications. In this industry, the AI utilizes a sophisticated machine imbued with a combination of historical data and medical intelligence that can analyze and detect diseases or signs of chronic conditions that ensure early diagnosis, identify cancer cells, and even discover new drugs. The reliance upon AI technology in the medical healthcare sector is saving a lot of lives.

Cambio Health Care, a healthcare organization based in Linköping, developed a clinical decision support system for stroke prevention. It can signal a warning to the physician when a patient is at risk of having a heart stroke. Another company called Coala life created a digitalized device that can discover and detect cardiac diseases.

1.3.3 Robotics

Robotics is a commonly used field for AI applications. Robots powered by AI use real-time updates to sense obstacles in its path and pre-plan its journey instantly. Applications for robotics are vast.

They can be used for carrying goods in hospitals, factories, and warehouses, cleaning offices and large equipment as well as Inventory management for the aforementioned industries.

Underwater discoveries and space expeditions also utilize AI-robotic technologies that always requires analyzing large amounts of data. NASA's Mars Rover, Perseverance (see **Figure SP-AI-SRB-3**), arrived on the red planet early this year (February 18, 2021) and is currently traversing the landscape, conducting an expedition of which according to NASA is scheduled for at least one Martian year (687 Earth days). The goal is to search for microscopic life by using an AI-powered device called the Planetary Instrument for X-ray Lithochemistry, or PIXL. PIXL is a lunchbox-size instrument carried at the end of Perseverance's 7-foot-long robotic arm.

AI will also be used aboard Perseverance for navigation on the planet's surface over the life of its mission, as well as for the landing and a variety of other scientific tasks.

Note

Again, Robotics will be covered in much detail in the next thema; Robotics.

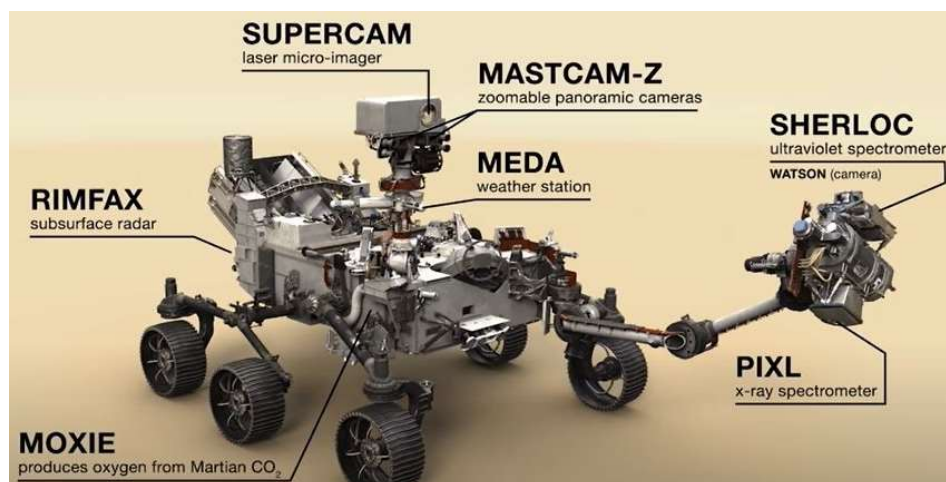


Fig.SP-AI-SRB-1.3: Image depicting major components of the Mars rover, Perseverance.

Using a coring drill on the end of the arm, the rover will collect core samples from the planet that will be left on the Mars surface for collection by a future mission.

(<https://www.youtube.com/watch?v=5qqsMjy8Rx0> - Mission Overview: NASA's Perseverance Mars Rover)

NOTE: For the upcoming topics, we will be focusing the following Branches and their concepts at a birds-eye view: (1) Machine Learning, (2) Neural Networks, and an application of AI in (3) Data Mining.

SAMPLE TOPIC ACTIVITIES

Activity 1: Exercises

- 1) Define Artificial Intelligence and compare it to Human Intelligence
- 2) Describe the different Branches of Artificial Intelligence and their functions in AI
- 3) List and explain the different applications of AI to date.

Topic 2: Machine Learning

Machine Learning is fairly a new thema introduced for STEM students. However, it will be taught in great details in the Technology phylum. Here, we will just look at the introduction of machine learning and elaborate on its different types.

2.1 Relation between AI and Machine Learning

Machine learning and deep learning are the two methods through which computers can learn artificial intelligence (see **Fig.SP-AI-SRB-2.1**).

2.1.1 Machine Learning

Machine learning is the science of teaching computers to think and act in human-like ways without having to be explicitly programmed. It makes use of algorithms to evaluate enormous amounts of data in order to make decisions and predictions.

2.1.2 Deep Learning

Deep Learning is an artificial intelligence function that mimics the neural networks of the human brain in order to understand patterns and make decisions. Deep learning sets up basic parameters about the data and educates the computer to learn on its own by spotting patterns utilizing several layers of processing, rather than structuring data to run through predefined equations.

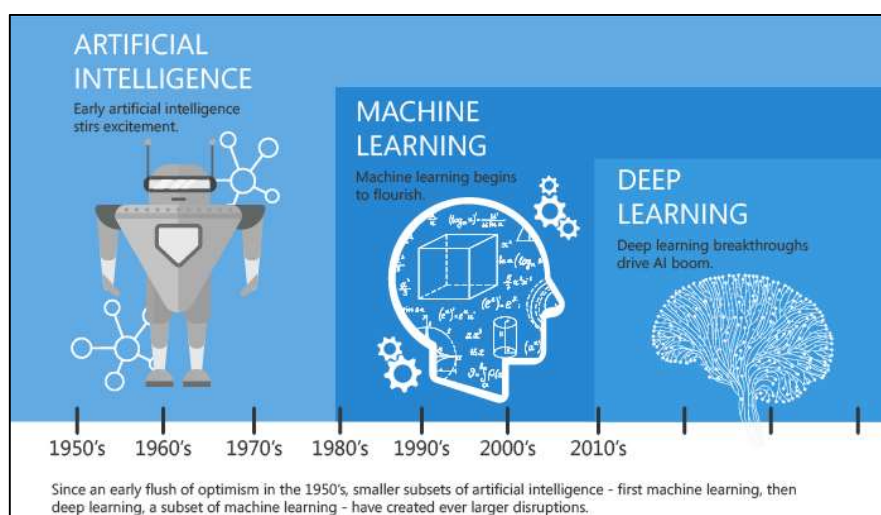


Fig.SP-AI-SRB-2.1: Machine Learning Vs Deep Learning (Source: LinkedIn)

As computers and algorithms process more data over a longer period of time, they learn and adapt their algorithms in the same way that humans do.

As we have already seen in *Topic 1*, Machine Learning is a branch of AI. In this topic, we will basically discuss about the fundamentals of Machine Learning and see how it is used in relation to AI.

2.2 Introduction

Machine learning is a word that has become more often used in recent years, thanks in part to technology advancements in our daily lives, such as Fitbits and Amazon Alexa.

2.2.1 What is Machine Learning?

Machine learning is a branch of artificial intelligence (AI). Machine learning's purpose, in a nutshell, is to learn from data and produce accurate outcome predictions without having to be explicitly programmed.

2.3 Uses of Machine Learning

In our day-to-day lives, machine learning is becoming more prominent. It can be utilized in a variety of ways, from providing intellectual insights to just executing tedious tasks. The following is a list of just a few of the applications of machine learning:

- Image recognition
- Voice recognition
- Predictions
- Finance and trading
- Medical diagnosis

Machine learning may be used in a variety of industries, and current technology is increasingly being integrated into enterprises.

2.4 Types of Machine Learning

Machine learning can be divided into three categories – Supervised, Unsupervised, and Reinforcement learning.

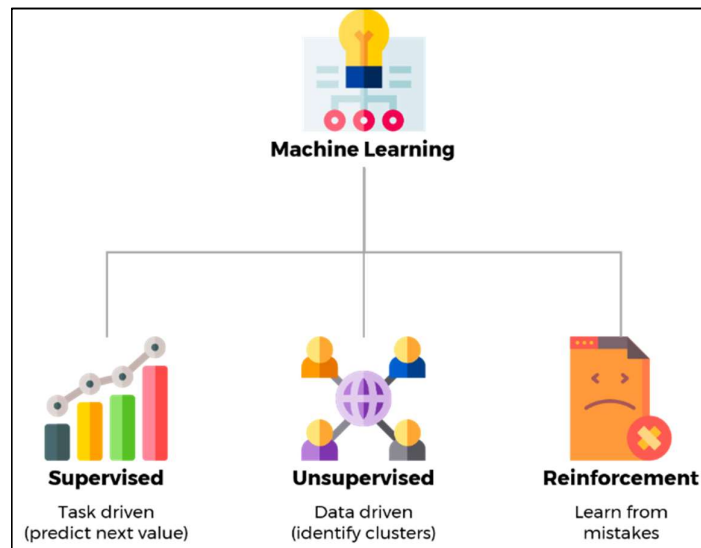


Fig.SP-AI-SRB-2.2 The 3 Types of Machine Learning

2.4.1 Supervised Learning

In supervised learning, an AI system is given data that has been labeled, meaning that each piece of data has been assigned the appropriate label.

The classification of spam emails as 'spam' or 'not spam' is an example of supervised learning in action.

The machine can use the data from earlier classifications to label future emails in the appropriate category based on the user's classifications.

2.4.2 Unsupervised Learning

An AI system is presented with unlabeled and uncategorized data in unsupervised learning. As a result, the system's algorithms act on the data without any prior training.

The outcome is determined by the algorithms that have been coded. Unsupervised learning is one method of putting AI to the test.

Let's imagine a baby is seeing his or her first bunch of cats and dogs. The baby has no concept what a cat or dog looks like, therefore they're 'unlabeled.' As a result, the newborn is unable to tell the difference between a cat and a dog right away.

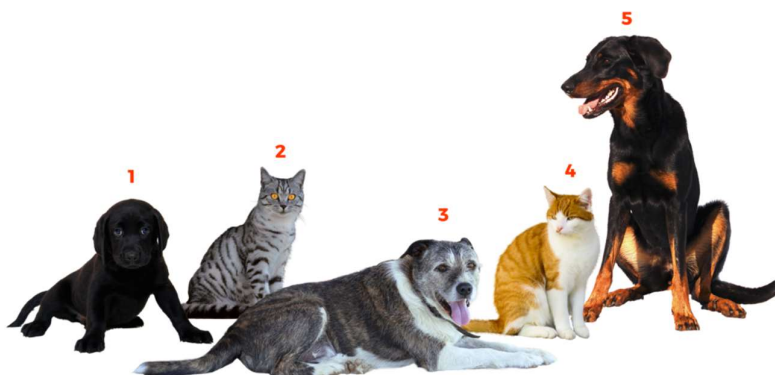


Fig.SP-AI-SRB-2.3: Group of Animals

The baby can identify those animals 1, 3, and 5 are similar to one another by looking at the group of animals, and that animals 2 and 4 are likewise similar to one another by looking at the group of animals. Although it is not possible to categorize the animals as cats or dogs, a pattern can be discovered and groupings formed based on that pattern. This is referred to as unsupervised learning.

2.4.3 Reinforcement Learning

Reinforcement learning is based on the fact that the machine in question (i.e., the agent) learns from its surroundings and interactions with them. When the agent performs wrong, he is penalized, and when he performs perfectly, he is rewarded.

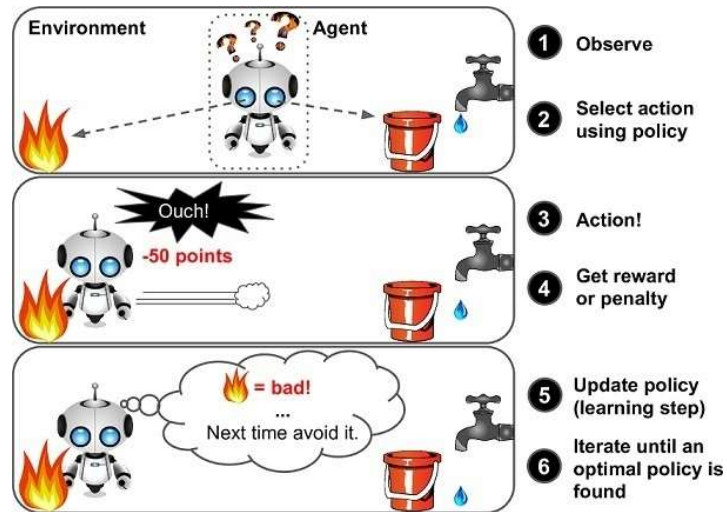


Fig.SP-AI-SRB-2.4: Reinforcement Learning (Source: RIP Tutorial: Reinforcement Learning)

There is no answer in reinforcement learning, but the reinforcement agent decides what to do to complete the job. It is obligated to learn from its experience in the absence of a training dataset.

For the next topic, we will look at Neural Networks.

SAMPLE TOPIC ACTIVITIES

Activity 1: Assignment

Title: Types of Machine Learning

For this assignment, individual students are to do research and answer the following points.

- List the different types of machine learning techniques.
- Explain each of these types.
- Give examples for each.

Activity 2: Group Discussion

- 1) Group Students
- 2) Task each group to explain in their own understanding what Supervised Learning, Unsupervised Learning and Reinforcement Learning is. Each group must also give examples regarding the applications of the types of Machine Learning above.
- 3) Each Group to have representatives present their discussions to the class.

Activity 3: Exercise

- 1) Define Machine Learning and Explain how it works in your own words
- 2) Define Deep Learning and Explain how it works in your own words
- 3) Explain the relationship between Machine Learning and AI
- 4) Give some example applications of Machine Learning
- 5) List the three types of Machine Learning

Topic 3: Neural Networks

3.1 Overview

3.1.1 Introduction

Neural Networks are systems made of many simple processing elements operating in parallel whose function is determined primarily by the pattern of connectivity.

3.1.2 Neurobiology

The design of neural networks draws heavily on developments in the field of neurobiology. Recent neurobiological studies have demonstrated neural plasticity with training in the somatosensory (sense of touch) cortex of mammalian brains and in the hippocampus (an area of the brain required to form long-term memories).

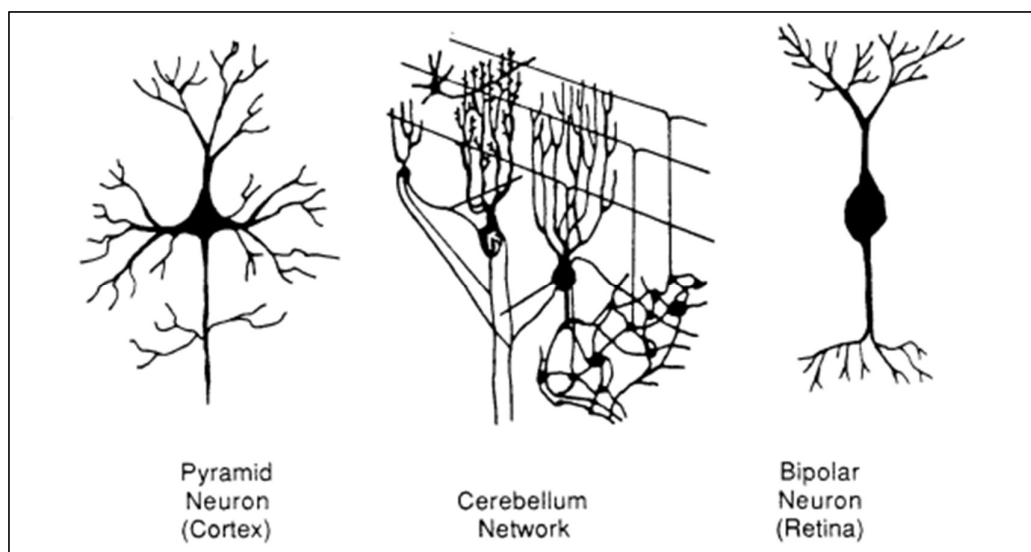


Fig.SP-AI-SRB-3.1: Small Biological Neural Network

Some neural network models are biologically unrealistic and use overly simplified neurons. Other more realistic and complex models are being developed and explored as simulation and analysis techniques improve and as results from neurobiological studies are disseminated.

3.1.3 Mathematical Theory

Mathematical theory in the field of neural networks builds on results from many other fields of science including pattern classification theory, information theory, nonlinear system theory, optimization theory, machine learning theory, automata theory, population dynamics, and evolution theory. Neural network researchers are, however, establishing a unique and strong theoretical foundation by exploring the limitations and capabilities of specific networks and algorithms and by exploring the theoretical limitations of learning in different contexts.

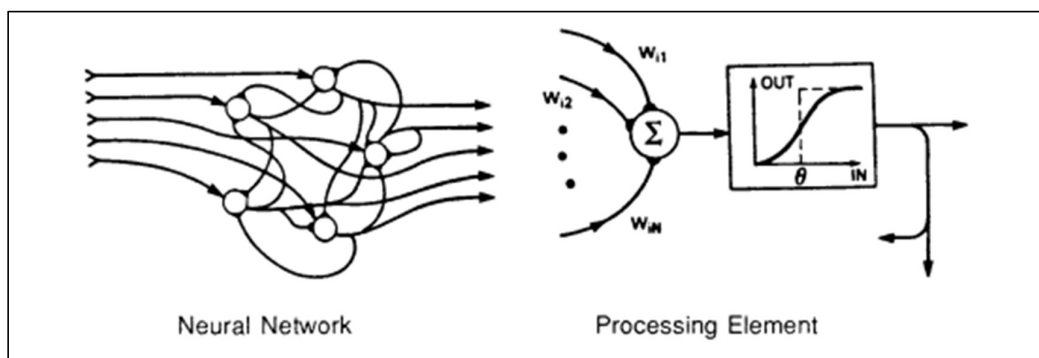


Fig.SP-AI-SRB-3.2: Example Neural Network

3.1.4 Theoretical Foundations of Neural Networks

A neural network is a computational structure modeled on biological processes. Interest in neural networks is prompted by two quests:

- the nervous system function of even a "lesser" animal can easily solve problems that are very difficult for conventional computers, including the best computers now available, and
- the ability to model biological nervous system function using man-made machines increases understanding of that biological function.

3.1.5 A Biological Inspiration

Fig.SP-AI-SRB-3.3 illustrates the sort of biological neuron which has influenced the development of "artificial," or computational, neural networks. **Fig.SP-AI-SRB-3.3** shows two neurons in synaptic contact. The synapse is the tissue connecting neurons and it is capable of changing a dendrite's local potential in a positive or negative direction, depending on the pulse it transmits. Note that these transmissions occur in very large numbers but they occur fairly slowly.

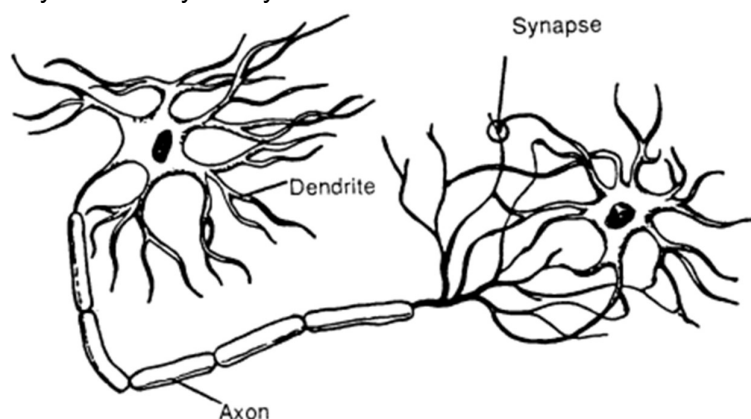


Fig.SP-AI-SRB-3.3: Biological Neurons

Notes from Fig.SP-AI-SRB-3.3

- The soma (or nerve cell) which is the large round central body of the neuron is anywhere from five to 100 microns in diameter.
- The axon is attached to the soma and is electrically active, producing the pulse which is permitted by the neuron.
- The electrically passive dendrites receive inputs from other neurons by means of a specialized contact – this is the synapse, which occurs where the dendrites of two different nerve cells meet.

3.1.6 Artificial Networks

"Artificial" neurons, as illustrated in **Fig.SP-AI-SRB-3.4**, are analogous to their biological inspirers. **Fig.SP-AI-SRB-3.4** presents the simplest artificial neuron configuration. Here neurons become processing elements, the axons and dendrites become wires, and the synapses become variable resistors carrying weighted inputs that represent data or the sums of weights of still other processing elements.

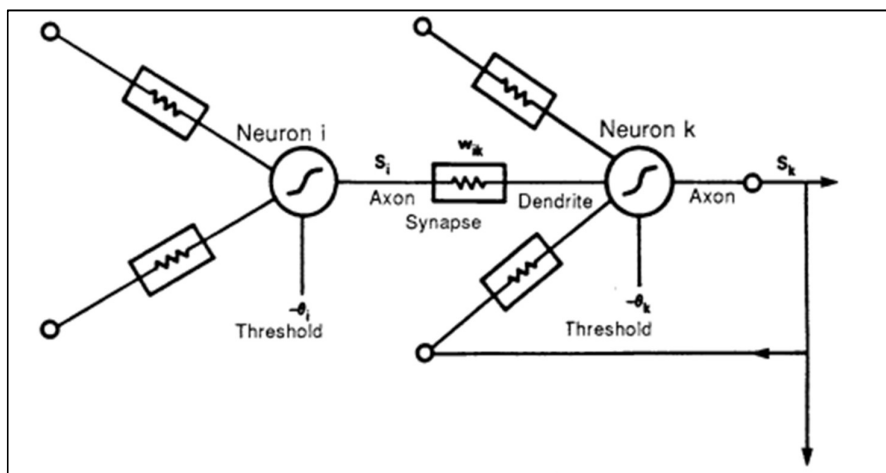


Fig.SP-AI-SRB-3.4: Artificial Neurons

These inputs - which are voltages that are proportional to weights that have been established - are summed across the resistor. The resistor is connected to an operational amplifier on which has been set a threshold so that when the sum of these voltages reaches a pre-set threshold, the neuron-or processing element-will fire. **Fig.SP-AI-SRB-3.4** assumes this processing element to be a hard-limiting device: that is, when the sum of the voltages is below threshold, the device will have a -1.0 output and not fire; when the sum reaches threshold, its output will be $+1.0$ and the device will fire. A simple neural net could thus be implemented using threshold logic technologies.

Processing elements can interact in many ways by virtue of the manner in which they are interconnected. **Fig.SP-AI-SRB-3.4** suggests a variety of possibilities:

- Processing elements which feed forward only; processing elements which have a feedback loop.
- Processing elements that are fully connected to all other processing elements; processing elements that are sparsely connected, linked only to a few others.

The nature and number of these feedback loops and connections depend on the architecture used to construct the neural network. The design of a neural network's feedback loops has implications for the nature of its adaptivity/trainability. The design of a network's interconnections has implications for its parallelism.

3.1.7 How a Neural Network Learns

There are many techniques used to train neural networks; they fall into three basic categories:

3.1.7.1 Supervised training

It requires the presence of an external teacher and labeling of the data used to train the network. The teacher knows the correct response wanted from the network and inputs an error signal when the network produces an incorrect response. The error signal "teaches" the network the correct response in a process that is explained below, and after a succession of learning trials the network consistently produces the correct response.

3.1.7.2 Unsupervised training

It uses unlabeled training data and requires no external teacher; data is presented to the network, which forms internal clusters that compress the input data into classification categories.

3.1.7.3 Self-supervised training

It is used by certain kinds of neural networks. The networks monitor performance internally, requiring no external teacher; an error signal is generated by the system and fed back to itself, and the correct response is produced after a number of iterations.

3.1.8 Layering of Neural Networks

In addition to learning, neural networks can be layered.

As noted in **Fig.SP-AI-SRB-3.5**, a network of 64×64 elements that are fully interconnected can undertake millions of simple parallel operations aimed at the dissection of a problem. **Fig.SP-AI-SRB-3.5** shows a neural network with layers M and N; each layer has 4,096 processing elements laid out in a 64×64 grid. If these two layers are fully interconnected—that is, if every processing element is linked to every other processing element—then this apparently simple structure sorts through no less than 16 million interconnects.

It is this pyramiding of many simple operations, to (automatically) provide complex classifications, that is the core of a neural network's technology promise.

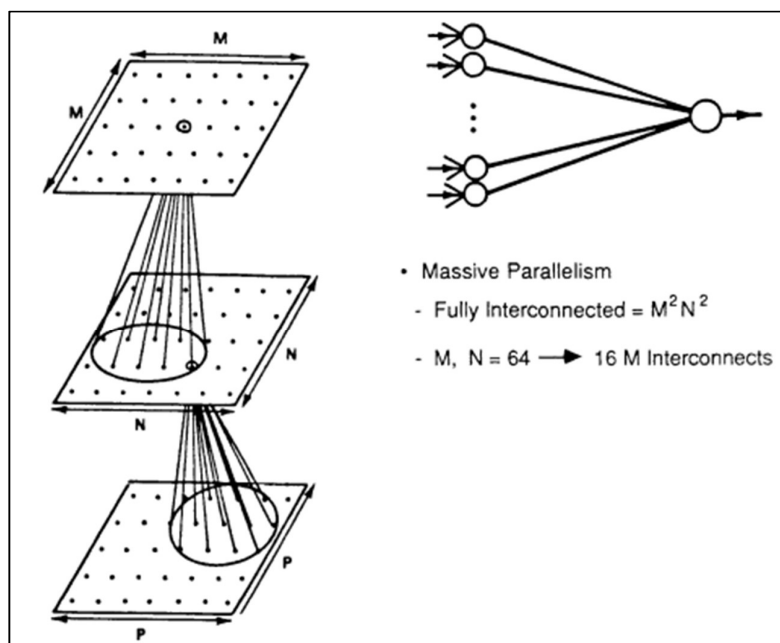


Fig.SP-AI-SRB-3.5: Parallelism in a Neural Network

3.2 Neural Networks and Other Information Processing Approaches

It is helpful to an understanding of neural networks and their implications to place them in the larger context of information processing approaches, for neural networks do not oppose other approaches as much as they extend them.

Table SP-AI-SRB-3.1 describes the history of neural network technology. Note that the current interest in neural nets is a result of the resurgence of a nearly dead technology.

Table SP-AI-SRB-3.1: Research History

Early Work 1940 - 1960 Fundamental Concepts	McCulloch & Pitts Hebb Farley & Clark Rosenblatt Steinbuch, Taylor	- Boolean Logic - Synaptic Learning Rule - First Simulation - Perceptron - Associative Memories
Transition 1960 - 1980 Theoretical Foundations	Widrow & Hoff Albus Anderson Von Der Malsburg Fukushima Grossberg	- LMS Algorithm - Cerebellum Model (CMAC) - Linear Associative Memory - Competitive Learning - Neocognitron - Art, BCS
Resurgence 1980 - Theory Biology Computers	Kohonen Feldman & Ballard Hopfield Reilly, Cooper et al Hinton & Sejnowski Rumelhart et al Rumelhart & McClelland Edelman, Reeke	- Feature Map - Connectionist Models - Associative Memory Theory - RCE - Boltzmann Machine - Back Propagation - PDP Books - Darwin III

Table SP-AI-SRB-3.2 delineates the major thrusts in information processing in a historical sense, beginning with system and information theory, which is really about database processing.

Table SP-AI-SRB-3.2: Comparison of Approaches

Approach	Method	Knowledge/ Acquisition	Form of Implementation
System & Information Theory	Model Data, Noise, Physical Constraints	Analyze Models To Find Optimal Algorithm	Hardware Implementation of Algorithm
AI Expert Systems	Emulate Human Expert Problem Solving	Observe Human Experts	Computer Program
Trainable Neural Nets	Design Architecture With Adaptive Elements	Train System With Examples	Computer Simulation or NM Hardware

This was followed by the development of artificial intelligence techniques for knowledgebase processing. It is worth noting that knowledgebase processing has not by any means eclipsed database processing; they are, in fact, quite complementary.

Neural networks have been touted as effective in tackling problems in, among other areas, machine vision, robotics, and speech understanding. It is highly unlikely that neural networks will replace either knowledgebase or database processing. They may find a niche within the whole scheme of information processing. The nature of the neural network niche becomes more apparent when neural networks as an information processing approach is compared to database and knowledgebase processing.

The expert system developer is interested in emulating a human expert's way of solving a specific set of problems, so he/she observes and analyzes a particular human expert(s), and then models that "intelligence," or expertise, in a computer program which explicates the rules (heuristic and otherwise) used by the human expert.

More generically, the system and information theorist models data, analyzes those models seeking the optimal algorithm, and then implements that algorithm in hardware.

If, for instance, the object of a hypothetical vision system is to find a cat in a backyard, current information processing techniques would require that the data containing the visual information be extracted from a digitizing preprocessor and entered into a computer; this data would be subject to a rather human-intensive analysis, followed by another human-intensive period of algorithm development. The algorithm would be implemented on yet another computer which would then be able to find the cat. The same system could not be used, if, perhaps, the goal is now to recognize a mouse; instead, the entire data-gathering, data analysis, algorithm development, and design implementation process must be repeated.

Neural network designers claim, by contrast, to place the "intelligence" of the network in its architecture and adaptation rules, which are optimized not to a single problem or application, but to an entire class of problems. The network is trained, either with supervision or in unsupervised mode, with examples. The neural network is then implemented either by simulating it on conventional computer hardware or in special-purpose neural network hardware.

Thus, a hypothetical neural network implementation of the system looking for the cat in the backyard may not even require the digitization of data; it can, perhaps, be entered directly from a sensor into the network in analog form. The neural network might then be trained by voice command to recognize the cat. Unlike its conventional information processing counterpart, the same neural network can also be trained to recognize a mouse.

This comparison suggests that neural networks can save substantial amounts of time and human effort which, in conventional information processing approaches, are now devoted to data analysis and algorithm development.

3.2.1 Overview of Brain Physiology

The brain consists of roughly 10 to 1000 billion neurons, cells with distinctive properties found only in the nervous system. There are hundreds of types of neurons, although they share a set of common features and functions.

The human cerebral cortex, for instance, is comprised of approximately 100 billion (10^{11}) neurons with each having roughly 1000 dendrites that form some 100,000 billion (10^{14}) synapses; given that this system operates at about 100 Hz, it functions at some 10,000 billion (10^{16}) interconnections per second. It weighs approximately three pounds, covers about 0.15 square meters, and is about two millimeters thick. This capability is clearly beyond anything which can be reconstructed or modeled; but it is perhaps, possible to understand how the brain performs information processing, and it is hoped that this understanding can be modeled and ultimately implemented in hardware.

A neuron is characterized by a cell body of approximately 30 microns in diameter, and thin (approximately 1 micron) branching extensions called dendrites and axons that are specialized for neuron-to-neuron communication. The dendrites receive inputs from other neurons and the axon provides output to other neurons. The neuron is imbedded in an aqueous solution of small ions, and its selective permeability to these ions establishes a negative electrical potential of some tens of millivolts.

Neurons receive electrochemical input signals from other cells at small (approximately 1 micron) discrete sites on their surface, as shown in **Fig.SP-AI-SRB-3.6**. These sites of neuron-to-neuron communication are called synapses, and number between 1000 to 100,000 for each neuron. The input signals are combined in various ways, triggering the generation of an output signal by a special region near the cell body under certain conditions. The output signal is a large (approximately 100 millivolt), brief (approximately .001 second), pulse that propagates without attenuation down the axon at velocities up to 120 meters/second. These pulses, called action potentials, can be produced at varying rates, up to several hundred per second. The axon branches many times, delivering the same pulse to all of its synapses. Some neurons generate sustained moderate pulse rates for seconds, while others produce short bursts of a few pulses in a few milliseconds.

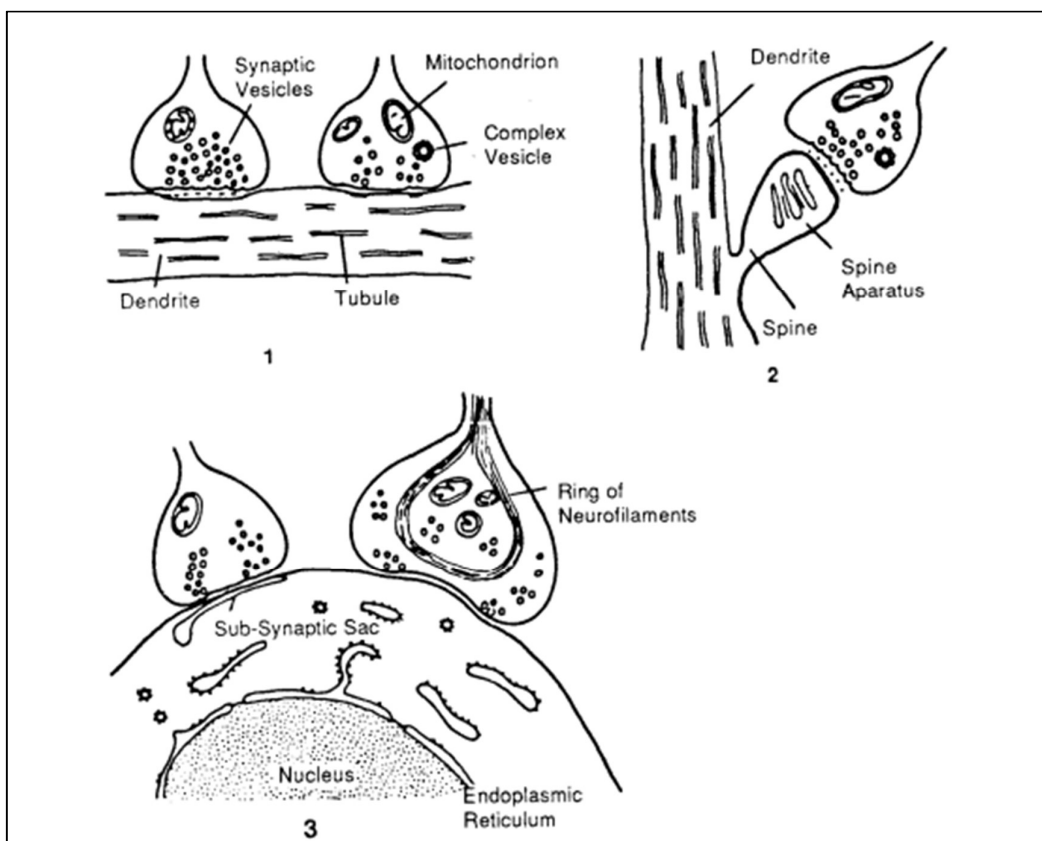


Fig.SP-AI-SRB-3.6: Three types of Neuron-to-Neuron Connections Called Synapses

It is tempting to think of a neuron, with its pulse-like output, a digital transistor with two states, pulse or no-pulse. However, this analogy requires clock-like synchronization of neuronal pulses on a time scale of the minimal inter-pulse interval—a few milliseconds. Such tight synchronization is not found in the brain (although brain regions are known to exhibit synchronized firing over larger spatio-temporal scales). Typically, many pulses must be averaged over tens of milliseconds to determine the average firing rate on an axon. Given this need for averaging, it is surprising that humans can respond to complex stimuli in fractions of a second when neural elements have computing times of tens of milliseconds. This response time indicates less than 100 sequential processing steps. The relative slowness of individual neurons is presumably countered by the use of massive parallelism in the nervous system. Parallel processing is an emerging field in computer science—however, the style and fine-grain scale of neuronal parallel processing is of an entirely different order.

On the transmitting or pre-synaptic side of the synapse, the pulse releases a specific chemical, called a neurotransmitter, that diffuses across a gap (approximately .01 microns) to the receiving side of the synapse. On the receiving or post-synaptic side of the synapse, the neurotransmitter binds to specific receptor molecules anchored in the membrane, opening ionic channels and changing the electrochemical potential. The magnitude of this change is determined by many factors local to the synapse, such as the amount of transmitter released on the pre-synaptic side and the number of receptor molecules on the post synaptic side. These factors can change with time, thus changing the effectiveness or "strength" of the synapse. It is not known how many of the thousands of synapses on a cell

are strong, or even functional at a given time. Estimates of the number of active synapses necessary to cause a cell to "fire" (generate an output pulse) range from a few to hundreds.

It has long been thought that the neuronal input/output function was relatively simple. Synaptic potentials are spatially and temporally filtered in accord with the cable properties and geometry of the neuron and integrated at the cell body. The resulting potential triggers action potentials only when it is greater than a fixed threshold. This description is reasonably accurate for a large number of neurons, but modern methods have revealed much more complicated mechanisms. Many neuroscientists now think of a neuron more as a microprocessor rather than as a leaky capacitor feeding a one-shot flip-flop circuit. In addition, there are neurons without axons, synapses that are bi-directional, synapses onto other synapses and onto axons, and non-chemical, electrical synapses.

The neurons of the brain are organized into thousands of discrete structures, each with its own particular types of neurons, patterns of neuron-to-neuron connections, and role in brain function. Shapes and sizes vary from small globular nuclei a few millimeters in diameter with tens of thousands of neurons, to large cortical sheets a few millimeters thick by one meter square with tens of billions of neurons. Large-scale information processing tasks, such as speech or vision, are performed by systems composed of many interconnected structures, each serving a small though specific subtask. This modular architecture is largely responsible for past successes in understanding the brain. Carefully controlled experiments can selectively study one brain system at a time, untangling its components from those of others, and elucidating their role in the systems function.

The growth and formation of brain structures is, of course, ultimately governed by genetic information. However, the equivalent of a schematic at the level of individual neurons does not exist. Rather, genetics specifies the types and numbers of neurons and the general patterns of connections, but leaves the details of neuronal wiring up to the adaptive processes of development. In a similar way, it is generally presumed that the acquisition of complex skills such as speech, vision, and movement-achieved with little supervision during the first years of life-is due to adaptive processes of neuronal re-organization operating within and upon the existing processing structures.

Present understanding is largely confined to the sensory and motor systems. The structures involved with more high-level cognitive functions-such as categorization, memory, decision making, and planning-have been tentatively identified, but much is still a mystery. For example, a multitude of specific structures in the visual system that analyze aspects of the elements in an image, such as their color, motion and depth, can be pointed out-but there is no knowledge of how those aspects are grouped into a representation of the physical objects in the scene, nor how the objects are recognized, categorized, and named. However, there are no presently foreseeable scientific obstacles barring an understanding of such cognitive functions, although this will likely require a greater interaction between theory, computational models, and equipment.

3.2.2 Recent Results in Neurobiology

Neural networks are based on relatively old biological knowledge, although at the time the field began, this knowledge was relatively current. Since then, knowledge of the nervous system has mushroomed. The old ideas of the importance of neurons, synapses, and learning are still true-i.e., the original ideas of neural networks are still valid, but due to new techniques, a great deal more is known about these and other structures and phenomena.

In this subsection important biological facts and principles that have been largely ignored in neural network research, and yet could significantly advance neural network technology, will be outlined. Emphasis will be on the critical role played by biologically oriented neural network models and theoretical analysis in translating biological knowledge into computational technology. Furthermore, the resulting computational models can greatly contribute to understanding the complex biological phenomena.

3.2.2.1 Neurons

It is now recognized that single neurons are capable of more functions than the standard one of linear summation followed by nonlinear thresholding employed in neural networks. These enhanced functions are in the analog domain and arise from nonlinear interactions between dendritic inputs and electrical state of the cell. This information is possible because of breakthroughs in intracellular recording technology, both electrical and optical. However, this information must be supplemented by mathematical models of membrane biophysics and computational models of how the analog functions are used in the larger computations of the system. Mathematical models of membrane biophysics are well advanced and rare now being extended to the network level in studies of the hippocampus. Computational models of analog retinal motion detectors have also been proposed. An understanding and implementation of such analog processes could lead to great increases in the computational power of individual nodes in neural networks. The technological feasibility of applying such analog processing to neural networks has been demonstrated by the construction of an artificial retina in silicon.

3.2.2.2 Synapses

The molecular mechanisms for changing synaptic efficacy, and hence neuronal signaling, are now becoming known. The work on long-term potentiation (LTP) and the NMDA receptor partially justifies the popular Hebb rule used by modelers, but significantly extends it as well. It is especially notable that the role of these synaptic mechanisms in behavioral learning is also being investigated. All these developments promise more powerful learning rules for neural networks-however, biologically oriented models relating synaptic changes to network phenomena and behavior are needed to guide the experimental work.

3.2.2.3 Map Representation

The importance of highly structured map-like representations in the brain is now very clear, especially in the visual system where the maps are linked together in a modular, hierarchical fashion. Neuroethologists are coupling physiology with ethological and behavioral studies to pinpoint the role of specific neuronal maps in solving specific real-world problems.

3.2.2.4 Self-Organization

It has long been known that the sensory systems go through brief critical periods in post-natal development in which the circuits are fine-tuned in response to environmental exposure. The environmental and physiological factors essential to this process have been extensively studied in the visual system. Recent work is determining the internal biochemical mechanisms involved. Network models are needed to explain how all these factors work together to produce the appropriate adult circuits, and there has been considerable work in this area. In addition to brief periods of self-organization during critical periods, there is strong evidence for reorganization in adult brains as well. This has long been predicted due to the neurological evidence of recovery after brain trauma. Recent work in the somatosensory system has demonstrated such processes of recovery at the neuronal level. The cortical topographic map of the skin is dynamically regulated by the

differential use of skin regions. The cortical map territory representing a given skin region will shrink or expand as that skin region is stimulated less or more than neighboring skin regions, with consequent decreases or increases of sensitivity, respectively. Models seeking to explain how this self-regulation arises from neuronal mechanisms are clearly needed, and several have been proposed. Understanding of neuronal self-organization phenomena could lead to neural networks whose internal representations of sensory information would be maximally efficient and adaptive to changes in the environment.

3.2.2.5 Network Circuitry

The knowledge of the circuitry of the brain is expanding rapidly due to new tracing methods. Special dyes are injected inside a cell, filling it and marking its inputs and outputs for microscopic analysis. It has always been clear that there is much more structure in the nervous systems than neural network models consider, but now neuroscience is in a position to actually determine what the circuitry is. Studies of very young animals have shown that much of the structural complexity of brains does not come from learning or synaptic weight modification within globally connected networks, as assumed by most neural networks. No matter how much sensory deprivation a developing cat is subjected to, it will still have a retina, a lateral geniculate nucleus, a striate cortex and so forth. They might not work very well, but the basic information for macroscopic development seems to come from genetic and developmental processes rather than environmental sources. It seems unlikely that the clever designs of local brain circuits will be discovered solely with the current methods of neural networks, which start with global connectivity and change the weights to optimize an input/output mapping. However, this method may provide helpful leads in more macroscopic mapping problems, as suggested by recent modeling work.

3.2.2.6 Network Dynamics

The availability of powerful computers and electronics has enabled the simultaneous recording of electrical activity at many closely spaced sites in the brain, both electrically and optically. This promises to reveal new modes of network dynamics and information processing. However, because of the tremendous data rates involved, severe data reduction methods must be employed to solve the "needle in the haystack" problem. Network models are needed to suggest what the important spatial and temporal scales might be in order to guide the data reduction process and the experimental design itself. Model based data reduction methods have been extensively applied in the olfactory bulb by Freeman and are under development by Shaw for the visual cortex.

3.2.2.7 Learning

Neurobiologists are extensively exploring the physiological correlates of learning in higher mammals. The response properties of single cells can now be determined while the animal is awake and learning a task or an association between stimuli. Pharmacological agents can be administered to block or aid the learning-related changes. These techniques, as well as others mentioned above, are determining the neuronal bases of learning and memory. Learning, of course, is one of the key features of neural networks. Most network models use a form of supervised learning in which the correct outputs for a large set of inputs are already known and is used to set the connection strengths to achieve the desired input/output mapping. However, there are equally important learning tasks that cannot include an omniscient teacher, such as any situation in which novelty is present. Biological research can aid neural networks in the design of such unsupervised learning paradigms and network models are expected to be an essential part of the discovery process itself. Recent efforts along these lines, in which there is a closed feedback loop between network modeling and experiment, has suggested a new mechanism for categorical memory. This

new memory model, unlike most neural network classifiers, explicitly uses time in its representation of stimulus categories. The output response pattern evolves in time, at first giving a classification response, followed by more and more of a stimulus-specific response. If two stimuli are presented at the same time, the output oscillates between the two correct outputs, thus solving the superposition problem that plagues categorizers which exclusively use spatial representations. Of course, for problems with many stimuli, further attentional methods are needed, as discussed below.

3.2.2.8 Attention

Animals are able to make sense of their complex sensory environments because they can attend to just those features or objects that are important to the task at hand, thus serializing their analysis of the parallel information flow provided by the senses. This essential function has been extensively studied by psychologists for the last 20 years, and is now under neurophysiological investigation. The ability to attend to one of many objects seems especially remarkable when considering the parallel and distributed manner in which objects are represented in the nervous system. Even within a single modality, features of a single object are represented on a multitude of neural structures, and yet the features of multiple objects are not mixed up with each other (the apples in a basket of oranges are seen as red and not orange). The problem of attention is thus related to the problem of sensory fusion.

3.2.2.9 Organizing Principles

Mountcastle has proposed a general organizing principle for information processing based on the modularity of brain structure and function, and the evidence that groups of neurons, rather than single neurons, are the irreducible processing elements. Edelman has put forward a complementary theory of higher brain function that posits that the brain is a selective system, sharing principles in common with the immune system and Darwinian evolution. This theory argues that the paradigm of information processing that is implicitly, if not explicitly, employed in both neuroscience and neural network research is the inadequate paradigm for understanding brain function and for building truly intelligent machines. These organizing principles are at present very controversial within neuroscience itself-however, neural network models based on these principles have been developed demonstrating new computational ideas and stimulating fresh insights into the theories.

3.3 Tasks Neural Networks Perform and Representative Models

3.3.1 Introduction

Neural network models can perform a wide variety of tasks useful for speech, vision, and robotics problems. The primary tasks performed include:

- Pattern classification,
- Self-organization or clustering,
- Associative memory storage and access,
- Vision and speech preprocessing,
- Computational tasks such as those required to solve combinatorial optimization problems,
- Nonlinear input/output mapping such as is required in robotics control,
- Sensory fusion of inputs from multiple sensors, and
- Recognition of time-varying patterns as is required for speech recognition.

3.3.2 Tasks that Neural Networks Perform

The field of neural networks includes many different models designed to address a wide range of problems in the primary application areas of speech, vision, and robotics.

Two separate classes of models have been explored. Neurobiological models closely model some aspect of biological brain function and/or animal or human behavior. Computational models perform some technically important function. Neurobiological models are judged by how well they summarize existing neurobiological and psychophysical data and on the accuracy of predictions. Computational models are judged primarily by their potential impact and by their performance and implementation efficiency.

Researchers developing computational models tend to focus on one or more of the following common problem areas:

- Designing neural network architectures to implement important conventional architectures.
- Developing new highly parallel neural network algorithms.
- Analyzing algorithms using simulation and theory.
- Implementing algorithms in hardware.

Most researchers focus on neural networks that perform those seven major tasks illustrated graphically in **Fig.SP-AI-SRB-3.7**.

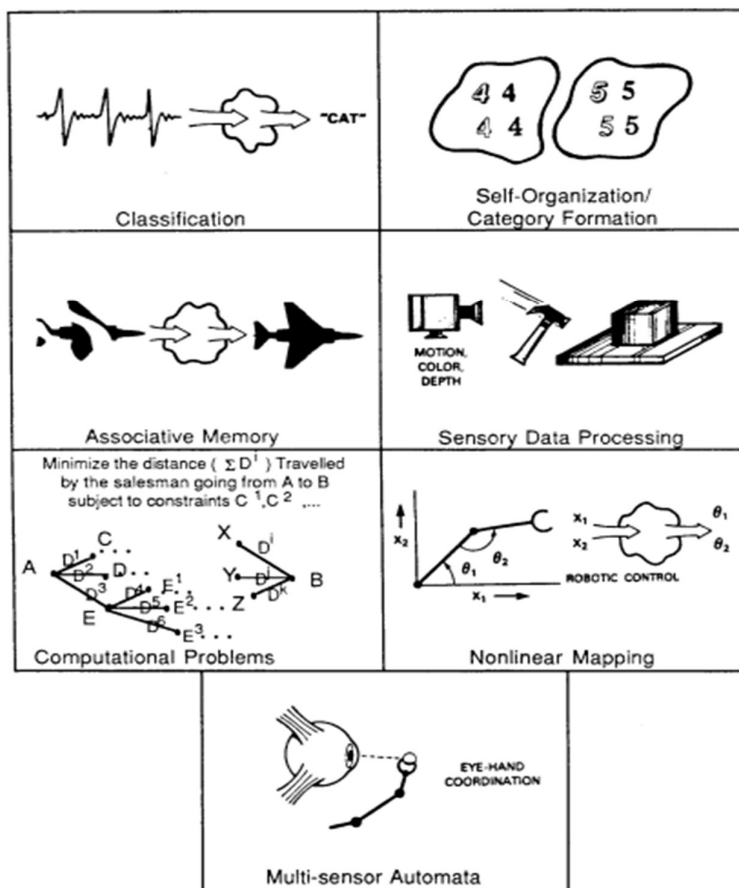


Fig.SP-AI-SRB-3.7: Tasks that can be performed

These tasks include:

- **Classification.**

Classifiers are trained with supervision using labeled training data to partition input patterns into a pre-specified number of groups or classes. These could represent different words for speech recognizer or different objects for a visual image classifier. Inputs to a classifier may be binary- or continuous-valued.

- **Self-organization/Category Formation.**

Self-organizing networks partition input examples into groups or clusters using unlabeled training data. This type of clustering or vector quantization is an efficient technique for reducing information that must be processed at higher levels with little loss in performance. It also makes good use of the large amount of unlabeled training data that is typically available in speech and vision problems. The number of clusters formed may be pre-specified or determined from the examples.

- **Associative Memory.**

An associative, or content-addressable memory provides a complete memory item from a key consisting of a partial or corrupted version of the memory. For example, it might return a complete article citation from only the author's name or a complete image of a face from only the bottom half.

- **Sensory Data Processing.**

An enormous amount of real time preprocessing is performed in the peripheral sensory vision and hearing centers. Neural networks can perform this function in real time using massive parallelism.

- **Computational Problems.**

Custom Neural Network architectures can be designed to solve specific computational problems, such as the traveling salesman problem and other constrained optimization problems, using nonlinear analog computation.

- **Nonlinear Mapping.**

Many neural networks can map a vector of analog inputs into an output vector using a nonlinear mapping function which can be learned from training data. These types of mapping are useful in many areas, including robot control and nonlinear signal processing

- **Multi-sensor Automata.**

A number of complex, multi-module neural network automata have been built with visual input and a robot arm to manipulate objects in an environment. These automata demonstrate how an eye or camera can learn to scan a scene using self-supervision, how control of a multi-jointed arm and hand can then be learned using self-supervision, and

then how the eye and hand can be coordinated to perform simple tasks. These automata also demonstrate how inputs from multiple sensors can be fused to provide classification performance better than could be achieved with a single sensor.

3.3.3 Illustrative Neural Network Models

Past work has led to many different networks which address the above problems and computational tasks. Some of these networks are presented in **Fig.SP-AI-SRB-3.8** and **Table SP-AI-SRB-3.3**. **Fig.SP-AI-SRB-3.8** emphasizes the biological inspiration and types of training supported by the networks and **Table SP-AI-SRB-3.3** emphasizes the tasks the networks perform.

Table SP-AI-SRB-3.3: Neural Network Models Arranged by Task

NEURAL NETS FOR A VARIETY OF TASKS					
Pattern Classification	Speech Recognition	Machine Vision	Robotics	Signal Processing	Optimization/ Computation
Adaline	Martin Speech Preprocessor	BCS/FCS	CMAC Cerebellum Model	Adaline	Boltzmann Machine
Art	Masking Field	Cellular Automata	Darwin III	Multi-Layer Perceptron	Cellular Automata
Boltzmann Machine	Multi-Layer Perceptron	Connectionist Models	Infant		Markov Random Field
Competitive Learning	Silicon Cochlea	Darwin II	Multi-Layer Perceptron		Talk-Hopfield Nets
Feature Map	Synaptic Triad	High-Order Nets	Tensor Nets		Winner-Take-All Net
Hamming Net	Time-Concentration Net	Markov Random Field	Topographic Maps		
Multi-Layer Perceptron (Back Propagation)	Time-Delay Net	Neocognitron			
Perceptron	Trace	Parameter Net			
Reduced Coloumb Energy (RCE)	Viterbi Net	Silicon Retina			

Networks in **Fig.SP-AI-SRB-3.8** are positioned vertically according to the type of adaptation used during training and horizontally according to the relationship between the network and biological nervous systems.

Networks in the lower third of **Fig.SP-AI-SRB-3.8** use fixed weights and are not adapted during use. Networks in the middle third use supervised training, and networks in the upper third use unsupervised training or are self-supervised.

Networks in the left of **Fig.SP-AI-SRB-3.8** are computational models with little connection to biological networks other than fine-grain parallelism. Networks in the middle have architectures modeled after early hearing and vision areas, and networks toward the right

are attempts to model higher cortical areas. Since present knowledge of cortical function is minimal, these rightmost networks are only rough guesses that remain to be verified.

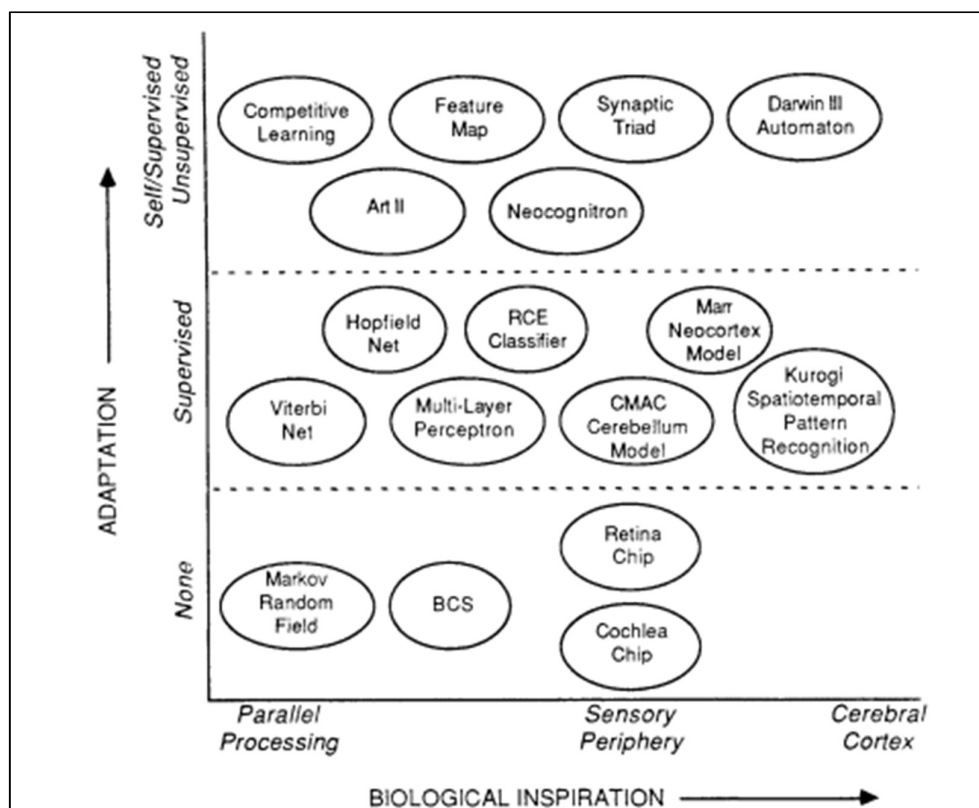


Fig.SP-AI-SRB-3.8: Neural Network Models

More detail on six important networks from **Fig.SP-AI-SRB-3.8** is provided in **Table SP-AI-SRB-3.4**.

Table SP-AI-SRB-3.4: Six Representative Neural Network Models

NETWORK	FUNCTION	INTER-CONNECTS	ADAPT	TYPE OF LEARNING	BIOLOGICAL INSPIRATION
Hopfield Network	CAM, Energy Minimization	Feedback, Full	Yes	Supervised	Parallel Processing
Multi-Layer Perceptron (Back Propagation)	Classification, Nonlinear Mapping	Feedforward, Full	Yes	Supervised	Parallel Processing
CMAC	Robotics, Non-linear Mapping	Feedforward, Full	Yes	Supervised	Cerebellum Model
Feature Map	Speech, Vector Quantization	Feedforward, Full	Yes	Unsupervised	Cortical Maps
Darwin III	Eye-Hand Coordination Automation	Complex	Yes	Self-supervised	Neuronal Groups
Retina Chip	VLSI Vision Front End	Grid, Local	No	None	Retina

Networks in the lower third of **Fig.SP-AI-SRB-3.8** are used for early vision processing (Markov random field, BCS, retina chip) and for speech preprocessing (cochlea chip). The Markov random field network represents a neural network architecture for an algorithm useful for early vision processing. The BCS, or boundary contour system, is a new neural network algorithm that forms boundaries used for early vision. The retina chip and cochlea chip are analog VLSI integrated circuits modeled after the retina and cochlea. The retina chip includes photodetectors with a logarithmic output and a wide dynamic range, circuitry to average inputs locally in space, and circuitry to enhance temporal changes in contrast. The size of the initial retina chip is small (48-by-48 pixels) and it requires primarily local connections as shown in **Table SP-AI-SRB-3.4**. It includes roughly 100,000 transistors, and has been demonstrated using live inputs. The retina and cochlea chips are good examples of the potential of integrating sensors and parallel neural processing in a compact package.

Networks in the middle third of **Fig.SP-AI-SRB-3.8** are trained with supervision. Hopfield introduced many current researchers to the field of neural networks through his work with a recurrent network commonly known as the Hopfield network. This is a fully-connected iterative network, as shown in **Table SP-AI-SRB-3.4**, which can be used as a content-addressable memory or to solve combinatorial optimization problems such as the traveling salesman problem.

The multi-layer perceptron trained with a new algorithm called backpropagation also introduced many researchers to the field of neural networks. It has been used successfully for many difficult problems, including forming text to phoneme rules, as a classifier in speech and vision problems to classify sonar targets and to form nonlinear mappings useful in nonlinear signal processing and robotics. As shown in **Table SP-AI-SRB-3.4**, a multi-layer perceptron is a layered feedforward network where weights are adjusted with supervision to provide the desired output during training.

The Viterbi network is a neural network architecture which implements a successful temporal decoding algorithm used for speech recognition using nonlinear analog processing. The reduced Coulomb energy (RCE) classifier differs from the other classifiers in **Fig.SP-AI-SRB-3.8** in that extra nodes are added when an error occurs due to the presentation of a novel input far from inputs which have already been seen. This allows rapid single-trial learning and the ability to form complex decision regions rapidly. The Marr/Albus cerebellum model, called the CMAC model in **Table SP-AI-SRB-3.4**, was originally a model of the cerebellum. This is a part of biological brains that is required to coordinate fine motor movements. The CMAC model forms complex nonlinear maps adaptively and has been used in a number of robotic applications.

The Marr neocortex model is an example of an attempt to determine the function of cell layers in the neocortex, as is the Kurogi spatiotemporal pattern recognition model. Kurogi's model is unique among high-level models in that it is one of the few to model internal neuron dynamics and specify how nerve action potentials are generated and processed on a spike-by-spike basis. Most other models include variables that represent the average cell firing rate. This is a neurobiological model which makes predictions whose details can be tested by neurobiologists.

The upper third of **Fig.SP-AI-SRB-3.8** includes models that are trained without supervision or with self-supervision. The three left-most networks vector-quantize the inputs and form

clusters. Competitive learning networks form internal clusters from binary inputs while Kohonen's feature map forms clusters from continuous valued inputs. Both networks form clusters using a fixed number of nodes and a sequential algorithm similar in purpose to the conventional k-means clustering algorithm. The feature map algorithm is unique in that nodes representing clusters become arranged in a two-dimensional spatial grid or map where nearby nodes respond to stimuli with similar characteristics. The adaptive resonance theory (ART) clustering algorithm forms a new cluster whenever an input is too far from an already existing cluster.

The synaptic triad network is the only network in **Fig.SP-AI-SRB-3.8** other than the Viterbi network that can recognize temporal pattern sequences. It self-organizes to form pattern sequence detectors that could be used in speech recognition applications or other applications, including vision, involving recognition of temporal patterns. The neocognitron is a multi-layer image classifier that can recognize handwritten characters and provides translation invariance. Feature detectors in each layer are created using competitive learning techniques. The right-most model, Darwin III, is a complex simulated automaton that learns to follow a moving target and touch the target with a multi-jointed arm. It is the most recent instantiation of a developing theory of brain function called Neural Darwinism.

3.3.4 Available Implementation Tools

The field of neural networks, as might be expected, has developed its own computational vernacular, some understanding of which is necessary to appreciate neural network simulation and implementation requirements. Some of the key concepts and terms (see **Fig.SP-AI-SRB-3.9**) are these:

- A typical neural network contains many more interconnects than neurons, or processing elements.
- Each interconnect requires one multiply/accumulate operation for summing.
- While digital computers are normally assessed in terms of storage or memory (where the unit of measure is words) and speed (instructions-per-second or floating-point-operations-per-second), the neural network vernacular defines storage as the value of the input weights and measures it in terms of interconnects; neural network speed is described in terms of interconnects-per-second within a layer or between layers. (This way of conceiving neural network storage is important only in the case of network simulation; in the case of implementation in special-purpose hardware, storage would be handled by resistive networks and would be defined differently.)

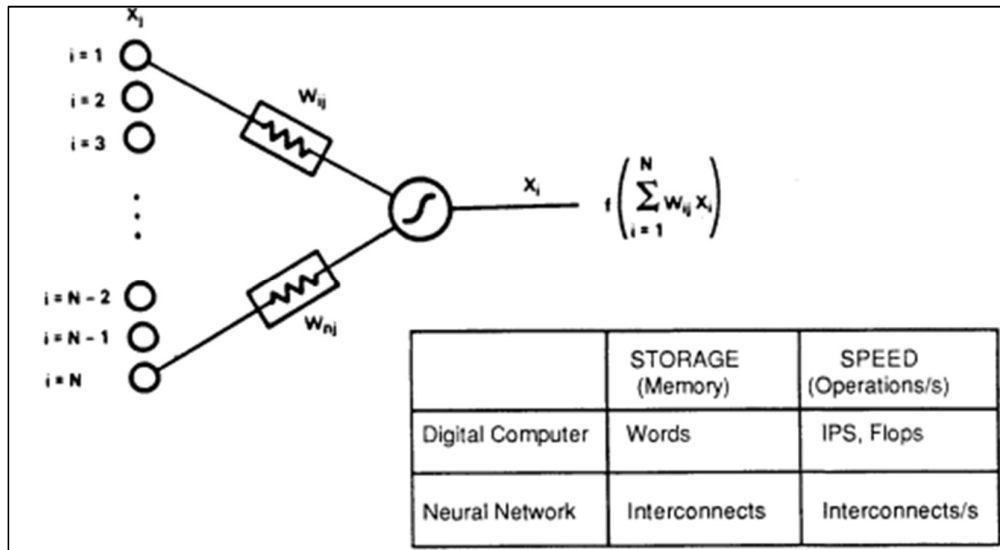


Fig.SP-AI-SRB-3.9: Neural Network Computation Units

To understand both this neural network computational vernacular and its implications for neural network tools and applications, DARPA has proposed to use the terms interconnects and interconnects-per-second to chart a set of coordinates, placing interconnects on the abscissa and interconnects-per-second on the ordinate. DARPA uses this environment chart to illustrate the computational capabilities of certain biological systems (**Fig.SP-AI-SRB-3.10**).

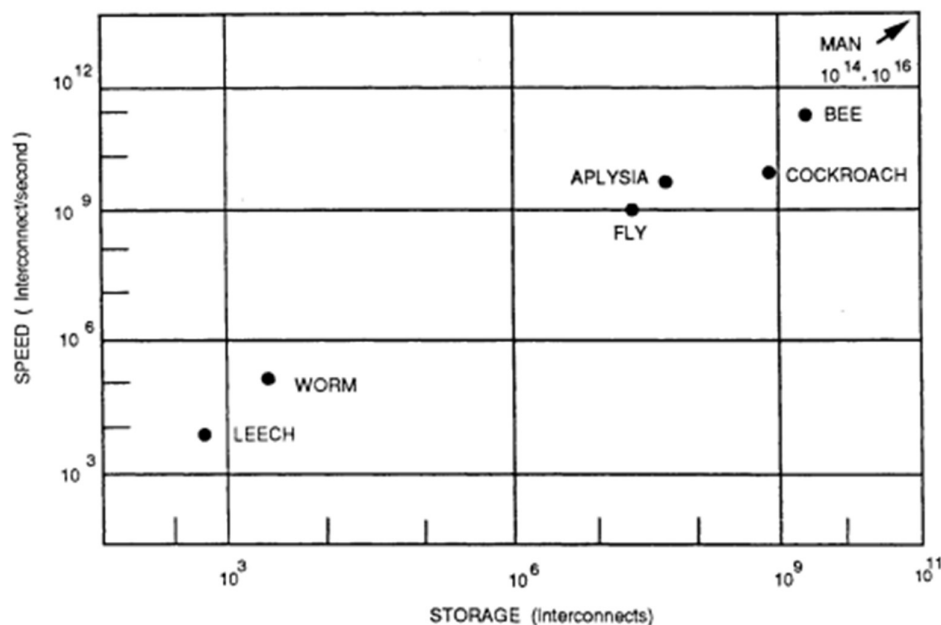


Fig.SP-AI-SRB-3.10: Computational capabilities of Biological Networks

Significantly, man - with the staggering capacity of 10^{14} interconnects and 10^{16} interconnects-per-second-is beyond the range of the chart for reasons which will shortly be apparent. Within the range of **Fig.SP-AI-SRB-3.10** are such creatures as the worm, the fly, and the bee. "Lowly" as a fly or a bee may seem, neural network researchers would be very pleased indeed to replicate their capabilities with a machine.

3.3.5 A Simulated World

That will be difficult with the tools now available. **Fig.SP-AI-SRB-3.11** delineates the variety of tools to which today's neural network researchers have access. They range from the low-priced microprocessor-based workstations, through attached processors and bus-oriented processors, to the more expensive massively-parallel machines, and finally to supercomputers at the very high end.

WORKSTATIONS		INTERCONNECTS	INTERCONNECTS/s
MICRO/MINI COMPUTERS	PC/AT	100K	25K
	Sun 3	250K	250K
	Vax	32M	100K
	Symbolics	10M	35K
ATTACHED PROCESSORS	Anza	500K	45K
	Delta 1 *	1M	10M
	Transputer *	2M	3M
BUS ORIENTED PROCESSORS	Mark III, V	1M	500K
	Odyssey	256K	20M
	MX - 1/16 *	50M	120M
MASSIVELY PARALLEL	CM-2 (64K)	64M	13M
	Warp (10)	320K	10K
	Butterfly (64)	60M	8M
SUPERCOMPUTERS	Cray XMP 1-2	2M	50M

* Projected Performance

Fig.SP-AI-SRB-3.11: Neural Network Simulation Hardware

Generally, the micro/minicomputers provide a very modest interconnects-per-second capability, though in some cases storage capacity is substantial. The speeds of these devices are limiting; neural network models take a very long time to run on them. Attached processors improve this situation somewhat, since they can boost interconnects-per-second into the millions from the hundreds of thousands.

Bus-oriented processors, in some cases, raise by an order of magnitude the number of interconnect-per-second available, but storage is not equivalently greater.

Supercomputers do not offer significantly more capability than systems which cost far less.

In addition, the programming necessary to run neural network models on these massively parallel machines is very complex-it is, in fact, a problem which limits the extent to which these systems can be used to conduct neural network research. Moreover, the architectural

limitations of these systems prevent researchers from stacking up several of them to significantly boost their storage or speed.

3.3.6 Typical Application: The Kanji Character Recognizer

Fig.SP-AI-SRB-3.12 illustrates Nestor Inc's Kanji Character Recognizer, which in this depiction, is translating handwritten Kanji characters for "neural network" into English.

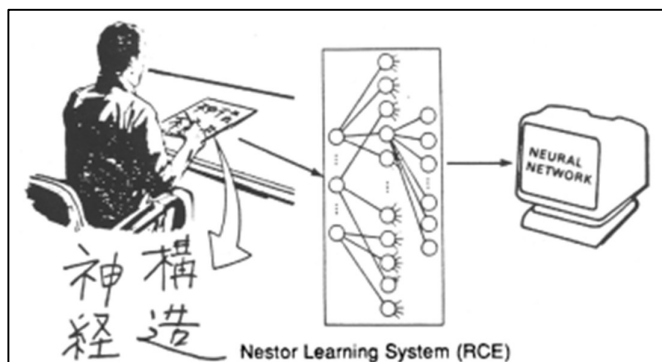


Fig.SP-AI-SRB-3.12: Kanji Character Recognizer

This system can accurately recognize approximately 2,500 handwritten Kanji characters better than 92% of the time. This compares favorably with the average Japanese reader, who is capable of recognizing between 1,800 and 3,000 characters; an intellectual has a grasp of perhaps 10,000 characters.

One useful feature of this system, thanks to the fact that it's a neural network, is that upon training it is capable of recognizing any alphabet--e.g., Cyrillic, Hebrew, etc.

The Kanji Character Recognizer is based on Nestor Inc's NLS (Nestor Learning System), a set of neural network modules based on the reduced Coulomb energy (RCE) architecture. Each of the NLS three-layer neural network modules is defined to process a different feature subspace of inputs; the RCE architecture allows each network to describe a class-separating mapping that can support any required degree of nonlinearity between pattern class territories. A controller module synthesizes the outputs of the individual neural networks and directs their training.

The Nestor Learning System is a software simulation which operates on an IBM PC/AT personal computer or workstations from Sun Microsystems or Apollo Computer.

3.3.7 Typical Application – Multilayer Perception Vowel Classifier

Fig.SP-AI-SRB-3.13 indicates how a multi-layer perceptron neural network has been used as a vowel classifier.

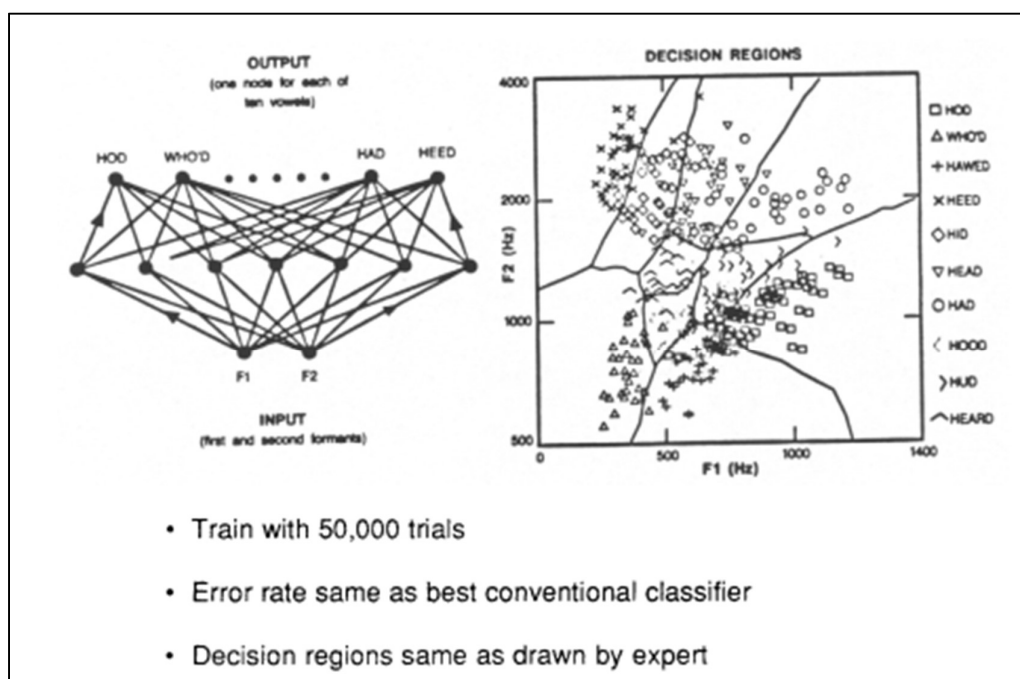


Fig.SP-AI-SRB-3.13: Multi-layer Perception Vowel Classifier

Conventional speech recognition systems do not perform well when their task is to recognize continuous speech from more than one talker because of several factors, including:

- the variability and overlap of information in the acoustic signal,
- the need for high computational rates (a human-like system must match inputs to 50,000 words in real time), and
- the multiplicity of analyses-phonetic, phonemic, syntactic, and pragmatic-that must be performed.

The input for the network shown in **Fig.SP-AI-SRB-3.13** consisted of the first and second formants (i.e., the regions of concentration of energy which combine to make up the frequency spectrum of a spoken sound) from roughly 330 tokens of ten different vowels spoken in context by 67 men, women, and children. Half the data was used for training the network; the other half was used to test it. The network was trained using backpropagation with 50,000 trials.

The network's error rate proved to be the same as the best conventional classifier, and the decision regions it formulated were the same as those which were hand-drawn by a human expert.

Note

Advanced topics that may be covered later on is Detailed Approaches – Single and Multi-Layer Perceptions, Recurrent Networks, and the Technologies and Tools for implementing Neural Networks.

SAMPLE TOPIC ACTIVITIES

Activity 1: Group Presentation

- 1) Group Students
- 2) Each group is to select a topic below to research on and prepare for Class Presentation on Neural Networks. Each student must include in their research a case study or example of research topic.
 - a. Explain how Neurobiology compares to Neural Networks.
 - b. How does Supervised Training in Neural Networks Work?
 - c. How does Unsupervised Training in Neural Networks Work?
 - d. How does Self-supervised Training in Neural Networks Work?

Topic 4: Data Mining

For this topic, we will be basically looking at the fundamentals of data mining.

4.1 Introduction

This topic is an introduction to a young and promising field called data mining and knowledge discovery from data. The material in this topic is presented from a database perspective, where emphasis is placed on basic data mining concepts and techniques for uncovering interesting data patterns hidden in large data sets. The implementation methods discussed are particularly oriented toward the development of scalable and efficient data mining tools.

4.2 What Motivated Data Mining? Why is it Important?

Data mining has attracted a great deal of attention in the information industry and in society as a whole in recent years, due to the wide availability of huge amounts of data and the imminent need for turning such data into useful information and knowledge. The information and knowledge gained can be used for applications ranging from market analysis, fraud detection, and customer retention, to production control and science exploration.

Data mining can be viewed as a result of the natural evolution of information technology. The database system industry has witnessed an evolutionary path in the development of the following functionalities (**Fig.SP-AI-SRB-4.1**): data collection and database creation, data management (including data storage and retrieval, and database transaction processing), and advanced data analysis (involving data warehousing and data mining).

For instance, the early development of data collection and database creation mechanisms served as a prerequisite for later development of effective mechanisms for data storage and retrieval, and query and transaction processing. With numerous database systems offering query and transaction processing as common practice, advanced data analysis has naturally become the next target.

Since the 1960s, database and information technology has been evolving systematically from primitive file processing systems to sophisticated and powerful database systems. The research and development in database systems since the 1970s has progressed from early hierarchical and network database systems to the development of relational database systems (where data are stored in relational table structures), data modeling tools, and indexing and accessing methods. In addition, users gained convenient and flexible data access through query languages, user interfaces, optimized query processing, and transaction management. Efficient methods for on-line transaction processing (OLTP), where a query is viewed as a read-only transaction, have contributed substantially to the evolution and wide acceptance of relational technology as a major tool for efficient storage, retrieval, and management of large amounts of data.

Database technology since the mid-1980s has been characterized by the popular adoption of relational technology and an upsurge of research and development activities

on new and powerful database systems. These promote the development of advanced data models such as extended-relational, object-oriented, object-relational, and deductive models. Application-oriented database systems, including spatial, temporal, multimedia, active, stream, and sensor, and scientific and engineering databases, knowledge bases, and office information bases, have flourished. Issues related to the distribution, diversification, and sharing of data have been studied extensively. Heterogeneous database systems and Internet-based global information systems such as the World Wide Web (WWW) have also emerged and play a vital role in the information industry.

The steady and amazing progress of computer hardware technology in the past three decades has led to large supplies of powerful and affordable computers, data collection equipment, and storage media. This technology provides a great boost to the database and information industry, and makes a huge number of databases and information repositories available for transaction management, information retrieval, and data analysis.

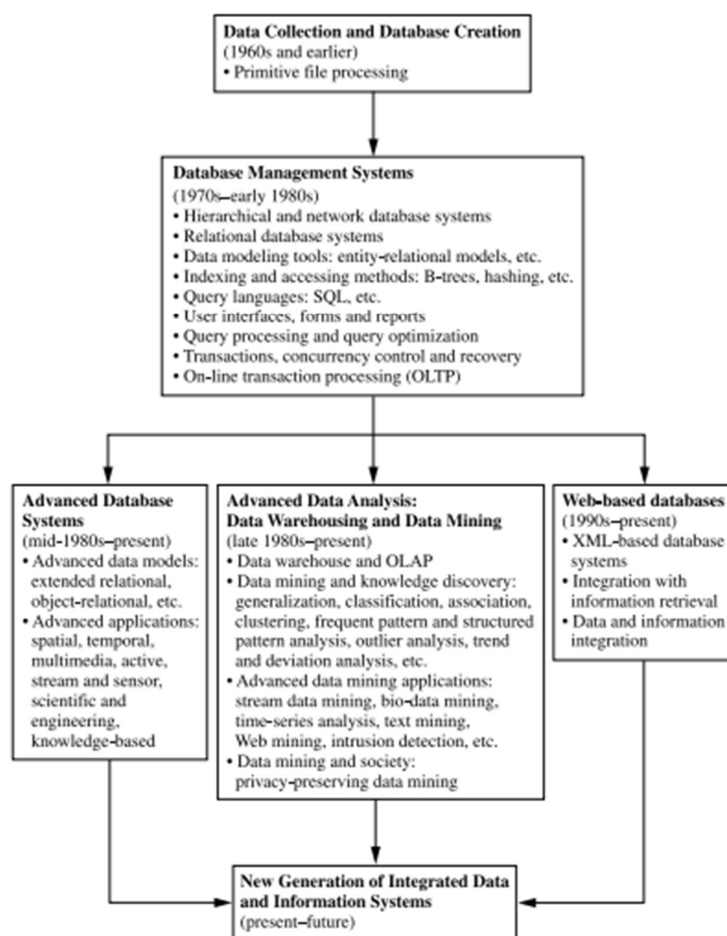


Fig.SP-AI-SRB-4.1: The evolution of database system technology

Data can now be stored in many different kinds of databases and information repositories. One data repository architecture that has emerged is the data warehouse, a repository of multiple heterogeneous data sources organized under a unified schema at a single site in order to facilitate management decision making. Data warehouse technology includes

data cleaning, data integration, and on-line analytical processing (OLAP), that is, analysis techniques with functionalities such as summarization, consolidation, and aggregation as well as the ability to view information from different angles. Although OLAP tools support multidimensional analysis and decision making, additional data analysis tools are required for in-depth analysis, such as data classification, clustering, and the characterization of data changes over time. In addition, huge volumes of data can be accumulated beyond databases and data warehouses. Typical examples include the World Wide Web and data streams, where data flow in and out like streams, as in applications like video surveillance, telecommunication, and sensor networks. The effective and efficient analysis of data in such different forms becomes a challenging task.

The abundance of data, coupled with the need for powerful data analysis tools, has been described as a data rich but information poor situation. The fast-growing, tremendous amount of data, collected and stored in large and numerous data repositories, has far exceeded our human ability for comprehension without powerful tools (**Fig.SP-AI-SRB-4.2**).



Fig.SP-AI-SRB-4.2: *We are data rich, but information poor*

As a result, data collected in large data repositories become “data tombs”—data archives that are seldom visited. Consequently, important decisions are often made based not on the information-rich data stored in data repositories, but rather on a decision maker’s intuition, simply because the decision maker does not have the tools to extract the valuable knowledge embedded in the vast amounts of data. In addition, consider expert system technologies, which typically rely on users or domain experts to manually input knowledge into knowledge bases. Unfortunately, this procedure is prone to biases and errors, and is extremely time-consuming and costly. Data mining tools perform data analysis and may uncover important data patterns, contributing greatly to business strategies, knowledge bases, and scientific and medical research. The widening gap between data and information calls for a systematic development of data mining tools that will turn data tombs into “golden nuggets” of knowledge.

4.3 What is Data Mining?

Simply stated, data mining refers to extracting or “mining” knowledge from large amounts of data. The term is actually a misnomer. Remember that the mining of gold from rocks or sand is referred to as gold mining rather than rock or sand mining. Thus, data mining should have been more appropriately named “knowledge mining from data,” which is

unfortunately somewhat long. “Knowledge mining,” a shorter term, may not reflect the emphasis on mining from large amounts of data. Nevertheless, mining is a vivid term characterizing the process that finds a small set of precious nuggets from a great deal of raw material (**Fig.SP-AI-SRB-4.3**).



Fig.SP-AI-SRB-4.3: Data mining – searching for knowledge (interesting patterns) in your data

Thus, such a misnomer that carries both “data” and “mining” became a popular choice. Many other terms carry a similar or slightly different meaning to data mining, such as knowledge mining from data, knowledge extraction, data/pattern analysis, data archaeology, and data dredging.

Many people treat data mining as a synonym for another popularly used term, Knowledge Discovery from Data, or KDD. Alternatively, others view data mining as simply an essential step in the process of knowledge discovery. Knowledge discovery as a process is depicted in **Fig.SP-AI-SRB-4.4** and consists of an iterative sequence of the following steps:

1. Data cleaning (to remove noise and inconsistent data)
2. Data integration (where multiple data sources may be combined)
3. Data selection (where data relevant to the analysis task are retrieved from the database)
4. Data transformation (where data are transformed or consolidated into forms appropriate for mining by performing summary or aggregation operations, for instance)
5. Data mining (an essential process where intelligent methods are applied in order to extract data patterns)
6. Pattern evaluation (to identify the truly interesting patterns representing knowledge based on some interestingness measures)
7. Knowledge presentation (where visualization and knowledge representation techniques are used to present the mined knowledge to the user)

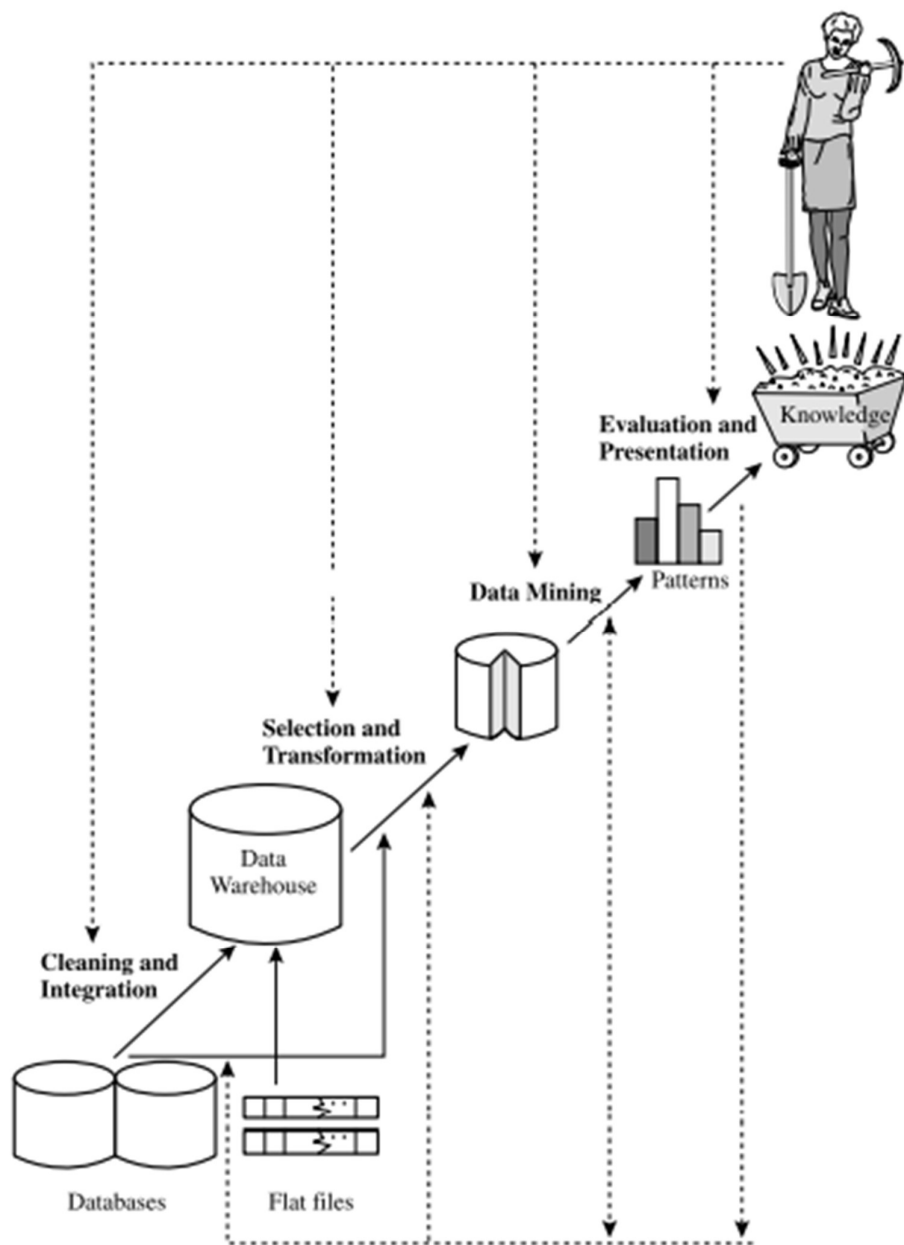


Fig.SP-AI-SRB-4.4: Data mining as a step in the process of knowledge discovery

Steps 1 to 4 are different forms of data preprocessing, where the data are prepared for mining. The data mining step may interact with the user or a knowledge base. The interesting patterns are presented to the user and may be stored as new knowledge in the knowledge base. Note that according to this view, data mining is only one step in the entire process, albeit an essential one because it uncovers hidden patterns for evaluation.

We agree that data mining is a step in the knowledge discovery process. However, in industry, in media, and in the database research milieu, the term data mining is becoming more popular than the longer term of knowledge discovery from data. Therefore, in this book, we choose to use the term data mining. We adopt a broad view of data mining functionality: data mining is the process of discovering interesting knowledge from large amounts of data stored in databases, data warehouses, or other information repositories.

Based on this view, the architecture of a typical data mining system may have the following major components (**Fig.SP-AI-SRB-4.5**):

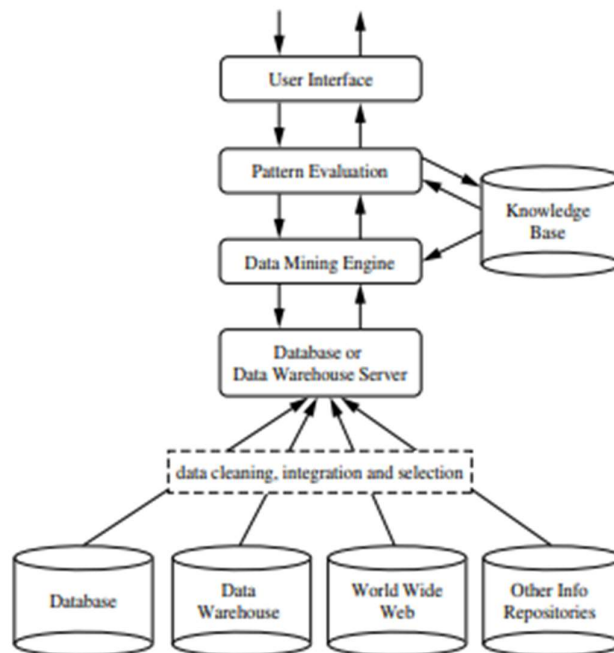


Fig.SP-AI-SRB-4.5: Architecture of a typical data mining system.

- Database, data warehouse, World Wide Web, or other information repository: This is one or a set of databases, data warehouses, spreadsheets, or other kinds of information repositories. Data cleaning and data integration techniques may be performed on the data.
- Database or data warehouse server: The database or data warehouse server is responsible for fetching the relevant data, based on the user's data mining request
- Knowledge base: This is the domain knowledge that is used to guide the search or evaluate the interestingness of resulting patterns. Such knowledge can include concept hierarchies, used to organize attributes or attribute values into different levels of abstraction. Knowledge such as user beliefs, which can be used to assess a pattern's interestingness based on its unexpectedness, may also be included. Other examples of domain knowledge are additional interestingness constraints or thresholds, and metadata (e.g., describing data from multiple heterogeneous sources).
- Data mining engine: This is essential to the data mining system and ideally consists of a set of functional modules for tasks such as characterization, association and correlation analysis, classification, prediction, cluster analysis, outlier analysis, and evolution analysis.
- Pattern evaluation module: This component typically employs interestingness measures (Section 1.5) and interacts with the data mining modules so as to focus the search toward interesting patterns. It may use interestingness thresholds to filter out discovered patterns. Alternatively, the pattern evaluation module may be integrated with the mining module, depending on the implementation of the data mining method used. For efficient data mining, it is highly recommended to push the evaluation of

pattern interestingness as deep as possible into the mining process so as to confine the search to only the interesting patterns

- **User interface:** This module communicates between users and the data mining system, allowing the user to interact with the system by specifying a data mining query or task, providing information to help focus the search, and performing exploratory data mining based on the intermediate data mining results. In addition, this component allows the user to browse database and data warehouse schemas or data structures, evaluate mined patterns, and visualize the patterns in different forms.

From a data warehouse perspective, data mining can be viewed as an advanced stage of on-line analytical processing (OLAP). However, data mining goes far beyond the narrow scope of summarization-style analytical processing of data warehouse systems by incorporating more advanced techniques for data analysis.

Although there are many “data mining systems” on the market, not all of them can perform true data mining. A data analysis system that does not handle large amounts of data should be more appropriately categorized as a machine learning system, a statistical data analysis tool, or an experimental system prototype. A system that can only perform data or information retrieval, including finding aggregate values, or that performs deductive query answering in large databases should be more appropriately categorized as a database system, an information retrieval system, or a deductive database system.

Data mining involves an integration of techniques from multiple disciplines such as database and data warehouse technology, statistics, machine learning, high-performance computing, pattern recognition, neural networks, data visualization, information retrieval, image and signal processing, and spatial or temporal data analysis. We adopt a database perspective in our presentation of data mining in this book. That is, emphasis is placed on efficient and scalable data mining techniques. For an algorithm to be scalable, its running time should grow approximately linearly in proportion to the size of the data, given the available system resources such as main memory and disk space. By performing data mining, interesting knowledge, regularities, or high-level information can be extracted from databases and viewed or browsed from different angles. The discovered knowledge can be applied to decision making, process control, information management, and query processing. Therefore, data mining is considered one of the most important frontiers in database and information systems and one of the most promising interdisciplinary developments in the information technology.

4.4 Data Mining – On What Kind of Data?

In this section, we look at a variety of various data repositories that can be mined. Data mining should, in theory, be applicable to any type of data repository, as well as transient data like data streams. As a result, relational databases, data warehouses, transactional databases, sophisticated database systems, flat files, data streams, and the World Wide Web will all be covered in our investigation of data repositories.

Object-relational databases and particular application-oriented databases, such as spatial databases, time-series databases, text databases, and multimedia databases, are examples of advanced database systems. Each repository system may have its own set of issues and approaches for mining.

4.4.1 Relational Databases

A database system, also known as a database management system (DBMS), is made up of a database, which is a collection of interconnected data, and a series of software applications that manage and access the data. The software programs include techniques for defining database structures, data storage, concurrent, shared, or distributed data access, and assuring the consistency and security of stored data despite system crashes or unwanted access attempts.

A relational database is made up of a number of tables, each with its own name. Each table has a set of characteristics (columns or fields) and can hold a huge number of tuples (records or rows). Each tuple in a relational table represents an item with a unique key and a set of attribute values to describe it. For relational databases, a semantic data model, such as an entity-relationship (ER) data model, is frequently built. The database is represented as a set of entities and their relationships in an ER data model.

Consider the example below on the next page.

Database queries defined in a relational query language, such as SQL, or graphical user interfaces can be used to access relational data. In the latter, the user can utilize a menu, for example, to select which properties should be included in the query and how these attributes should be constrained. A query is first turned into a series of relational operations, such as join, selection, and projection, and then optimized for speed.

A query allows you to retrieve specific subsets of data. Assume you're in charge of analyzing the AllElectronics data. You can ask something like "Show me a list of all items that were sold in the last quarter" using relational queries. Aggregate functions in relational languages include sum, avg (average), count, max (maximum), and min (minimum). These allow you to ask questions like "Show me the total sales for the last month, broken down by branch," "How many sales transactions occurred in December?" and "Which salesperson had the most sales?"

When data mining is used on relational databases, we can look for trends or patterns in the data. Data mining systems, for example, can evaluate customer data to anticipate prospective customers' credit risk based on their income, age, and previous credit history. Data mining algorithms can also spot anomalies, such as items whose sales are significantly lower than predicted compared to the prior year. Such variances can then be studied further (e.g., has the packaging of such items changed, or has the price increased significantly?).

Since relational databases are one of the most widely available and rich information repositories, they represent a major data type in our data mining research.

Example

AllElectronics' relational database. The following relation tables define the AllElectronics company: customer, item, employee, and branch. **Fig.SP-AI-SRB-4.6** shows fragments of the tables provided below.

customer

<u>cust_ID</u>	name	address	age	income	credit_info	category	...
C1	Smith, Sandy	1223 Lake Ave., Chicago, IL	31	\$78000	1	3	...
...	...	...	...	...	...	...	...

item

<u>item_ID</u>	name	brand	category	type	price	place_made	supplier	cost
I3	hi-res-TV	Toshiba	high resolution	TV	\$988.00	Japan	NikoX	\$600.00
I8	Laptop	Dell	laptop	computer	\$1369.00	USA	Dell	\$983.00
...	...	...	...	...	...	...	...	...

employee

<u>empl_ID</u>	name	category	group	salary	commission
E55	Jones, Jane	home entertainment	manager	\$118,000	2%
...	...	...	...	...	...

branch

<u>branch_ID</u>	name	address
B1	City Square	396 Michigan Ave., Chicago, IL
...	...	...

purchases

<u>trans_ID</u>	<u>cust_ID</u>	<u>empl_ID</u>	date	time	method_paid	amount
T100	C1	E55	03/21/2005	15:45	Visa	\$1357.00
...	...	...	...	...	...	...

items_sold

<u>trans_ID</u>	<u>item_ID</u>	qty
T100	I3	1
T100	I8	2
...	...	...

works_at

<u>empl_ID</u>	<u>branch_ID</u>
E55	B1
...	...

Fig.SP-AI-SRB-4.6: Fragments of relations from a relational database for AllElectronics

- A unique customer identity number (cust ID), customer name, residence, age, occupation, annual income, credit information, category, and other factors make up the customer relationship.
- In the same way, each relation item, employee, and branch have a set of characteristics that describe their properties.
- Relationships between or among numerous relation tables can also be represented using tables. These include purchases (a customer purchases products, resulting in a sales transaction that is handled by an employee), items sold (a list of the items sold in a specific transaction), and works at (a customer purchases items, resulting in a sales transaction that is handled by an employee) (employee works at a branch of AllElectronics).

4.4.2 Data Warehouses

Assume that AllElectronics is a thriving international corporation with locations all over the globe. Each branch has its own database system. The president of AllElectronics has requested that you analyze the company's third-quarter sales by item category and branch. This is a difficult task, especially because the essential data is dispersed across multiple databases and physical locations.

This task would be simple if AllElectronics had a data warehouse. A data warehouse is a collection of data from various sources that is kept in a consistent structure and is usually located at a single location. Data warehouses are built through a series of steps that include data cleansing, data integration, data transformation, data loading, and data refreshing on a regular basis. The basic foundation for building and using a data warehouse for AllElectronics is shown in **Fig.SP-AI-SRB-4.7**.

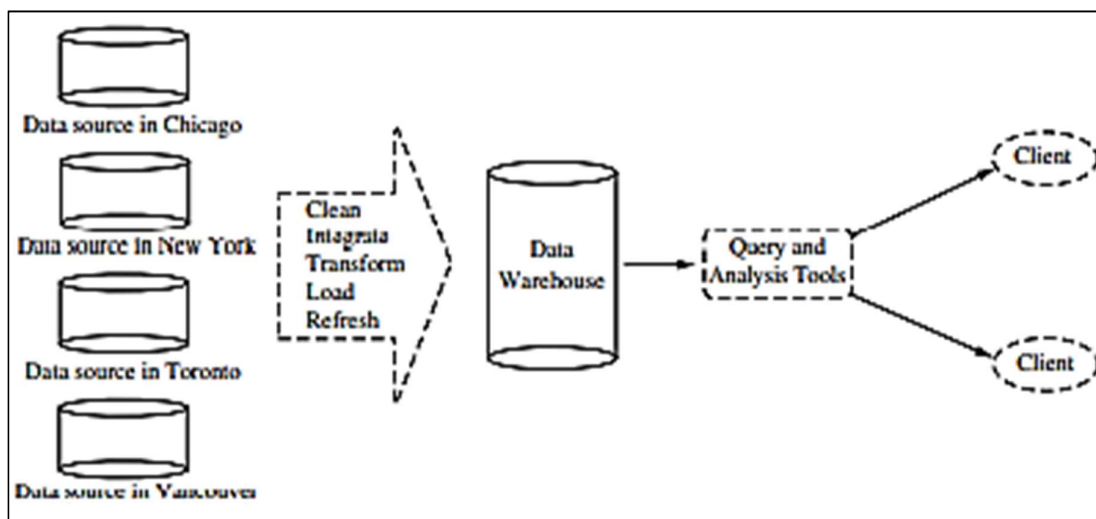


Fig.SP-AI-SRB-4.7: Typical framework of a data warehouse for AllElectronics

The data in a data warehouse is structured around important subjects like customer, item, supplier, and activity to make decision-making easier. The data is often aggregated and saved to offer information from a historical viewpoint (such as the previous 5–10 years). Instead of recording the details of each sales transaction, the data warehouse might keep a summary of transactions per item type for each store or, at a higher level, for each sales region.

A multidimensional database structure is typically used to model a data warehouse, with each dimension corresponding to an attribute or collection of attributes in the schema and each cell storing the value of some aggregate measure, such as count or sales amount. A data warehouse's physical structure can be either a relational data storage or a multidimensional data cube. A data cube gives you a multidimensional perspective of your data and allows you to pre-calculate and obtain summarized data quickly.

You might wonder, "What's the difference between a data warehouse and a data mart?" The scope of a data warehouse is enterprise-wide since it collects information on subjects that cover a whole firm. A data mart, on the other hand, is a subset of a data warehouse that is organized by department. It concentrates on a few themes and so has a department-wide scope.

Data warehouse systems are highly suited for on-line analytical processing, or OLAP, because they provide multidimensional data views and precomputation of summarized data. In order to present data at different levels of abstraction, OLAP procedures employ background knowledge about the domain of the data being investigated. Different user perspectives are accommodated by such processes. Drill-down and roll-up are two OLAP processes that allow the user to examine data at different levels of summarization, as shown in **Fig.SP-AI-SRB-4.8 (b)**. For example, we can dig down into quarterly sales data to get month-by-month sales statistics.

Although data warehouse technologies can aid with data analysis, more in-depth and automated analysis requires additional data mining tools.

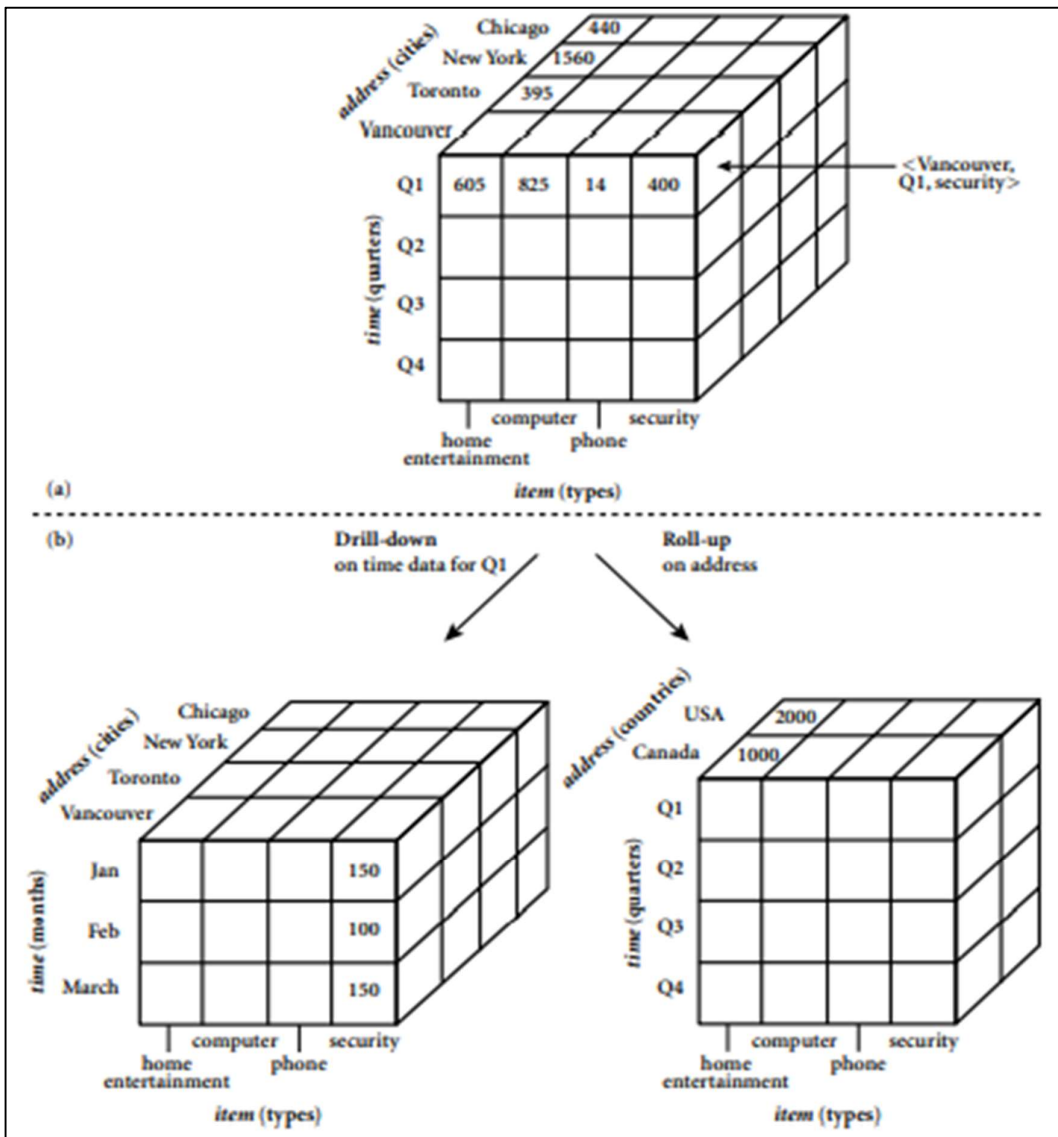


Fig.SP-AI-SRB-4.8: A multidimensional data cube, commonly used for data warehousing, (a) showing summarized data for AllElectronics and (b) showing summarized data resulting from drill-down and roll-up operations on the cube in (a). For improved readability, only some of the cube cell values are shown.

4.4.3 Transactional Databases

A transactional database, in general, is made up of a file with each record representing a transaction. A transaction often comprises a unique transaction identity number (trans ID) as well as a list of the transaction's components (such as items purchased in a store).

Extra tables may be associated with the transactional database, which hold additional information about the sale, such as the transaction date, the client ID number, the ID number of the salesperson and the branch where the sale occurred, and so on.

Example: A transactional database for All Electronics

A table can be used to hold transactions, with one entry per transaction. **Table SP-AI-SRB-4.1** depicts a portion of an All-Electronics transactional database. Because the attribute list of item IDs contains a set of items, the sales table in **Table SP-AI-SRB-4.1** is a nested relation in relational databases. Because most relational database systems do not support nested relational structures, the transactional database is typically saved in a flat file, similar to the table in **Table SP-AI-SRB-4.1**, or unfurled into a conventional relation, similar to the items sold table in **Fig.SP-AI-SRB-4.6**.

Table SP-AI-SRB-4.1: *Fragment of a transactional database for sales at AllElectronics*

<i>trans_ID</i>	<i>list of item_IDs</i>
T100	I1, I3, I8, I16
T200	I2, I8
...	...

"Show me all the things purchased by Sandy Smith," you would ask as an analyst of the AllElectronics database, or "How many transactions include item number I3?" Such queries may necessitate a full scan of the transactional database.

If you wanted to delve deeper into the data, you could question, "Which items sold well together?" This type of market basket data analysis would allow you to combine similar things together as a sales tactic. Given the fact that printers are frequently purchased in conjunction with computers, you may, for example, offer an expensive model of printer at a discount to consumers purchasing specific computers in the hopes of selling more expensive printers. A standard data retrieval system would be unable to respond to questions like the one above. Data mining algorithms for transactional data, on the other hand, can do this by detecting frequent itemsets, or groups of items that are frequently purchased together.

4.4.4 Advanced Data and Information Systems and Advanced Applications

In corporate applications, relational database systems are commonly employed. As database technology advances, new types of advanced data and information systems have evolved and are being developed to meet the demands of new applications.

Handling spatial data (such as maps), engineering design data (such as building, system, or integrated circuit designs), hypertext and multimedia data (including text, image, video, and audio data), time-related data (such as historical records or stock exchange data), stream data (such as video surveillance and sensor data, where data flows in and out like streams), and the World Wide Web are among the new database applications (a huge, widely distributed information repository made available by the Internet). For managing complicated object structures, variable-length records, semi structured or unstructured data, text, spatiotemporal, and multimedia data, and database schemas with complex structures and dynamic changes, these applications require efficient data structures and scalable methodologies.

Advanced database systems and specific application-oriented database systems have been created in response to these needs. Object-relational database systems, temporal and time-series database systems, spatial and spatiotemporal database systems, text and multimedia database systems, heterogeneous and legacy database systems, data stream

management systems, and Web-based global information systems are just a few of the options.

While sophisticated facilities are required to efficiently store, retrieve, and update massive volumes of complicated data, such databases or information repositories also provide fertile ground for data mining research and implementation. Each of the advanced database systems listed above is described in this section.

4.4.5 Advanced Database Systems

4.4.5.1 Object-Relational Databases

An object-relational data model is used to build object-relational databases. This approach improves on the relational model by adding a rich data type for dealing with complicated objects and object orientation. Object-relational databases are becoming increasingly prevalent in industry and applications since most sophisticated database applications require the handling of complex objects and structures.

The object-relational data model inherits the fundamental notions of object-oriented databases, in which each entity is treated as an object in general. Individual employees, customers, or merchandise, in the case of AllElectronics, are examples of objects. The data and code for an item are combined into a single entity. The following information is connected with each object:

- Objects are described by a set of **variables**. In the entity-relationship and relational models, these relate to characteristics.
- A set of **messages** that an object can employ to communicate with other objects or the database system as a whole.
- A collection of **methods**, each of which contains the code for implementing a message. The method returns a value in response to receiving a message. For example, the `get photo(employee)` method will obtain and return a photo of the provided employee object.

An object class is a collection of objects that have a common set of properties. Each object is an instance of the class it belongs to. Object classes can be arranged into hierarchies of class/subclasses, with each class representing properties that are shared by all objects in that class. Variables like name, address, and birthdate can be found in an employee class, for example. Assume that the salesperson class is a subclass of the employee class. All of the variables associated with its superclass of employee would be passed down to a salesperson object. It also includes all of the factors that are unique to being a salesperson (e.g., commission). Information exchange is aided by such a class inheritance feature.

Techniques for dealing with complicated object structures, complex data types, class and subclass hierarchies, property inheritance, and methods and procedures are all needed for data mining in object-relational systems.

4.4.5.2 Temporal Databases, Sequence Databases, and Time-Series Databases

- A temporal database is a type of relational database that stores data with time-related properties. These characteristics may include many timestamps, each with its own semantics.

- A sequence database is a database that contains ordered sequences of occurrences, with or without a meaningful sense of time. Customer purchase sequences, Web click streams, and biological sequences are all examples.
- A time-series database keeps track of values or events that occur over a period of time (e.g., hourly, daily, weekly). Data from the stock exchange, inventory control, and natural phenomenon observation are just a few examples (like temperature and wind).

The features of object evolution, or the trend of changes for items in the database, can be discovered using data mining techniques. Such data can aid in decision-making and strategy development. Mining banking data, for example, could aid in the scheduling of bank tellers based on the volume of customer activity. Stock exchange data can be mined for trends that can aid in the planning of investment strategies (for example, when is the best time to buy AllElectronics stock?). Typically, such analyses necessitate the definition of various time granularities. Time can be divided into fiscal years, academic years, or calendar years, for example. Years can be broken down even further into quarters or months.

4.4.5.3 Spatial Databases and Spatiotemporal Databases

Spatial databases store data that is related to space. Geographic (map) databases, very large-scale integration (VLSI) or computer-aided design databases, and medical and satellite imaging databases are only a few examples. Raster data, which consists of n-dimensional bit maps or pixel maps, can be used to represent spatial data. A 2-D satellite image, for example, could be represented as raster data, with each pixel registering rainfall in a specific location. Roads, bridges, buildings, and lakes can all be represented as unions or overlays of basic geometric constructions like points, lines, polygons, and the partitions and networks generated by these components in vector format.

Geographic databases can be used for a variety of purposes, including forestry and ecological planning, as well as giving public service information on the location of telephone and electric wires, pipes, and sewage systems. Geographic databases are also utilized often in automobile navigation and dispatching systems. A city map with information on one-way streets, proposed routes for going from region A to region B during rush hour, the location of restaurants and hospitals, as well as the present location of each driver, would be stored in such a system for taxis.

You might wonder, "What kind of data mining can be done on geographical databases?" Data mining can reveal patterns that describe the characteristics of dwellings near a specific type of site, such as a park. Other patterns could be used to explain the climate of mountainous places at varying elevations, or the trend in metropolitan poverty rates based on city distances from main roadways. It is possible to study the relationships between a set of spatial objects in order to determine whether subsets of objects are spatially auto-correlated or connected. Spatial cluster analysis can be used to identify clusters and outliers. Furthermore, based on the appropriate set of attributes of the spatial objects, spatial classification can be used to build models for prediction. Additionally, "spatial data cubes" can be built to organize data into multidimensional structures and hierarchies on which OLAP operations (such drill-down and roll-up) can be done.

A spatiotemporal database is a spatial database that records spatial objects that change over time and may be mined for useful information. For example, based on the geographic distribution of a disease over time, we may be able to group the trends of moving items

and spot some suspiciously moving cars, or distinguish a bioterrorist strike from a routine flu outbreak.

4.4.5.4 Text Databases and Multimedia Databases

Text databases are collections of item descriptions in the form of words. Long sentences or paragraphs, such as product specifications, error or bug reports, warning messages, summary reports, notes, or other papers, are commonly used as word descriptions. It's possible that text databases are excessively unstructured (such as some Web pages on the World Wide Web). Some text databases are semi-structured, such as e-mail messages and many HTML/XML Web pages, while others are well-structured (such as library catalogue databases). Relational database technologies are often used to implement text databases with extremely regular structures.

"What may text database data mining reveal?" Text data mining can be used to uncover generic and concise descriptions of text documents, keyword or content relationships, and text object clustering behavior. Standard data mining methods must be combined with information retrieval techniques, the building or usage of hierarchies designed for text data (such as dictionaries and thesauruses), and discipline-specific term classification systems to do this (such as in biochemistry, medicine, law, or economics).

Image, audio, and video data are stored in multimedia databases. They're utilized in things like image content retrieval, voicemail systems, video-on-demand systems, the Internet, and speech-based user interfaces that recognize spoken commands. Because data objects like video can demand gigabytes of storage, multimedia databases must accommodate huge objects. It's also necessary to use specialized storage and search strategies. Continuous-media data refers to video and audio data that must be retrieved in real time at a constant and predetermined pace to avoid picture or sound gaps and system buffer overflows.

Storage and search algorithms must be linked with typical data mining methods for multimedia data mining. The generation of multimedia data cubes, the extraction of various characteristics from multimedia data, and similarity-based pattern matching are all promising approaches.

4.4.5.5 Heterogeneous Databases and Legacy Databases

A heterogeneous database is made up of multiple interconnected, self-contained component databases. To exchange information and respond to requests, the components communicate. Objects in one component database may differ significantly from those in other component databases, making assimilation of their semantics into the overall heterogeneous database challenging.

As a result of the long history of information technology development, many businesses acquire legacy databases (including the application of different hardware and operating systems). A legacy database is a collection of disparate data systems that includes relational or object-oriented databases, hierarchical databases, network databases, spreadsheets, multimedia databases, and file systems. Intra- or inter-computer networks can connect the disparate databases of a legacy database.

4.4.5.6 Data Streams

Many applications require the creation and analysis of a new type of data known as stream data, in which data flows in and out of an observation platform (or window) on a continuous

basis. Large or possibly infinite volume, dynamically changing, flowing in and out in a defined order, enabling only one or a small number of scans, and requiring fast (sometimes real-time) response time are all characteristics of such data streams. Scientific and engineering data, time-series data, and data generated in other dynamic contexts, such as power supply, network traffic, stock market, telecommunications, Web click streams, video surveillance, and weather or environment monitoring, are all examples of data streams.

Because data streams are typically not retained in any form of data repository, researchers have significant hurdles in managing and analyzing stream data. Many academics are currently looking at various aspects of the development of data stream management systems. The continuous query model is a typical query paradigm in such a system, in which predefined queries continuously assess incoming streams, collect aggregate data, present the current status of data streams, and respond to their changes.

The efficient detection of broad patterns and dynamic changes within stream data is the goal of data mining. For example, we might want to detect computer network intrusions based on anomalies in message flow, which can be detected by grouping data streams, dynamically constructing stream models, or comparing current frequent patterns to those from a past time. The majority of stream data is at a low level of abstraction, but analysts are usually interested in higher and numerous levels of abstraction. Stream data should also be subjected to multilevel, multidimensional on-line analysis and mining.

4.4.5.7 The World Wide Web

The World Wide Web and its accompanying distributed information services, such as Yahoo!, Google, America Online, and AltaVista, provide rich, global on-line information services in which data objects are connected together to allow for interactive access. Users seeking relevant information navigate from one thing to another using links. Data mining potential and challenges abound in such systems.

Understanding user access patterns, for example, leads to better marketing decisions (e.g., by placing advertisements in frequently visited documents, or by providing better customer/user classification and behavior analysis), as well as better system design (by providing efficient access between highly correlated objects). Web usage mining is the process of capturing user access patterns in such distributed information systems (or Weblog mining).

Human readers may perceive Web sites as fancy and instructive, although they might be highly unstructured and lack a set schema, type, or pattern. As a result, computers struggle to comprehend the semantic content of various Web pages and organize them in a logical manner for systematic information retrieval and data mining. Users can only get limited aid from web services that give keyword-based searches without comprehending the context of the Web pages. A Web search based on a single keyword, for example, may yield hundreds of Web page links containing the keyword, but the majority of the points will be unrelated to what the user is looking for.

Web search services aren't always enough, and data mining can help. For example, authoritative Web page analysis based on Web page links can aid in the ranking of Web pages based on their importance, influence, and themes. Automated Web page clustering and classification assist in the multidimensional grouping and arrangement of Web pages based on their contents. Web community analysis aids in the discovery of hidden Web

social networks and communities, as well as the tracking of their evolution. The development of scalable and effective Web data analysis and mining methods is known as web mining. It has the potential to help us better understand the distribution of information on the Web in general, characterize and classify Web pages, and expose Web dynamics such as the association and other relationships between distinct Web pages, users, communities, and Web-based activities.

4.5 Data Mining Functionalities – What kind of Patterns can be mined?

We've seen a variety of databases and information repositories that can be used for data mining. Let's have a look at the different types of data patterns that can be discovered.

We've seen a variety of databases and information repositories that can be used for data mining. Let's have a look at the different types of data patterns that can be discovered. Data mining functionalities are used to define the types of patterns that will be discovered during data mining jobs.

Data mining tasks are generally divided into two types: descriptive and predictive.

- Descriptive mining tasks describe the general characteristics of the database's data.
- In order to produce predictions, predictive mining activities make inferences about existing data.

Users may have no notion what kinds of patterns in their data are interesting in some circumstances, thus they may want to search for multiple different kinds of patterns at the same time. As a result, having a data mining system that can mine several types of patterns to satisfy various user expectations or applications is critical.

Data mining systems should also be able to find patterns at multiple degrees of granularity (i.e., different levels of abstraction). Users should be able to specify tips in data mining systems to help guide or focus the search for intriguing patterns. Because some patterns may not hold for all of the data in the database, each discovered pattern is usually given a level of certainty or "trustworthiness."

The functionalities of data mining, as well as the types of patterns that can be discovered, are detailed here.

4.5.1 Concept/Class Description: Characterization and Discrimination

Data can be linked to concepts or classes. Computers and printers, for example, are sold in the AllElectronics store, and clients are divided into two categories: big spenders and budget spenders. Individual classes and concepts can be described in simplified, succinct, and precise language. Class/concept descriptions are such descriptions of a class or idea.

These descriptions can be derived through

1. data characterization, which involves summarizing the data of the understudied class (often referred to as the target class) in broad terms,

2. data discrimination, which involves comparing the target class to one or more comparative classes (often referred to as the contrasting classes), or
3. a combination of the two.

A summary of the general characteristics or attributes of a target class of data is called data characterization. A database query is often used to acquire data pertaining to the user-specified class. For instance, to investigate the characteristics of software goods whose sales increased by 10% in the previous year, data pertaining to such products can be gathered using a SQL query.

Effective data summary and characterization can be accomplished using a variety of ways. The OLAP roll-up procedure, which is based on data cubes, can be used to execute user-controlled data summarization along a chosen dimension. Without requiring step-by-step user involvement, an attribute-oriented induction technique can be used to conduct data generalization and characterization.

The data characterization result can be presented in a variety of ways. Pie charts, bar charts, curves, multidimensional data cubes, and multidimensional tables, including crosstabs, are just a few examples. The descriptions that result can be expressed as generic relations or as rules (called characteristic rules).

Example: Data characterization

A data mining system should be able to generate a summary of the characteristics of AllElectronics customers who spend more than \$1,000 per year. The end result could be a generic customer profile, such as 40–50-year-olds who are employed and have outstanding credit ratings. Users should be able to dig down on any dimension, such as occupation, in order to see these customers according to their line of work.

The general features of target class data items are compared to the general features of objects from one or more opposing classes in data discrimination. The user can provide the target and contrasting classes, and the appropriate data items can be retrieved via database queries. For example, the user might want to compare the general features of software items whose sales climbed by 10% in the previous year vs those whose sales declined by at least 30% in the same time period. Data discrimination strategies are similar to those used for data characterization.

"How are discriminatory descriptions generated?" says the researcher. Although discrimination descriptions should include comparative measures that help discriminate between the target and contrasting classes, output presentation is similar to that of characteristic descriptions. Discriminant rules are discrimination descriptions stated as a set of rules.

Example: Data discrimination

A data mining system should be able to compare two groups of AllElectronics consumers, such as those who buy for computer products on a frequent basis (more than twice a month) vs those who don't (i.e., less than three times a year). The resulting description provides a general comparative profile of the customers, such as the fact that 80% of customers who frequently purchase computer products are between the ages of 20 and 40 and have a university education, whereas 60% of customers who infrequently purchase such products are seniors or youths with no university education. Drilling down on a dimension, such as occupation, or adding new dimensions, such as income level, may aid in the discovery of even more distinguishing characteristics between the two classes.

4.5.2 Mining Frequent Patterns, Associations, and Correlations

Frequent patterns are patterns that appear frequently in data, as the name implies. Itemsets, subsequences, and substructures are only a few examples of common patterns. A frequent itemset is a group of items that appear together frequently in a transactional data collection, such as milk and bread. A (frequent) sequential pattern is a subsequence that occurs frequently, such as when clients buy a PC first, then a digital camera, and finally a memory card. Different structural forms, such as graphs, trees, or lattices, can be joined with itemsets or subsequences to produce a substructure. A (frequent) structured pattern is one in which a substructure appears frequently. Within data, mining common patterns leads to the finding of interesting relationships and correlations.

Example: Association analysis

Assume you're a marketing manager at AllElectronics who wants to know which things are usually purchased together in the same transaction. The following is an example of a rule extracted from the AllElectronics transactional database.

$$\text{buys}(X, \text{computer}) \rightarrow \text{buys}(X, \text{software}) [\text{support} = 1\%, \text{confidence} = 50\%]$$

X is a variable that represents a customer. A 50 percent confidence, or certainty, suggests that if a consumer buys a computer, there's a 50 percent probability she'll also buy software. A support of 1% indicates that computer and software were purchased jointly in 1% of all transactions under consideration. This association rule has a single repeating property or predicate (i.e., buys). Single-dimensional association rules are association rules that contain only one predicate. The above rule can be expressed simply as "computer software [1%, 50%]" if the predicate notation is removed.

Let's pretend we've been handed the AllElectronics relational database for purchasing. Association rules such as this may be discovered using a data mining system.

$$\begin{aligned} & \text{age}(X, "20 \dots 29") \\ & \text{age}(X, "20 \dots 29") \wedge \text{income}(X, "20K \dots 29K") \Rightarrow \text{buys}(X, "CD \text{ player} ") \\ & [\text{support} = 2\%, \text{confidence} = 60\%] \end{aligned}$$

According to the rule, 2% of AllElectronics customers are 20 to 29 years old, have an income of 20,000 to 29,000 dollars, and have purchased a CD player from AllElectronics. A CD player is likely to be purchased by a customer in this age and income group 60% of the time. It's important to note that this is a relationship between multiple attributes or predicates (i.e., age, income, and buys). The above rule can be referred to as a multidimensional association rule if you adopt the terminology used in multidimensional databases, where each attribute is referred to as a dimension.

If an association rule does not meet both a minimal support and a minimum confidence criterion, it is usually dismissed as uninteresting. Additional research can be done to find statistical relationships between the connected attribute-value pairs.

The simplest form of frequent pattern mining is frequent itemset mining.

4.5.3 Classification and Prediction

The process of finding a model (or function) that explains and separates data classes or concepts in order to use the model to predict the class of objects whose class label is unknown is known as classification. The model is derived by the examination of a collection of training data (i.e., data objects whose class label is known).

"Can you tell me how the derived model is presented?" The resulting model can take many different forms, including classification (IF-THEN) rules, decision trees, mathematical equations, and neural networks (**Fig.SP-AI-SRB-4.9**). A decision tree is a tree structure that looks like a flowchart, with each node representing a test on an attribute value, each branch representing the test's outcome, and tree leaves representing classes or class distributions. Classification rules can be simply converted from decision trees.

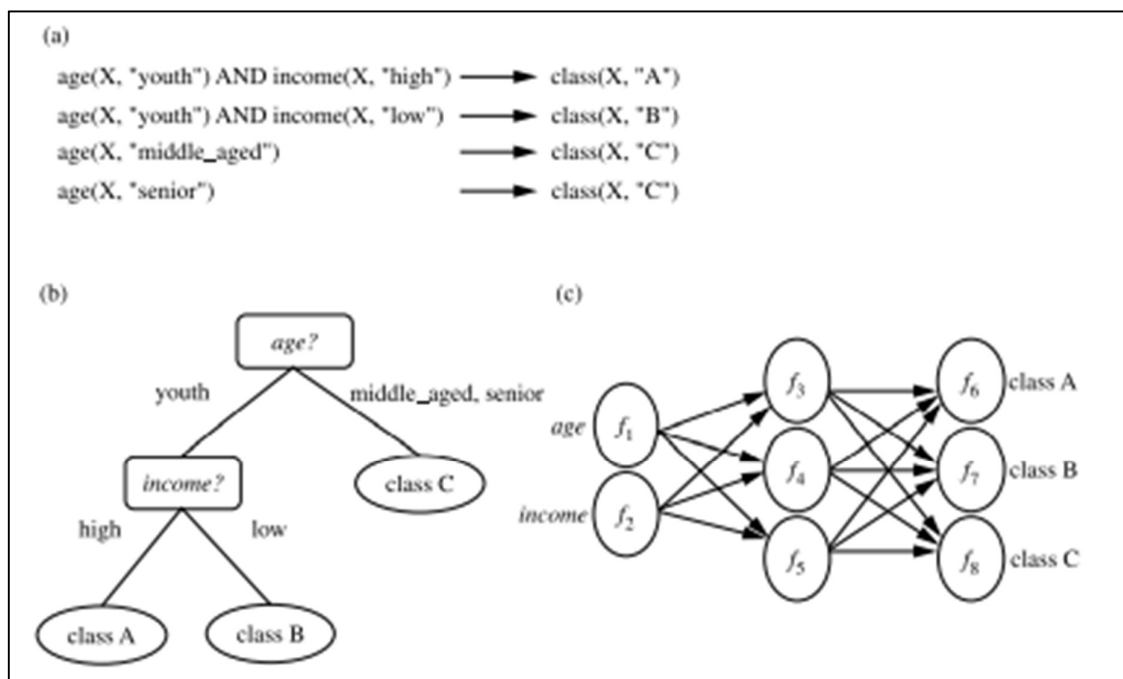


Fig.SP-AI-SRB-4.9: A classification model can be represented in various forms, such as (a) IF-THEN rules, (b) a decision tree, or a (c) neural network

When used for classification, a neural network is often made up of neuron-like processing units with weighted connections between them. There are a variety of additional classification approaches, including naive Bayesian classification, support vector machines, and k-nearest neighbor classification.

Prediction models continuous-valued functions, whereas classification predicts categorical (discrete, unordered) labels. That is, rather than class labels, it is used to forecast missing or unavailable numerical data values. Although the term prediction can refer to both numeric and class label prediction, we use it largely to refer to numeric prediction in this work. Although alternative methods exist, regression analysis is a statistical tool that is most typically employed for numeric prediction. Prediction also includes determining distribution trends based on the facts given.

Relevance analysis, which aims to identify features that do not add to the classification or prediction process, may be required before classification and prediction. These characteristics can then be removed.

4.5.4 Cluster Analysis

"What is cluster analysis?" you might wonder. Clustering analyzes data objects without consulting a known class label, unlike classification and prediction, which study class-labeled data objects.

The class labels are generally absent from the training data because they are unknown to begin with. Such labels can be generated via clustering. The items are clustered or categorized according to the maximization of intraclass similarity and minimization of interclass similarity principle. That is, object clusters are generated in such a way that items inside a cluster have a high degree of resemblance to one another but are quite distinct to objects in other clusters. Each created cluster can be thought of as a class of objects from which rules can be deduced. Taxonomy formation, or the arrangement of observations into a hierarchy of classes that group comparable events together, can be aided by clustering.

Example: Cluster analysis

Customer data from AllElectronics can be used to perform cluster analysis in order to find homogeneous subpopulations of customers. Individual target groups for marketing could be represented by these clusters. A 2-D plot of consumers in relation to customer locations in a city is shown in **Fig.SP-AI-SRB-4.10**. There are three distinct clusters of data points.

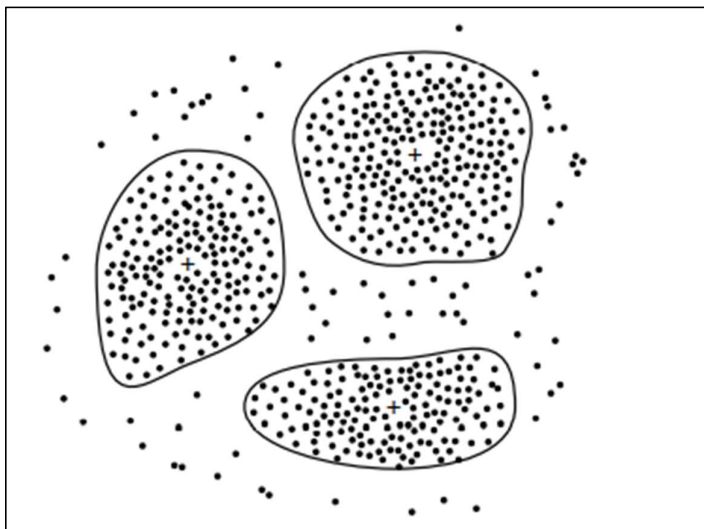


Fig.SP-AI-SRB-4.10: A 2-D plot of customer data with respect to customer locations in a city, showing three data clusters. Each cluster “center” is marked with a “+”.

4.5.5 Outlier Analysis

A database may contain data objects that deviate from the data's overall behavior or model. Outliers are data objects that are out of the ordinary. Outliers are typically discarded as noise or exceptions by most data mining algorithms. Rare occurrences, on the other hand, can be more intriguing than more often occurring ones in some applications, such as fraud detection. Outlier mining is the process of analyzing outlier data.

Outliers can be found using statistical tests that presume a distribution or probability model for the data, or by employing distance measures, in which objects that are far apart from any other cluster are termed outliers. Deviation-based approaches identify outliers by assessing deviations in the basic properties of items in a group, rather than using statistical or distance metrics.

Example: Outlier Analysis

By recognizing purchases of exceptionally large sums for a specific account number in relation to regular charges incurred by the same account, outlier analysis can help discover fraudulent credit card use. Outlier values can also be found in terms of buy location and kind, as well as purchase frequency.

4.5.6 Evolution Analysis

Data evolution analysis is a technique for describing and modeling regularities or patterns in the behavior of things that vary over time. Time-series data analysis, sequence or periodicity pattern matching, and similarity-based data analysis are various elements of such an analysis, which may involve characterization, discrimination, association and correlation analysis, classification, prediction, or clustering of time-related data.

Example: Evolution Analysis

Assume you have access to the New York Stock Exchange's significant stock market (time-series) data for the last several years and want to invest in shares of high-tech industrial enterprises. A stock exchange data mining investigation can uncover stock evolution regularities for both general stocks and specific businesses' stocks. Such regularities may aid in the prediction of future stock market trends, assisting you in making stock investing decisions.

4.6 Are all of the Patterns Interesting?

A data mining system has the capability of producing thousands, if not millions, of patterns or rules. These are some questions we consider;

1. "Are all of the patterns interesting?" you might wonder. Typically, no—only a small percentage of the patterns that could be formed are of interest to any individual user. This raises severe concerns about data mining.
2. "What makes a pattern interesting?" you might think. Is it possible for a data mining system to generate all of the intriguing patterns?
3. Is it possible for a data mining system to produce just interesting patterns?"

We answer these questions below.

To address the first question, a pattern is intriguing if it is;

- easily understood by people,
- reliably valid on fresh or test data,
- possibly beneficial, and
- original.

A pattern is also intriguing if it confirms a notion that the user was trying to prove. Knowledge is represented by an intriguing pattern.

There are several objective metrics of pattern interest. These are based on the structure of patterns observed and the statistics that support them. Rule support, which represents the percentage of transactions from a transaction database that the provided rule fulfills, is an objective metric for association rules of the form $X \Rightarrow Y$. Another objective metric for association rules is confidence, which quantifies the degree of certainty of the discovered link.

In general, each metric of interest is coupled with a threshold that the user can modify. Rules below the threshold are likely to represent noise, exceptions, or rare cases, and are therefore of lower value.

Although objective metrics aid in the identification of interesting patterns, they are insufficient until supplemented with subjective indicators that reflect a user's requirements and interests. Furthermore, many patterns that are interesting by objective criteria may actually be dull

because they represent common information. Subjective measures of data interest are based on user perceptions of the data. Patterns are intriguing to these measures if they are unexpected (contradicting a user's belief) or if they provide strategic information on which the user can act. Such patterns are deemed to as actionable in the latter situation.

Expected patterns can be intriguing if they confirm a hypothesis that the user wanted to test, or if they mirror a user's hunch.

The second question, "Can a data mining system generate all of the intriguing patterns?" concerns the algorithm's completeness. Data mining techniques are frequently unrealistic and inefficient in their attempts to generate all conceivable patterns. Instead, the search should be focused using user-provided limitations and interestingness measurements. For some mining tasks, like as association, this is typically enough to assure the algorithm's completeness. The use of restrictions and interestingness metrics helps ensure the completeness of mining in association rule mining, for example.

Finally, the final question is an optimization problem in data mining: "Can a data mining system generate only interesting patterns?" Only intriguing patterns are generated by data mining systems, which is highly desirable. Users and data mining systems would both benefit from this because neither would have to sift through the resulting patterns to find the ones that are actually interesting. Although progress has been made in this regard, data mining optimization remains a difficult problem.

Pattern interestingness measures are critical for finding patterns of value to a given user quickly. After the data mining process, such measurements can be used to rank the discovered patterns according to their interestingness, filtering out the ones that aren't. More crucially, such measurements can be used to guide and constrain the discovery process, reducing search time by pruning out subsets of the pattern space that do not meet predefined interest constraints.

With respect to each type of pattern that can be mined, methods to measure pattern interestingness and their usage to improve data mining efficiency are explored throughout the book.

4.7 Classification of Data Mining Systems

Database systems, statistics, machine learning, visualization, and information science all come together in data mining, which is an interdisciplinary field (**Fig.SP-AI-SRB-4.11**). Techniques from other fields, such as neural networks, fuzzy and/or rough set theory, knowledge representation, inductive logic programming, or high-performance computing, may also be used, depending on the data mining approach used.

The data mining system may incorporate techniques from spatial data analysis, information retrieval, pattern recognition, image analysis, signal processing, computer graphics, Web technology, economics, business, bioinformatics, or psychology, depending on the types of data to be mined or the given data mining application.

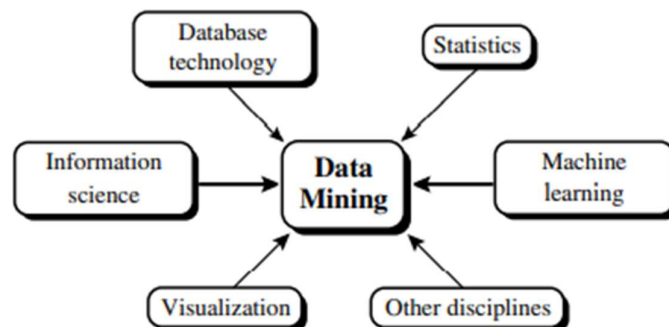


Fig.SP-AI-SRB-4.11: Data mining as a confluence of multiple disciplines

Because of the number of disciplines that contribute to data mining, it is projected that data mining research will result in a wide range of data mining systems. As a result, a precise classification of data mining systems is required, which may aid potential users in distinguishing between such systems and identifying those that best meet their needs. The following are some of the criteria that can be used to classify data mining systems:

- **Classification based on the types of databases mined:** A data mining system can be categorised based on the databases it mines. Distinct criteria (such as data models or the types of data or applications involved) can be used to categorize database systems, each of which may need a different data mining technique. As a result, data mining systems can be categorised properly.
- **Classification based on the types of knowledge mined:** Data mining systems can be classified based on the types of knowledge they mine, which includes characterization, discrimination, association and correlation analysis, classification, prediction, clustering, outlier analysis, and evolution analysis. Multiple and/or integrated data mining functionalities are usually provided by a complete data mining system.
- **Classification based on the kind of procedures used:** The underlying data mining techniques used can be used to characterize data mining systems. The degree of user interaction (e.g., autonomous systems, interactive exploratory systems, query-driven systems) or the data analysis methods used (e.g., database-oriented or data warehouse-oriented techniques, machine learning, statistics, visualization, pattern recognition, neural networks, and so on) can be used to categorize these techniques. A sophisticated data mining system will frequently employ many data mining techniques or devise an effective, integrated strategy that incorporates the advantages of several different approaches.
- **Classification based on adapted applications:** Data mining systems can be classified based on the applications they adapt. Data mining systems, for example, could be customized to finance, telecommunications, DNA, stock markets, e-mail, and other industries. Various applications frequently necessitate the incorporation of application-specific procedures. As a result, a general-purpose data mining system might not be appropriate for domain-specific mining activities.

Note

Advanced topics can be looked at after covering this topic includes; Data Mining Task Primitives, Integration of a Data Mining System with a Database (or Data Warehouse System), Major issues in Data Mining, and other advance Data Mining topics.

SAMPLE TOPIC ACTIVITIES

Activity 1: Group Research and Presentation

- 1) Group Students
- 2) Each group is to select one of the research topics and explain in simple terms how Data Mining is used under that topic.
 - a. Financial Data Analysis
 - b. Retail Industry
 - c. Telecommunication Industry
 - d. Biological Data Analysis
 - e. Other Scientific Applications
 - f. Intrusion Detection

THEMA: ROBOTICS

Introduction

Robotics is a specific yet versatile field of study and is a branch of AI. It utilizes robots (hence the name robotics) that are endowed with the equivalence of human senses including vision, touch, and the ability to sense surrounding conditions such as temperature and pressure to perform simple to complicated tasks.

Robots possess the capabilities to exceed human limitations such as to work longer hours, be accurate and precise in calculations and actions, and to work in hazardous conditions. Much like AI, robotics has a widespread of applications including manufacturing industries, medical healthcare, and military purposes for security and defense intelligence.

There are robots that are designed to accomplish pre-programmed instructions and there are some programmed to make simple decisions. Current robotics research is focused on improving this decision-making process for robots to become self-sufficient; allowing them to make decisions in unconstructed environments.

AI and machine learning algorithms have grown more efficient because of advancements in computing, data transfer, storage, and computer power, making it easier to encode programs for robots, design innovative uses for applications, and minimize energy consumption. Additionally, the improvements in sensor technology and material science improve the hardware of robotics to interface with AI capabilities, ultimately improving the field of robotics.

Although much focus is directed towards product technology (e.g., smartphones, commercial jets, automobiles, solar panels, etc.), process technology (which involves the usage of machines to improve how a good or service is produced, enabling automation) will influence significant changes that will later redefine the scope of modern technology.

MODULE 12.1: ROBOTICS AND ELECTRONICS

4 weeks

Robotics is a multidisciplinary field combining computer science and engineering. Robotics is the study of the design, manufacture, operation, and use of robots. Robotics aims to create devices that can support and help humans. Mechanical engineering, electrical engineering, information engineering, mechatronics, electronics, bioengineering, computer engineering, control engineering, software engineering, and mathematics are just a few of the subjects that robotics encompasses.

Robots can be employed in a variety of scenarios and for a variety of objectives, but many are now used in hazardous areas (such as radioactive material inspection, bomb detection and deactivation), manufacturing operations, or in situations where humans are unable to live (e.g., in space, underwater, in high heat, and clean up and containment of hazardous materials and radiation). Robots can take on any shape, but some are designed to seem like people.

This module introduces students to Robotics; it highlights the different aspects of the module electronics and describes the different applications of robotics in the real world.

Topic 1: Introduction to Robotics and Electronics

From the previous *Thema: Artificial Intelligence*, we have learnt that Robotics is the branch of AI that employs scientific and engineering methods to design, construct, and operate intelligent robots.

In this topic, we will elaborate further on robots, their practical applications and focus on understanding the basic electronics behind this mechanical machine. Whilst covering this Topic, students are to think about how to design a robot and what type of task or (problem) that robot will do or solve.

1.1 Definition of Robot

A robot is a programmable mechanical device that can perform tasks and interact with its environment, without the aid of human interaction. We can see two examples of robots in Figure SC-R-SRB-1.



Figure SP-R-SRB- 1: Robots in action – (a) MARS Rover, and (b) Robot arms in production.
(Modified Source: Bing Images)

1.2 Robotics in AI

Robots are often deployed for conducting tasks that might be difficult for humans to perform steadily. They often imbue Machine Learning and Deep Learning models to function. There are many applications for Robotics: some applications include the assembly for automobile manufacturing, underwater and space expeditions, and agricultural practices such as seeding, weed and pest control, and harvesting.

Robotics is commonly used field for AI applications. Robots that are powered by AI use real-time updates to sense obstacles in its path and pre-plan its journey instantly. In this sense, the applications for robotics in AI are vast. They can be used for carrying goods in hospitals, factories, and warehouses, cleaning offices and large equipment as well as inventory management for the aforementioned industries. Underwater discoveries and space expeditions also utilize AI-robotic technologies that always require analyzing large amounts of data.

NASA's Mars Rover, Perseverance (see Fig.SP-R-SRB-2), arrived on the red planet on February 18, 2021 and is currently traversing the landscape. According to NASA, it is

conducting an expedition that is scheduled for at least one Martian year (687 Earth days). The goal is to search for microscopic life by using an AI-powered device called the Planetary Instrument for X-ray Lithochemistry (PIXL). PIXL is a lunchbox-size instrument carried at the end of Perseverance's 7-foot-long robotic arm.

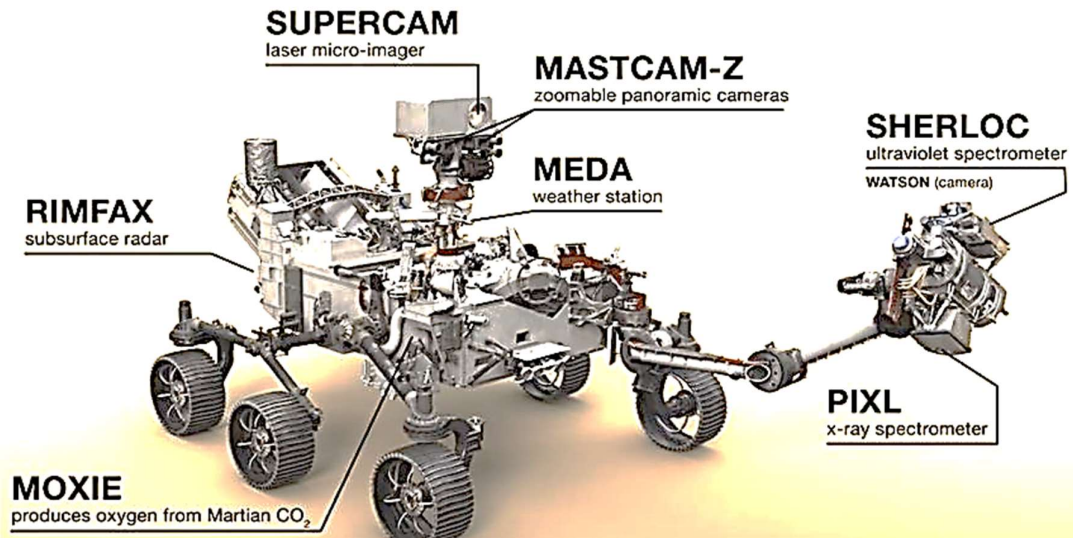


Figure SP-R-SRB- 2: Image depicting major components of the Mars rover, Perseverance. Using a coring drill on the end of the arm, the rover will collect core samples from the planet that will be left on the Mars surface for collection by a future mission.

(<https://www.youtube.com/watch?v=5qqsMjy8Rx0> - Mission Overview: NASA's Perseverance Mars Rover)

1.3 Basic Components of a Robot

Most people are comfortable with human-sized and shaped robots that they have seen in movies, but the majority of actual robots look nothing like humans. Typically, robots are designed more for function than appearance. Thus, their appearance is often machine or tool-like rather than human-like. The main components of a robot are; the body (or frame), control system, actuators, sensors, and drivetrain.

1.2.1 Body

The body or frame can be of any shape and size. Essentially, the body/frame provides the structural support for most of the other components.

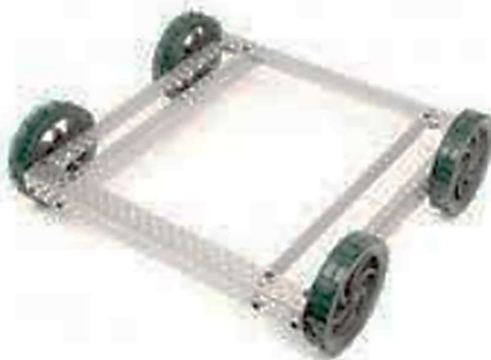


Figure SP-R-SRB- 3: Frame of a Robot (Modified Source: Bing Images)

1.2.2 Control System

The control system of a robot is equivalent to the central nervous system of a human. It coordinates and controls all aspects of the robot. Sensors provide feedback based on the robot's surroundings, which is then sent to the Central Processing Unit (CPU) within the microcontroller. The CPU filters this information through the robot's programming and makes decisions based on logic. The same can be done with a variety of inputs or human commands.



Figure SP-R-SRB- 4: Control System of a Robot (Modified Source: Bing Images)

1.2.3 Actuator

Actuators (such as motors, pistons, grippers, wheels, gears etc.) are used to make the robot move, grab, turn, or lift by producing mechanical movements of the robot elements. Electricity, hydraulic fluid, or pneumatic pressure are the most common sources of energy used by actuators to create motion.

1.2.4 Sensors

The sensors provide the robots with awareness of their surroundings as well as other vital information. The sensing devices communicate measurements and data such as sizes, forms, and distance between objects, direction, and so on to the controller, which processes these and then sends back the control signal to activate the actuators, allowing the robot to move appropriately.

1.2.5 Drivetrain

Although some robots are able to perform their tasks from one location (such as fixed robotic arms), it is often a requirement of robots that they are able to move from location to location. For this task, a drivetrain is required. Drivetrains consist of a powered method of mobility. Humanoid style robots use legs, while most other robots will use some sort of wheeled solution.



Figure SP-R-SRB- 5: *Drivetrain of a Robot*

Activity 1: Student Task

Using the mentioned components in this section, students are to construct a diagram showing how these components work together in a robot.

1.4 Uses and Examples

Robots have a variety of modern day uses. These uses can be broken down into three major categories – Industrial Robots, Robots in Research, and Robots in Education.

1.4.1 Industrial Robots

In the technical industry, there are numerous jobs that require high degrees of speed and precision. For many years, humans were responsible for all these jobs. With the advent of robotic technology, it became evident that many industrial processes could be sped up and performed with a higher degree of precision by the use of robots. Such jobs include packaging, assembly, painting, and palletizing.

1.4.2 Robots in Research

Robots come in very handy in the world of research, as they often can be used to perform tasks or reach locations that would be impossible for humans. For decades, NASA has utilized probes, landers, and rovers with robotic characteristics to study outer space and planets in our solar system.

1.4.2.1 Pathfinder and Sojourner

The Mars Pathfinder mission, which was an ambitious NASA space project, developed a unique technology that allowed the delivery of an instrumented lander and a robotic rover, Sojourner, to the surface of Mars. It was the first robotic roving vehicle to be sent to the planet, Mars. Sojourner weighs 11 kg (24.3 lbs.) on Earth (about 9 lbs. on Mars) and is about the size of a child's wagon. It has six wheels and could move at speeds up to 0.6 m (1.9 ft) per minute. The mission landed on Mars on July 4th, 1997. Pathfinder not only accomplished this goal but also returned an unprecedented amount of data and outlived its primary design life.



Figure SP-R-SRB- 6: Building the Pathfinder Robotic Rover

1.4.2.2 Space Shuttle Robotic Arm – Canadarm

When NASA scientists first began the design for the space shuttle, they realized that there would have to be some way to get the enormous, but fortunately weightless, cargo and equipment into space safely and efficiently. The Remote Manipulator System (RMS), or Canadarm, made its first flight into space on November 13, 1981. The arm has six joints, designed to simulate the joints of the human arm. Two are in the shoulder, one is at the elbow, and three are in the highly dexterous wrist. At the end of the wrist is an end effector which can grab or grapple the desired payload. In the weightless environment of space, it can lift more than 265,800 kg (586,000 lbs) and place it with incredible accuracy. Its total weight on earth is 450.871 kg (994 lbs).

The RMS has been used to launch and rescue satellites and has proven itself invaluable in helping astronauts repair the Hubble Space Telescope. The Canadarm's final shuttle mission took place in July of 2011, marking the 90th time it was used on a shuttle mission.



Figure SP-R-SRB- 7: *Robotic Arm of the Space Shuttle*

A similar device to the RMS, the Mobile Servicing System (MSS) or otherwise known as Canadarm2 was designed to provide manipulation functions for the International Space Station.

The MSS is responsible for servicing payloads and instruments attached to the International Space Station, while also assisting with the transport of supplies and equipment around the station.



Figure SP-R-SRB- 8: *Canadarm2 in Space*

1.4.2.3 Special Purpose Dexterous Manipulator – Dextre

As part of the Space Shuttle mission STS-123 in 2008, the shuttle Endeavour carried the final part of the Special Purpose Dexterous Manipulator, or “Dextre”.

Dextre is a robot with two smaller arms. It is capable of handling the delicate assembly tasks currently performed by astronauts during spacewalks. Dextre can transport objects, use tools, and install and remove equipment on the space station. Dextre is also equipped with lights, video equipment, a tool platform, four tool holders and several sensors that help to manipulate the objects it is dealing with and automatically react to movements or changes. Four mounted cameras enable the crew to observe what is going on.

Dextre's design somewhat resembles a person. The robot has an upper body that can turn at the waist and shoulders that support arms on either side.



Figure SP-R-SRB- 9: *Dextre in Space*

1.4.3 Robots in Education

As AI and Robotics technologies continue to grow, the integration of robots in the field of education will experience an innovative method of teaching and will be used as an exciting and accessible tool for STEM learning, such as its design principles and problem solving.

Robotics enables students to use their hands and minds to create like an engineer, artist, and technician does, all at once. It allows for instantaneous application of scientific and mathematical principles.

It is also predicated by an educational expert, namely Sir Anthony Seldon, that within the next ten years, teaching robots will be able to read children's expressions, and maybe even their brains.

1.4 Importance of Robotics

Robotics is an emerging field with applications in many facets of our lives. It is important for all members of society to have an understanding of the technology that surrounds us. However, robotics is important for more than that reason. Robotics provides a unique combination of the pillars of STEM: science, technology, engineering and math. When taught in schools, it allows students to experience a true interdisciplinary lesson while studying a cutting edge and exciting topic. Also, the aesthetics which go into the design and creation of robots allow students to experiment with an artistic side, while working through technical principals. This combination rewards participants on a plethora of different learning levels.

In the next topic, we will look at the design of robots, particularly the control system.

SAMPLE TOPIC ACTIVITIES

Activity 2: Group Discussion

1. Group Students
2. Task each group to come up with an innovative solution using robots to solve a problem of their choice in Papua New Guinea.
3. Each group must choose a representative to present in class their chosen problem and their innovative solution using robotics, elaborating as clearly as they can.

Activity 3: Group Presentation

1. Group Students
2. Task each group to select one of the applications below and select a robot to discuss on.
 - a. Robots used in Industrial Setting.
 - b. Robots used in Research.
 - c. Robots used in Education.
 - d. Robots in space expeditions.
 - e. Robots used in deep sea exploration.
 - f. Robots used in mining industries.
3. Each group is to create a PowerPoint Presentation to show and explain to the class.

Activity 4: Self-Test-Exercises

- 1) Define Robotics
- 2) Define Robots
- 3) List the main components that a robot should have and describe their respective functions.
- 4) Justify why Robots are important in Society.

Topic 2: Control Systems

In this topic, we will look at robotics control system which is the component that enables the control of robots to perform a wide range of tasks with speed and precision in order to assist humans.

Robotics is now used in a wide range of applications, including material handling, hauling large objects, automating manufacturing processes, self-driving cars, health care applications, and many others. The notion of robotics control will be discussed here as an introduction to how control systems efficiently program a robot's actions such as physical motions, rotations, path planning, and other activities.

Robotics control, which contributes to robot movement, can be accomplished in a number of methods, including manual control, wireless control, and fully automatic control using AI to allow the robot to move on its own. In the operation of Robotics, automation is an essential component. The methods of robotics control continue to improve and advance as technology advances.

Previously, we learnt how each main component is assembled to operate a fully functioning robot. We will now discuss and understand the robot's control system.

2.1 Control system in Robotics

To achieve a desired result, the robot control system manages commands, directs, or regulates the movement and function of several sections of the robot. The automatic control of robot motion is a must-have for any robotic installation. A controller, which is a sort of feedback control system, is built into every robot. It continuously reads data from sensors and modifies actuator controls to accomplish the desired robot behavior. All parts of the robot's operation are coordinated and controlled by the Robot controller.

Two main types of control systems: Closed and open loop systems.

2.1.1 Open Loop System

The open loop system is the simplest and less expensive type where the input parameter is fed into the controller that gives instructions to the actuator which will then perform the required action (process) to generate a desired output variable. It is basically a forward system without any feedback.



Figure SP-R-SRB- 10: A flow diagram of an open control system

2.1.2 Closed Loop Systems

The control system that uses feedback is called a closed loop system or a feedback control system. This system compares the output variable with an input parameter, and any difference between the two is used to drive the output into agreement with the input.

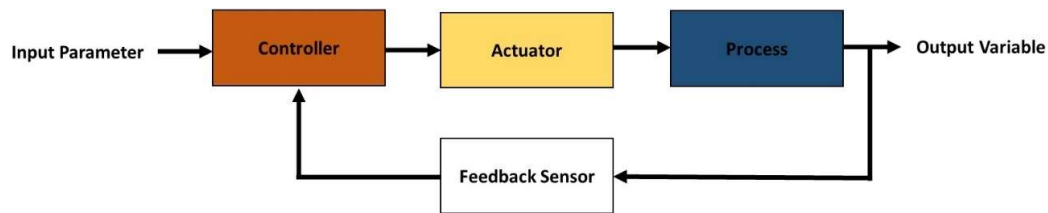


Figure SP-R-SRB- 11: A flow diagram of a closed control system

To build a high-performance control system, feedback from the joint sensors is crucial. Typically, the feedback is used to compute errors by finding such as the difference between the desired and actual position and that between the desired and the actual velocity. The control system can then compute how much torque is required as a function to reduce servo errors. In order to ensure that the system reduces these errors and is performing satisfactorily, certain assumptions, mathematical models and empirical results from simulations and experiments need to be conducted.

The robotic control system uses a closed-loop control system with sensor feedback to attain high levels of performance, as shown below.

1. Input;
2. Process being controlled;
3. Output;
4. Sensing elements;
5. Controller and actuating devices.

2.2 What is a closed loop control system?

The following is an explanation of the closed loop control system's underlying theory:

- Closed loop control refers to a control system that makes advantage of feedback.
- The controller in a closed loop feedback control compares a process' measured output value to the desired set value, which is the reference or set point.
- The 'sensing elements' feedback the output value for comparison with the reference value.
- The 'error' signal is the difference between the actual output value and the process's desired set value.
- This error signal is given into the controller, which generates a control action by changing the input signal to the 'process' under control in order for the process's output value to reach the intended set value.

The concept of a closed loop robot control system is depicted in the diagram below.

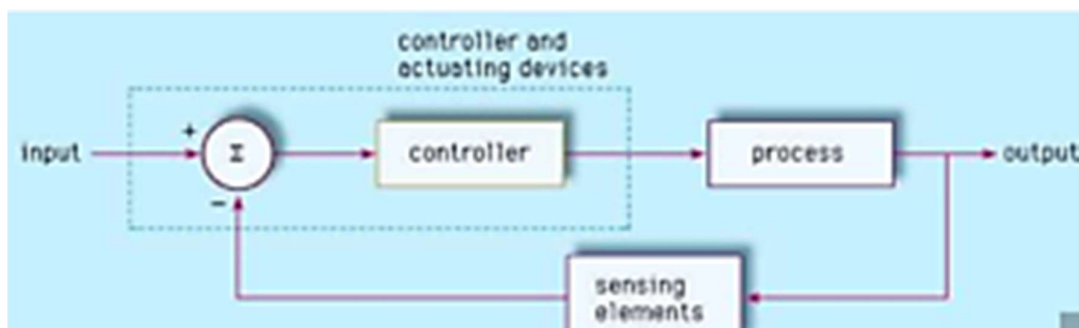


Figure SP-R-SRB- 12: Closed Loop Robot Control System

- The robot's brain can be thought of as the controller that regulates the robot's movements.
- The basic premise behind a robot control system is that an input signal is generated in response to a program command for attaining a specified objective, and the controller produces an output signal that engages an actuator to perform mechanical movements of the robot elements.
- The more intelligently the control algorithm is programmed, the more reliable and efficient the control system becomes.
- 'Sensors' measure the robot's movements and forces and relay the information back to the robot controller.
- The robot controller uses the reference set point and sensor feedback to make necessary modifications to the control signal in order to reduce system error and return the system's output to the desired set value.

Let's look at some additional essentials about robotics control systems now that we've learned about the closed loop control system in robotics and how robots work.

There are four main types of control systems that are used in Robots:

1. Limited Sequence control – pick-and-place operations using mechanical stops to set positions.
2. Playback with point-to-point control – records work cycles as a sequence of points, then plays back the sequence during program execution.
3. Playback with continuous path control – greater memory capacity and/or interpolation capability to execute paths (in addition to points)
4. Intelligent control – exhibits behaviour that makes it seem intelligent, e.g., responds to sensor inputs, makes decisions, communicates with humans.

Each of these control systems feed instructions to the actuators and can also be called the joint-drive systems. They can either be:

1. Electric: Uses electric motors to actuate individual joints and is the most preferred drive system used in most robots today.
2. Hydraulic: Uses hydraulic pistons and rotary vane actuators. This type of joint drive system is able to provide a high power and lift capacity.
3. Pneumatic: Often used for smaller robots and simple material transfer applications.

Hierarchical control structure of a robot microcomputer controller:

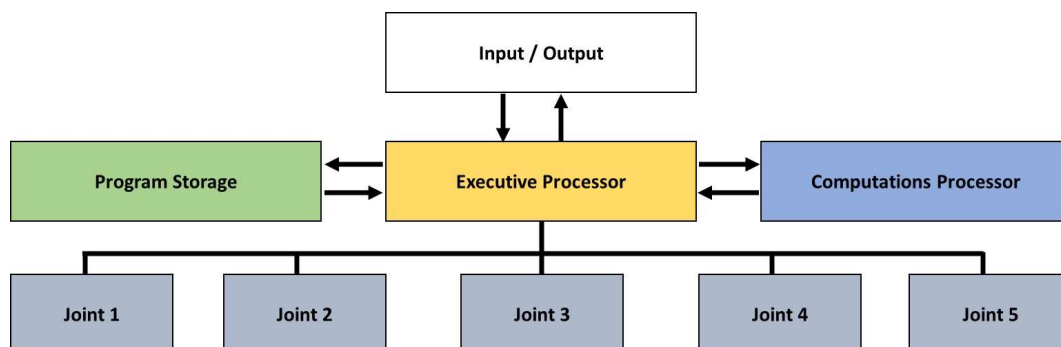


Figure SP-R-SRB- 13: Hierarchical Control Structure of a Robot Microcomputer Controller

Activity 1: Video Learning

Title: Open and Closed Loop Systems in Robotics

Setup a Projector and have the students watch a video explaining open and closed loop systems in Robotics. (Encourage for visual learners). Link: <https://youtu.be/f32SDx3C6sA>

Control of manipulators - Linear control of manipulators

The initial and the final location of end-effectors in cartesian space specified by the transformation matrices T_{initial} and T_{final} , serve as the input

The inverse kinematics model computes the desired end-effector location in joint space. Then, trajectory generator computes the joint-position time histories, based on the joint space algorithm.

Depending on the servo error computed from the base reference values and the sensor measurements, the control system commands the individual actuators to achieve the desired motion.

The use of linear control techniques for the manipulator is usually modelled mathematically using linear differential equations. For dynamic movements, it is more practical and is properly represented by a non-linear differential equation.

2.3 What are major types of robotics control systems?

Robotics control can be divided into two categories:

1. *Point-to-point system*

With this control system, the robot can determine where to pick up and release a part. This system is operational during the loading and unloading of applications.

2. *Continuous path control system*

For jobs where the robot must follow a precise course, such as welding or painting, this control system is employed.

2.4 How can we control robots?

Manual control, wireless control, and fully automatic control utilizing Artificial Intelligence are all options for controlling robots.

2.5 What is the difference between robotics and robots?

Robotics is a technology discipline that deals with actual robots, whereas robots are programmable machines that do tasks automatically or semi-automatically.

2.6 Can robotics replace human development?

Humans can be replaced by robots in a variety of jobs. Machine learning algorithms are increasingly being used to control robots on factory floors so that they can adapt to people working alongside them.

2.7 Conclusion

A robot control system, in summary, manages commands, directs or regulates the movement and functions of various robot parts, and executes a certain set of forces and motions. The actuating error signal, which is the difference between the actual output value and the process's desired set value, is sent into the robot controller in a closed-loop control system to reduce the error and stabilize the system's output in order to accomplish the desired robot behavior.

For the next topic, we will look at circuit boards.

SAMPLE TOPIC ACTIVITIES

Activity 2: Class Exercise

Title: Closed Loop Control Systems

For this class exercise, students are to do their personal research at home and discuss with their teachers in class about the fundamentals of a closed loop control system.

Activity 3: Assignment - Individual Students

Title: Robotic Control Systems

For this assignment, each student is to do their research and answer the following points:

- List the two categories for the control systems of robots
- Explain each of them by highlighting the mechanics and physics involved in their design and functions

Topic 3: Circuit Boards

In the past, all printed circuit boards (PCBs) had traces on only one side. Today, printed circuit boards have traces on both sides and are referred to as double trace layers or double-sided boards. Complex boards may have four or more trace layers with each layer insulated from the other by board material. Special multilayer boards may have eight or more trace layers. This chapter focuses on how simple single and double-sided printed circuit boards are fabricated.

The PCB is a crucial element in electronics. People frequently forget to explain what a PCB is since it is so basic. This topic will go over the components of a PCB as well as some of the terminologies commonly used in the PCB industry.

Prerequisites

You might wish to brush up on some of the concepts we'll be covering in this topic before getting started:

- What is electricity?
- What is a circuit?
- Voltage, current, resistance and Ohm's law
- Connector basics
- Soldering
- Signals

3.1 Fundamentals

A printed circuit board (PCB) consists of an insulating base material that supports the copper traces, as shown in **Fig.SP-R-SRB-14**. The insulating base material is typically epoxy fiberglass, but ceramic and Teflon are also used as base material in special cases.

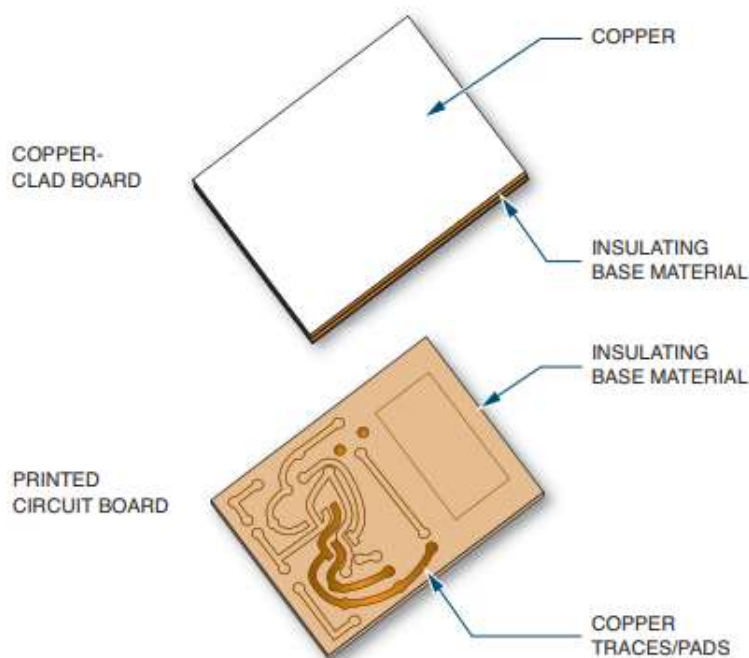


Figure SP-R-SRB- 14: Copper clad board which is made into a PCB

Phenolic base material was used in the past for applications in which performance was not critical because it was inexpensive. The standard base thickness was typically 1/16" or 0.062" (**Fig.SP-RB-SRB-15**).

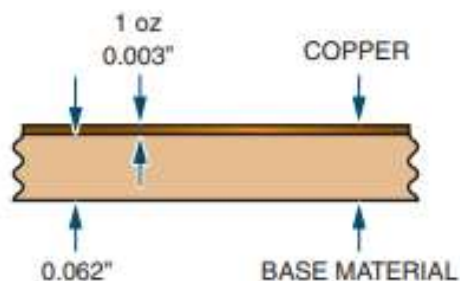


Figure SP-R-SRB- 15: Standard board thickness

A majority of printed circuit boards made in the United States start with a solid copper foil that is bonded to the insulating base material, which is typically epoxy fiberglass (**Fig.SP-RB-SRB-16**:).



Figure SP-R-SRB- 16: Example of epoxy fiberglass PCB material

The thickness of the copper on a printed circuit board is designated in ounces. One ounce refers to the weight of copper spread over one square foot. One ounce copper plating is approximately 0.003" thick. Two-ounce copper is 0.006" thick, and so on.

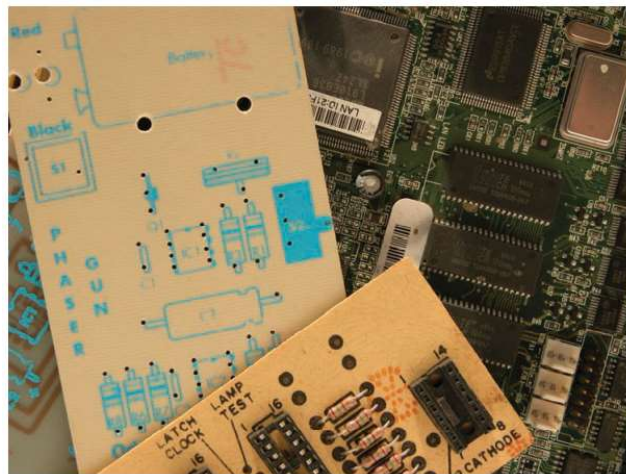
The desired traces are printed onto the copper with a material called resist. The board is then placed in a chemical tray with a chemical for etching, referred to as an etchant, to remove the copper not covered by the resist, as shown in **Fig.SP-R-SRB-17(a)**. The etchants often used for etching small quantities of printed circuit boards are either ferric chloride or ammonium persulphate. A major concern with etchants is disposal according to current environmental standards.

Commercial printed circuit board fabricators generally use different chemicals that are less expensive, such as ammonium chloride. They heat the bath and add agents to improve performance. After washing and cleaning the etched board, the result is a copper circuit pattern on an insulating base—a printed circuit board (**Fig.SP-R-SRB-17(b)**).

Prior to the arrival of surface mount technology (SMT), there were two designated layers for through-hole printed circuit boards: the component side and the solder side. With the implementation of SMT, components are now placed on both sides of the printed circuit

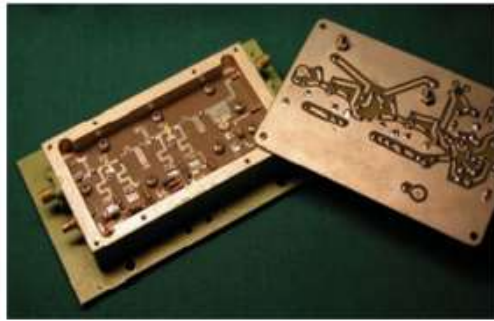
(a) (b) (c)

There are also two additional layers of a printed circuit board after it is etched. These additional layers are nonconductive and form the component layout layer and the solder mask layers. The component layout layer acts as a map when assembling and troubleshooting the printed circuit board. This layout layer consists of text and graphic symbols that identify each component with its proper orientation. A good component layout layer can save a tremendous amount of time (**Fig.SP-R-SRB-18**).



The solder mask controls or indicates where the solder will be placed on the printed circuit board pads during a process called wave soldering. This solder will only connect the components to the solder pads. When making only one or two boards with a simple design a solder mask may not be needed. Additionally, the solder mask adds corrosion protection to the printed circuit board traces.

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(a)



(b)

Figure SP-R-SRB- 19: (a) Single-sided PCB, and (b) Ground plane on a PCB

Most radiofrequency (RF) printed circuit boards have traces on only one side, although a second layer is used as a ground plane to improve performance, as shown in **Fig.SP-R-SRB-19(b)**. Standard integrated circuits pin outs used with analog ICs—such as op-amps, timers, and so on—can get by with a single layer. This is because there is plenty of room for traces to pass between the leads of the passive components.

When digital logic or other designs that use high-pin-count chips are used, two trace layers are required. Digital IC chips with a large number of pins are impossible to connect to other chips with a different layout without crossing and shorting the traces. Therefore, a second layer is required. In theory, only two layers are ever necessary with high pin count components.

In practice, more than two layers are used for dense, highly populated boards or for performance considerations. Two layers are inadequate with densely populated printed circuit boards because the parts are so close together and the traces and holes take up too much space. There simply isn't room for all the traces, holes, and vias on two sides. Vias are connections between one layer and another (**Fig.SP-R-SRB-20**). If the density could be reduced, the two-layer theory would work.

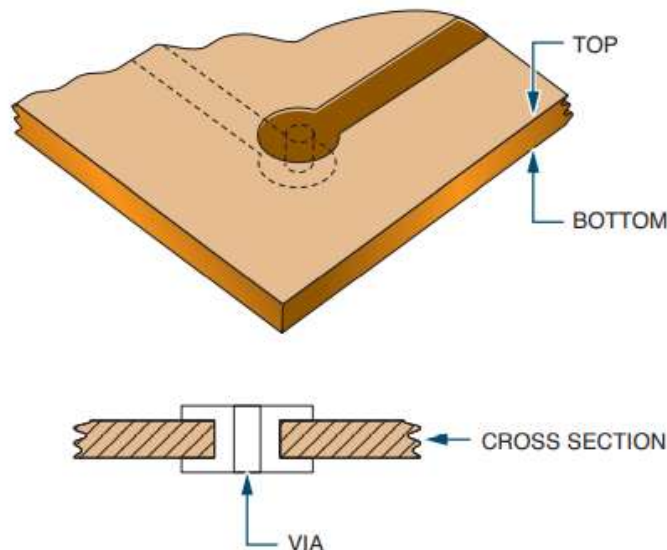


Figure SP-R-SRB- 20: V/As connect the top to the bottom trace of a PCB

For high-speed or high-precision designs, more than two layers are used to improve performance. With four trace layers, the two middle ones are usually power and ground. This evens out the flow of power and reduces voltage fluctuations at different points on the printed circuit board. The ground plane layer provides controlled impedance for high-speed signals. The close proximity of the ground and power planes also acts as a capacitor to reduce noise. One problem with multilayer printed circuit boards is reliable connections between the layers. It is sometimes difficult to detect which layer is defective.

Two sides can be etched at the same time; therefore, the cost of a **double-sided board** is only slightly higher than that of a **single-sided board**. **Multilayer boards** are built up by laminating layers together. A four-layer board will cost about twice as much as a double-sided printed circuit board. The printed circuit board is often the most expensive part of a design.

3.2 Schematic Diagram

The key to an effective printed circuit board design is a schematic diagram that becomes the basic reference for a circuit and gives the necessary specifications needed to design a printed circuit board. The diagram should exhibit good signal flow and complete reference data.

The best way to preview a schematic diagram is to lay out or view a block diagram of the circuit. The block diagram will represent how the component blocks are connected. As an example, one block may represent a stage of signal amplification with one signal entering and one signal leaving and proceeding to the next block (**Fig.SP-R-SRB-21**).



Figure SP-R-SRB- 21: **Block diagram representing components blocks are connected**

Analog circuits use discrete components such as resistors, capacitors, and semiconductor devices. Analog schematics present a different appearance than digital schematics

Discrete analog components may require close physical proximity to the printed circuit board surface to maintain signal paths and to prevent coupling or delays between stages. Tight component grouping is important to show in the schematic diagram so the printed circuit board layout can follow it precisely.

Digital schematic diagrams appear quite different from analog schematic diagrams due to differing interconnection patterns. Digital circuits tend to have many common signals that occur throughout the schematic. This results in many long signal lines that transverse the length and width of the drawing. To avoid confusion, it is better to label the common signals rather than drawing every connection as a line, as shown in **Fig.SP-R-SRB-22**.

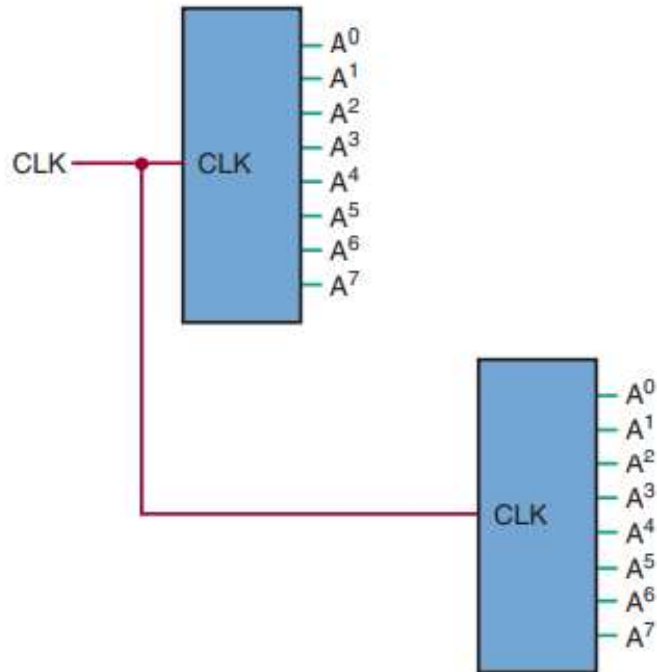


Figure SP-R-SRB- 22: Label common signals in digital circuits

A schematic diagram should show the entire circuit in as few drawings as possible and should use the following technique:

1. Group sub circuits components together in the drawing.
2. Signal flow should proceed from left to right with input on the left and output on the right (**Fig.SP-R-SRB-23**).

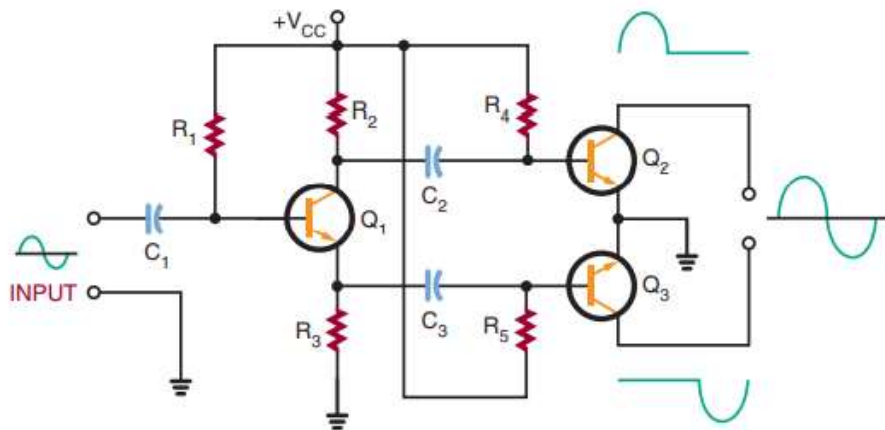


Figure SP-R-SRB- 23: Signals flow from left to right, voltage potential has highest potential at top

3. Voltage potentials are drawn with the highest voltage at the top of the diagram and the lowest at the bottom, as shown in **Fig.SP-R-SRB-23**.
4. Signal lines should cross each other as little as possible. On digital schematic diagrams label the connections with the appropriate signal abbreviation rather than drawing a complete maze of lines (**Fig.SP-R-SRB-24**).

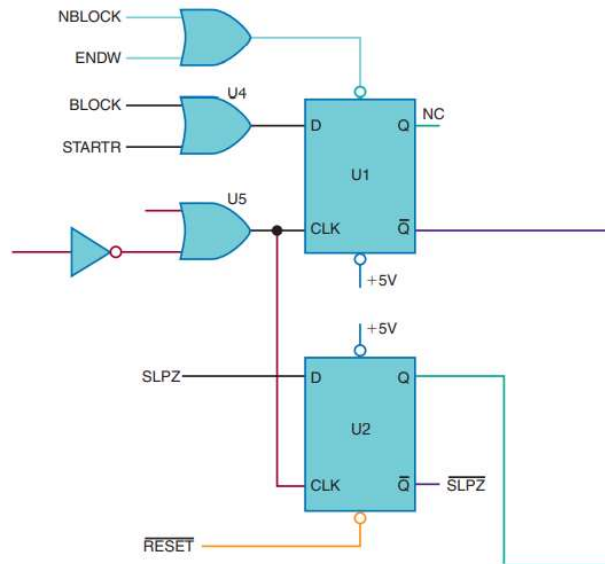


Figure SP-R-SRB- 24: Use signal abbreviations rather than draw a maze of lines

5. Label components starting on the left or input side of the schematic diagram and move down and then back to the top, repeating this process across the schematic diagram (**Fig.SP-R-SRB-25**).

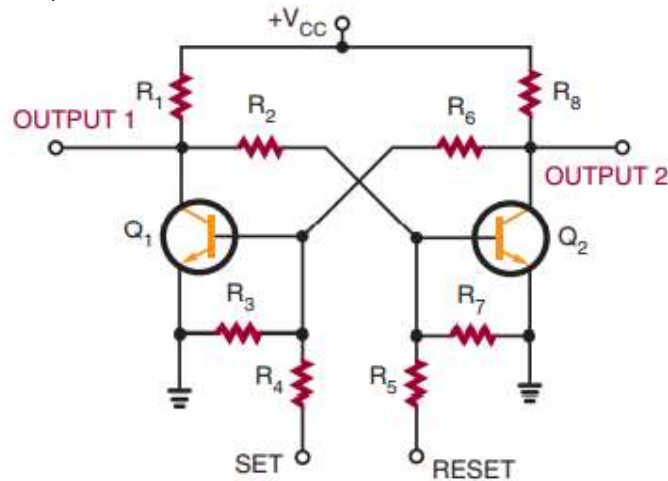


Figure SP-R-SRB- 25: Signal flows from left to right, voltage potential has highest potential at top

6. Critical leads should be short or isolated from other signals.
7. All external connectors and components should be clearly indicated.
8. All integrated circuit pins should be labeled, including the power supply inputs.
9. Any unused integrated circuit logic gates or extra subcircuits' inputs should be tied to the appropriate power supply level to provide stability in the circuit.
10. Extra components, such as bypass capacitors, that are installed during the printed circuit board construction process should be added to the final schematic.

Fig.SP-R-SRB-26 shows the schematic symbol for common components used in the electronics field. Keep in mind the following points while reviewing the symbols:

1. Components that are variable include an arrow as part of their symbol. For example, a potentiometer is a variable resistor.
2. Arrows also indicate the emitting or receiving of various forms of energy. Arrows pointing away from a symbol indicate it is giving off energy, such as a light-emitting diode (LED). If the arrow is pointing toward a symbol, it indicates it is receiving energy, such as a photocell.
3. A component's lead may be identified with letter symbols. For example, SCR leads are identified A (anode), K (cathode) and G (gate). Transistor leads are identified as C (collector), E (emitter), and B (base).

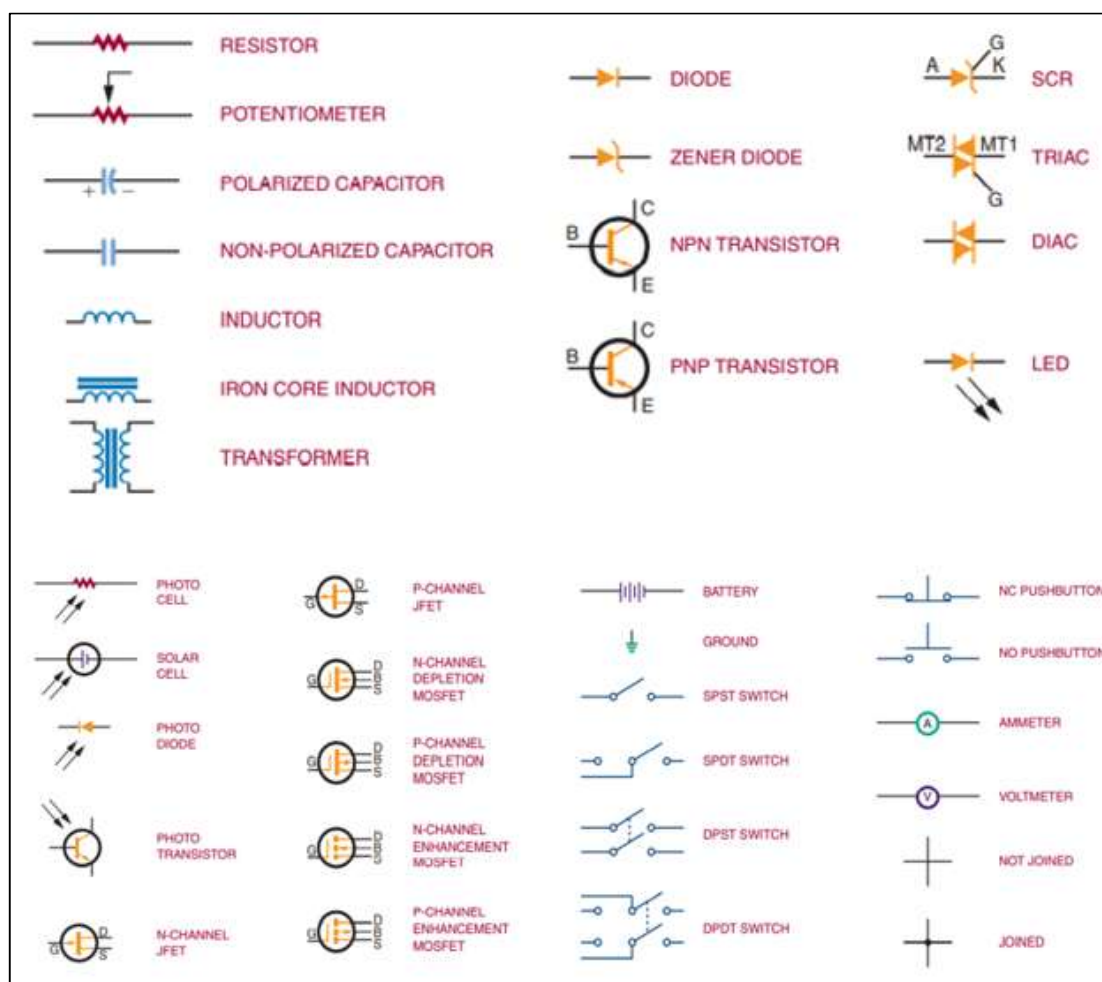


Figure SP-R-SRB- 26: Schematic symbols for common electronic components

Fig.SP-R-SRB-27 shows two schematic symbols for ground. The most commonly used symbol is shown in **Fig.SP-R-SRB-27(A)**. It represents chassis or earth ground, may be used multiple times in the same schematic diagram, and represents a common return point for a circuit. The ground symbol show in **Fig.SP-R-SRB-(B)** represents chassis ground only. If the schematic drawing uses this symbol it is connected to the ground prong of an AC line cord and the chassis will be at earth ground potential.

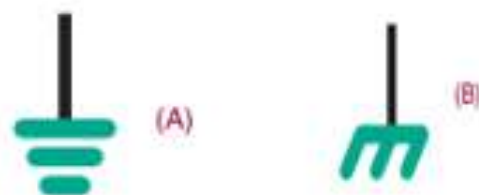


Figure SP-R-SRB- 27: **Ground symbols**

Schematic drawing lines represent wires or component leads. Letting lines cross on a diagram does not mean a connection occurs (**Fig.SP-R-SRB-28(A)**). If a connection is intended, as shown in **Fig.SP-R-SRB-28(B)**, a dot is placed at the junction. A connection dot is also used when three or more lines come together.

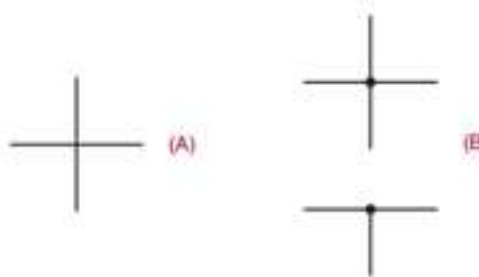


Figure SP-R-SRB- 28: **Schematic drawing lines**

Common drafting tools are used in the generation of schematic diagrams. Using schematic symbol templates can speed up the process. Computer programs can also speed up the generation of a schematic diagram. In addition to computer aided design (CAD) programs, specialty programs are also available. Specialty programs include computer programs such as Electronic Workbench MultiSIM and New Wave Livewire. These programs not only allow the schematic diagram to be drawn but also simulate current through the circuit and allow the user to troubleshoot on the computer using virtual test equipment. Additionally, the computer program Livewire shows animated current flow throughout the circuit represented by the schematic diagram.

3.3 Breadboarding

Breadboarding is essential to proving a circuit design works. The most popular technique involves the use of solderless breadboards (**Fig.SP-R-SRB-29**). Solderless breadboards were developed to allow circuits to be assembled and altered quickly without soldering. A printed circuit board is not easy to modify once the circuits are assembled. Printed circuit boards only allow minor alterations following final assembly, such as changing resistor or capacitor values or replacing integrated circuits with the same type of devices.

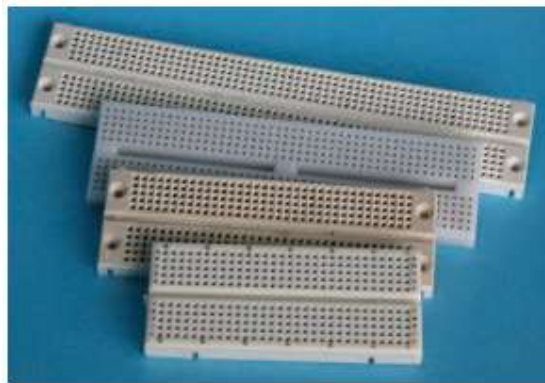


Figure SP-R-SRB- 29: Solderless breadboards

Note

Software's Micro-Cap 7 allow virtual breadboarding, or testing of a circuit and troubleshooting using virtual test equipment. Livewire also shows the current flow in a circuit to aid in understanding how a circuit performs. These programs are the wave of the future.

3.4 Laying out PCBs

Once a good schematic diagram is confirmed through breadboarding, either physical or virtual, the printed circuit board can be laid out. The necessary electronic components should be collected for sizing the design layout. Leave the components from the breadboard together and use them only as a last resort. A working circuit on the breadboard can be used for troubleshooting in the event there is a problem with the printed circuit board design.

Once the decision to design a printed circuit board has been made, a number of questions must be addressed before the printed circuit board is laid out:

1. What is the power source? Will it be part of the board design or external to it?
2. What oddly shaped or sized devices will be mounted? Devices such as relays, switches, and large capacitors need to be identified.
3. How are the discrete components attached? They need to be identified along with sockets, pins, connectors, and mounting devices.
4. What is the testing and troubleshooting considerations? They must be taken into account to make the board accessible for test measurements of all signals.
5. Will the board be mounted in an enclosure? How much space it will use and how it will be attached must be taken into consideration.

Once the questions have been answered, identify which components will be mounted on the printed circuit board and which ones will be off the board.

One approach to a printed circuit board design layout is to use paper dolls of each component that is to be mounted on the printed circuit board (**Fig.SP-R-SRB-30**). Each paper doll is the exact size of the base of the component being used and permits correct component placement.

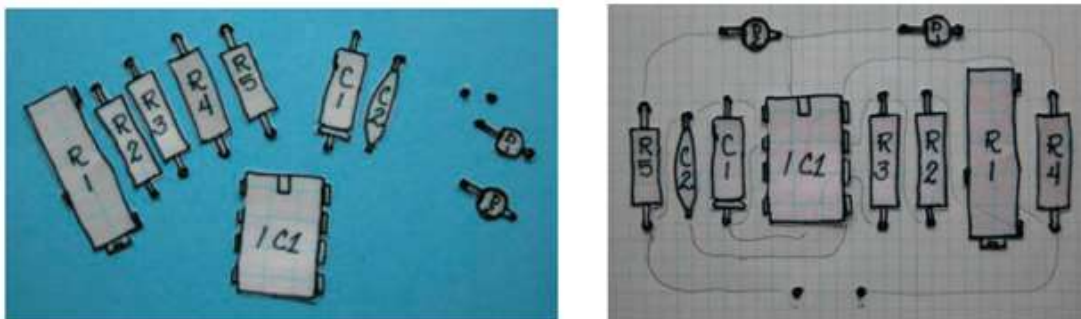


Figure SP-R-SRB- 30: Using paper dolls for laying out a PCB design

Using the paper dolls, lay out the printed circuit board in such a way that the paper dolls duplicate the schematic diagram. Exact component and lead placement are not critical at this stage. Next, consider the trace placements for proper electrical connections. During the layout process, the printed circuit board will be viewed from the component side or top.

Three issues address the best component placement:

1. Wasted board space—Keep components in a horizontal or vertical plane, parallel to the board edge, to achieve a tighter layout. Avoid diagonal placements.
2. Insertion of polarized components—Try to group polarized components facing the same direction to aid assembly.
3. Ease of troubleshooting—Components should be easy to find; try to keep input on the left side of the board, output on the right, and power from top to bottom of the board.

With the exception of discrete logic gates and multiple op-amp packages, arrange the IC pins in physical order, as viewed from the top. Start with the IC that has the most pins and then place the other parts that follow the simple schematic. The goal is to lay out a printed circuit board with no crossed traces that resembles the physical relationship of the schematic diagram, which becomes useful in troubleshooting.

Transfer the paper doll components to a 0.100" grid graph paper (ten lines per inch). Component orientation should be horizontal or vertical to the edge of the board. Identify the top and bottom layers with text component side and solder side. The solder side is viewed from the top like an X-ray view, as shown in **Fig.SP-RB-SRB-31**. Traces are then laid out left to right or top to bottom, with exceptions permitted. Traces may be routed between component leads but not between the leads of semiconductor devices. At this stage it may be necessary to reposition parts on the layout to avoid crossing traces. Once the paper doll components are placed and the traces are laid out, the decision to make a single-side or double-sided printed circuit board can be identified. If no traces cross, a single-sided printed circuit board is optimal; otherwise, a double-sided printed circuit board will be used.

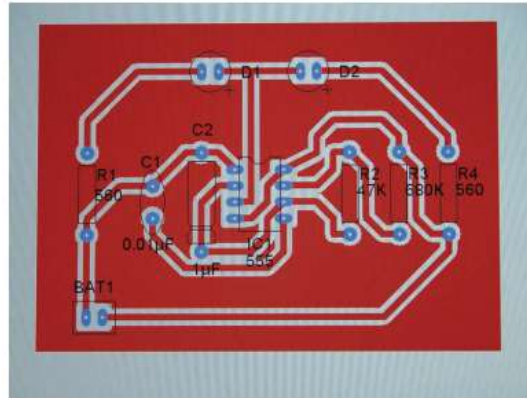


Figure SP-R-SRB- 31: Solder side as viewed from the top

Another choice for crossed traces is to make a single-side board with jumpers, as shown in **Fig.SP-R-SRB-32**, but keep the jumpers to a minimum.

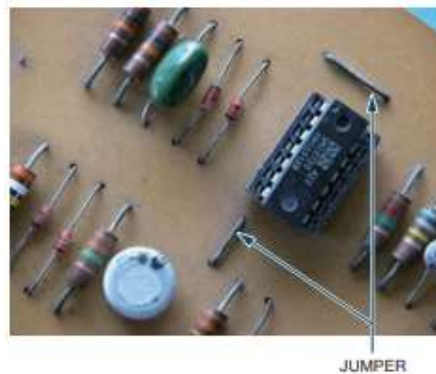


Figure SP-R-SRB- 32: **Single sided PCB using jumpers**

Rearranging artwork can often eliminate the need for a jumper wire. Some guidelines for do's and don'ts for trace layouts are shown in **Fig.SP-R-SRB-33**.

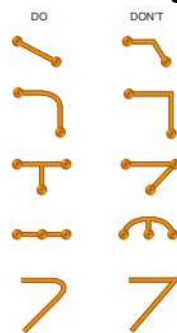


Figure SP-R-SRB- 33: *Guidelines for trace layouts*

Rules are made to be broken, even when designing printed circuit boards. When a printed circuit board is designed that has a connection on one edge, the input and output connections are on the same edge of the board. When this occurs, separate them on opposite ends of the connection. If this is not possible, separate those by putting other traces in the space between them.

Pads are placed on the printed circuit board when a connection is to be made. A pad is used to connect a lead from a component or connect a wire from an off-board component. Generally, each lead has its own mounting pad. Two leads connected to the same point will each have their own mounting pad. Multiple pads are permitted (**Fig.SP-R-SRB-34**).



Figure SP-R-SRB- 34: *Two leads connecting to the same point will have a pad for each lead
Each component should have enough mounting room. It is difficult to modify the board or
component for errors after a printed circuit board is made.*

Double-sided printed circuit boards should always have some identification on the top and bottom layer to aid in proper orientation. In the past, optical targets or registration marks were used to aid in differentiation between the top and bottom traces. Presently, this step is omitted by identifying pin 1 of an IC with a square pad.

Printed circuit boards are easier to manufacture if the traces are wide and far apart—this also makes the board more reliable and provides better performance. Most printed circuit board manufacturers will work with 0.007" traces and spaces (**Fig.SP-R-SRB-35**).



Figure SP-R-SRB- 35: *0.007 "trace vs. 0.020 "trace*

Traces that are too thin are fragile and have higher resistance than thick traces and handle less current. Estimating how much current a trace can handle is important. **Table SP-RB-SRB-3.1** shows a table comparing trace width and current capacity for a 1-ounce-per-square-foot copper foil. These values are reasonably linear for different trace cross-section areas.

Table SP-RB-SRB-3.1: PCB trace information for 1-ounce copper clad board (based on a trace length of 10 inches)

TRACE WIDTH IN INCHES	TRACE WIDTH IN MILLIMETERS	RESISTANCE /MM IN $\mu\Omega$	1 OZ COPPER 0.00135" THICK		CURRENT IN AMPERES AT 20°C
			cm ²	AWG	
0.02	0.508	989.7	0.000174	35	3.00
0.04	1.016	494.9	0.000348	32	5.00
0.06	1.524	329.9	0.000523	30	6.50
0.08	2.032	247.4	0.000697	29	7.00
0.10	2.540	197.9	0.000871	28	8.00
0.12	3.048	165.0	0.001045	27	9.25
0.14	3.556	141.4	0.001219	26	10.00
0.16	4.064	123.7	0.001394	26	11.00
0.18	4.572	110.0	0.001568	25	11.75
0.20	5.080	99.0	0.001742	25	12.50

Traces that are placed close together may short out and can electrically short to each other. The traces can also act as antennas and capacitors. High-speed digital lines next to high-impedance analog lines will cause problems. Long traces have the same problems as long wires. They pick up noise and have noticeable impedance. Even though a digital design only runs at 1 MHz, the changing state will cause transients in the GHz range.

Guidelines on laying out traces are shown in **Fig.SP-R-SRB-35**. Try to limit using acute angles of less than 90° in traces. They leave a point that causes problems by acting like a one-turn inductor. Additionally, the point will not be well supported and will easily lift—it's not good to have traces peeling off a printed circuit board.

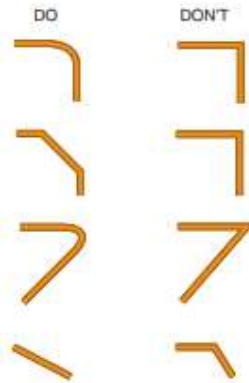


Figure SP-R-SRB- 36: *Trace layout Do's and Don'ts*

Using two 45° angles instead of one 90° angle will save space. This is especially true when there are a lot of traces running parallel to each other. The more space saved, the smaller and less expensive the printed circuit board will be. Printed circuit boards are purchased by the square inch.

Proper grounding is critical. A ground and power plane are ideal, but often it's simply too expensive as it doubles the cost of a printed circuit board.

In laying out a printed circuit board it is best to start with a wide ring of 0.100" to 0.250" around the printed circuit board to form a ground. This provides large ground areas around the printed circuit board mounting holes, which are typically located near the edge of the printed circuit board. This setup allows a star washer and a metal screw to the chassis make a good chassis-ground connection. It also means that ground traces only have to go to any board edge, simplifying the layout. If there is any concern regarding ground loops with more than one ground current flowing in one conductor, it can be solved by cutting the loop.

In this matter the current flows in the trace can be controlled. If a design is fairly simple, spend the extra time to make it a single-sided printed circuit board and use the second side as a ground plane. There are always more connections to ground than to any other circuit node.

When the layout is finished, increase the width of the power and ground traces as much as possible. In large areas without traces, fill them with solid copper connected to power or ground. More copper often helps and very rarely hurts printed circuit board performance and reduces etching time.

Proper placement of traces is something that is learned through practice. The only way to do it is to actually hand-route printed circuit boards. A common and effective technique is to run horizontal traces on one layer and vertical traces on the other layer. Obviously, this doesn't work for single-layer boards. This tends to create boards with a lot of vias. A via is a plated through-hole used to pass a conductor from one side of the printed circuit board to the other.

Basic rules for laying out a printed circuit board include:

- Avoid running parallel traces closer than necessary.
- Segregate analog and digital areas.
- Put ground traces next to sensitive analog lines to act as shields. Do the same for highspeed clock lines to reduce electromagnetic interference (EMI) and crosstalk.
- Pay close attention to parts placement. It's amazing how this can simplify routing.

It is useful to use a color-coded system to simplify the presentation of a double-sided printed circuit board during the design process. This process is standard with computer-generated designs. It can also be done with colored pencils. A typical color-code system is as follows:

- Black—board outline, holes, pads, and hardware.
- Green—component electrical symbols and outlines.
- Red—connecting traces on the component side.
- Blue—connecting traces on the trace side.
- Yellow—through holes.

Surface mount technology (SMT) has added more facets to printed circuit board layout. The parts are very small and the lead pitch or spacing between the centers of adjacent leads can be very small. It is not possible to run any traces between the leads. This makes routing more difficult and increases the number of vias.

It's virtually impossible to use a single-side layout with high-pin-count SMT ICs. The pad sizes and shapes are different for different parts. You will need to refer to the manufacturer's specifications. With SMT designs, testing and repair need more consideration because it's hard to probe tiny, closely spaced pins. Boards should be laid out to make servicing easy.

Autorouters programs are designed to use the computer to lay out a printed circuit board. Autorouters should not be relied on unless they are "smart." Typical low-cost routers only connect points together. They don't consider the length of the trace, which traces are analog and which are digital, which lines are sensitive, which are ground loops, and so on. Subtle points can make a significant difference in how a printed circuit board performs. It's easier to lay out a printed circuit board manually than to root out elusive errors made by a program.

Getting a printed circuit board made commercially requires a design to be sent by the Internet to a service. It will require the following files in Gerber format for a standard double-sided printed circuit board:

1. Top trace layout
2. Bottom trace layout
3. Silkscreen (if used)
4. Soldermask (if used)
5. Aperture file
6. Drill file

The Gerber format is a standard printed circuit board format supported virtually all layout software. The Gerber files define the trace placement, but they don't specify the physical sizes.

The aperture file specifies the physical sizes with a short list of "flash codes" Additionally, the Gerber files don't specify the actual size of the holes to be drilled in the printed circuit board. The drill file informs the manufacturer of where the holes are and what size they are supposed to be. Drilling and etching are two separate procedures. When these files are sent, they include a short "read me" file that relates the file names to the physical parts of the printed circuit board to save time and lessen confusion.

Proper printed circuit board layout requires common sense, patience, and attention to detail.

3.5 Transferring designs

There are several techniques used to transfer the artwork to the copper-clad board. The simplest technique is to hand-draw the design on the board with a resist pen or any other permanent marker with a very fine tip.

Take the original artwork and transfer it to the back of the graph paper. Place the component side of the artwork to the surface of a light table or hold the paper to a window with light on the opposite side. Start by placing Xs where each pad is located and then connect the pads with the trace lines. On the side just drawn, write "BOTTOM."

Wrap the artwork around a piece of copperclad board that has been properly cut to size. Clean the board with steel wool or with powder cleanser and water prior to wrapping the artwork around the board. The copper side of the board should be toward the artwork with the side identified as bottom facing outward.

Using a scribe, locate the center of each pad where an X was placed on the bottom side. Prior to removing the artwork, run a finger over each pad to ensure it was marked with the scribe. Remove the artwork, being careful not to get fingerprints on the copper.

With a resist pen or other fine-tip permanent marker, such as a black Sharpie, draw the pads at each of the scribe marks. Using the bottom-view artwork, connect the pads with traces. Go over each of the traces and pads several times to build up a thick layer of resist.

Another technique for creating a resist involves using the computer to make the artwork with a CAD or drawing program. Draw the design from the component side and then reverse it to form the trace or bottom side. Print out a positive image 1:1 on transparency or special film that can then be transferred to a clean, properly sized copperclad board with an iron and pressure.

A second technique involves printing out two positive copies of the artwork 1:1 on transparency film. Align the two pieces of film and tape them together. They form a positive film transparency. If a higher density is needed from the film, align and tape additional pieces of film. Sensitize a clean copper-clad board with positive resist according to manufacturer instructions. Using a contact print frame, sandwich the film and sensitized board together with the bottom side outward. Expose the sandwiched board and film to an ultraviolet light source or sunlight. Develop the board in the appropriate developer; rinse and dry the board. This technique forms a very good quality resist

A third technique involves using the positive film transparency developed in the previous discussion to create a screen for screenprinting the printed circuit board. A good graphic arts book will cover the making of a screen and the screenprinting process in detail. Use resist inks or oil base inks when screenprinting the layout on the boards.

3.6 Etching PCBs

There are two major techniques to remove copper from a copper-clad board to form a printed circuit board. The first technique involves using a mild acid such as ferric chloride or ammonia persulphate. The other technique uses a CAD-type program and a CNC (computer numerical control) machine to remove the copper and is referred to as a non-chemical process. Environmental and safety issues are favoring the use of nonchemical copper removal techniques.

Using either ferric chloride or ammonia persulphate acid the copper can be removed by the tray method or with a spray etcher.

Caution

When Etching, Always Wear Safety Glasses, Rubber Gloves, and an Apron.

To use the tray method, the copper-clad board with applied resist is etched in a glass tray by dropping the board onto the surface of the acid. The copper side of the board should face toward the acid. Float it on the acid for approximately fifteen to twenty minutes. The acid laden with copper sinks to the bottom and fresh acid replaces it. In this fashion no agitation is required.

A different technique uses a one-inch square of sponge to dab the acid on the printed circuit board. The acid is absorbed in the sponge by pouring a small amount on it. The sponge is dabbed continuously on the copper on the board until the copper is removed. Boards with thinner copper will etch faster.

Caution

Always Wear Safety Glasses, Rubber Gloves, and an Apron When Using This Technique.

Another method to remove the excess copper from the board is to use a commercial or shopbuilt spray etcher (**Fig.SP-R-SRB-37**). These etchers can etch the board in approximately two minutes. A spray etcher is the best way to etch double-sided boards.



Figure SP-R-SRB- 37: *Examples of commercial spray etchers*

Computer programs such as New Wave's PCB Wizard provide the appropriate codes to drive a CNC machine to remove the copper to form a printed circuit board. **Fig.SP-R-SRB-38** shows a CNC machine.



Figure SP-R-SRB- 38: *CNC used for PCB fabrication (Courtesy of T-tech)*

3.7 Preparing the Etched PCB

After the printed circuit board has been etched the resist needs to be removed. Remove the printed circuit board from the acid, rinse it thoroughly in running water, and dry it. The resist can be removed by soaking the board in a solvent, scrubbing it with powder cleanser, or using steel wool to rub it off. After the resist is removed from the printed circuit board, be sure to avoid getting fingerprints on the board.

The next step is optional in fabricating a printed circuit board. The printed circuit board can be tin-plated to stop oxidation of the copper using a tin-plating solution, as shown in **Fig.SP-R-SRB-39**. The printed circuit board is immersed in the solution for a set amount of time identified by the manufacturer.



Figure SP-R-SRB- 39: *Example of tin-plating*

Any defects resulting from the etching process should be taken care of at this time. Types of defects include traces that are bridged or shorted together or traces that are open (**Fig.SP-R-SRB-40**). Traces that are shorted together can be separated using a knife. Traces that are open need to be bridged with wire and solder.

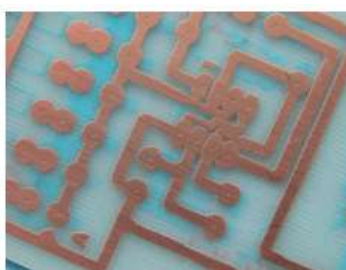


Figure SP-R-SRB- 40: *Example of traces shorted together*

Use a high-speed drill for drilling all holes (**Fig.SP-R-SRB-41**). Use a backer board under the printed circuit board to avoid breakout of the backside.



Figure SP-R-SRB- 41: High speed drill for drilling holes in a PCB

A number 59 drill bit is used for all holes. Holes that need be drilled larger are then drilled. Burrs that form around the holes can be removed by hand using a larger drill bit and a twisting motion (**Fig.SP-R-SRB-42**).



Figure SP-R-SRB- 42: *Removing burrs from drilled holes*

3.8 Material Safety Data Sheet (MSDS)

OSHA (the Occupational Safety and Health Administration) requires that a material safety data sheet (MSDS) be supplied when a chemical or hazardous substance is purchased. The law requires that copies of MSDSs be maintained and available on request by the person(s) using this material.

When unpacking a shipment of chemicals look for the MSDS in the box. File the MSDS in a folder in the location where the chemical is being used.

- 1 To help protect personnel from injury and exposure hazards.
- 2 It is required by law.

Read each MSDS before using hazardous material. Everyone who uses the chemical product must understand its dangers and the precautions to be taken while using it.

- Appendix A provides a brief description of the sections of an MSDS.
- Appendix B shows an MSDS for ferric chloride.

Recipients of material safety data sheets should consult the OSHA Safety and Health Standards for general guidance on control of potential occupational health and safety hazards.

In the next topic, we will look at how to design a simple PCB.

SAMPLE MODULE ACTIVITIES

Activity 1: Practical or Project

Title: Assembling a PCB

Objective: To assemble a PCB

Note

An electronics technician uses a variety of tools to build and repair electronic circuits on printed circuit boards. These tools coupled with electronic test equipment allow the technician to analyze any electronic circuit for proper operation or identify faulty circuit components and repair them. The right tools make the job easier for an electronics technician.

Introduction

Electronics Technician Toolbox

The installation and repair of components in a printed circuit board may be accomplished through the use of tools designed specifically for the task. Appendix C shows some of the tools in an electronics technician's toolbox.

Fig.SP-RB-SRB-4.1 shows two types of printed circuit board holders for holding a printed circuit board while it is being assembled or repaired. The fixtures should provide good support and should be rotatable



Fig.SP-RB-SRB-4.1: Examples of PCB holders

An illuminated magnifier, shown in Fig.SP-RB-SRB-4.2, provides a means of detecting improper connections or broken traces on the printed circuit board. It also helps when soldering SMT components to a printed circuit board.



Fig.SP-RB-SRB-4.2: Illuminated magnifier

Fig.SP-RB-SRB-4.3 shows special tools used to assist with the forming of leads of resistors for installing in a printed circuit board.



Fig.SP-RB-SRB-4.3: Component lead former tool

Electronic Test Equipment

Electronics technicians use a variety of test equipment for determining proper operation, aligning components, and troubleshooting a circuit. Appendix D a sample of test equipment available to electronics technicians and explains their function.

An isolate power supply is important when powering a circuit to prevent it from generating a lethal shock potential. Power supplies are available that generate DC voltage and/or AC voltage. Both fixed and variable voltage power supplies are available. The power supply is used to supply a circuit with the correct voltage for testing and troubleshooting.

Analog and digital multimeters are the first pieces of equipment used to analyze a circuit. To prevent possible circuit damage the meter should have a sensitivity of more than 20,000 ohms per volt on the voltage scales. The meter should also not pass more than one milliampere of current on the resistance scales. Static DC and resistance checks will usually locate power failures or defects that exhibit large deviations from the normal on the printed circuit board. However, these methods are time consuming and inadequate when the defect is intermittent in the circuit.

An oscilloscope is one of the most useful pieces of test equipment available to the electronics technician. It can identify defects in a circuit while it is powered up; this includes changing parameters of a circuit. It displays the instantaneous voltage on the vertical axis against time on the horizontal axis. The oscilloscope can detect both DC and AC signals. Some oscilloscopes allow the comparison of two waveforms of the same frequency to determine the phase shift. The oscilloscope probe on most oscilloscopes will allow switching between settings of 1X and 10X (**Fig.SP-RB-SRB-4.4**). When the signal exceeds the capability of the oscilloscope on the 1X setting, the 10X setting multiplies the voltage setting of the oscilloscope by ten and makes viewing the signal possible.



Fig.SP-RB-SRB-4.4: Oscilloscope probe

Frequency counters are useful for measuring the frequency and period of a recurring signal. They provide a digital readout of the frequency. Frequency measurements are an important part of preventive and corrective maintenance of electronic equipment.

Signal generators are used in the testing and aligning of radio transmitters, receivers, and amplifiers. They are also used for troubleshooting various electronic devices that require an input signal. Signal generators produce an alternating voltage of a desired frequency and amplitude for testing or measuring a circuit. Signal generators are classified by their frequency into the following categories: audio generator, video signal generator, radiofrequency generator, frequency-modulated RF generators, and special types of signal generators that combine several or all of the frequency ranges.

A function generator is a type of signal generator and is a common piece of electronic test equipment. The function generator is capable of generating fundamental waveforms such as sine waves, triangular waveforms, and square waveforms. The voltage, symmetry, and time characteristics of a waveform can be adjusted to meet a specific application.

A logic probe contains LEDs that produce a visual indication of what is occurring at a specific point in a digital circuit. It can indicate a high, low, clock, or lack of a signal. It can be used with a logic pulser, which looks similar to the logic probe. The pulser injects a logic level into a circuit, which can be picked up by the logic probe. The logic probe is useful for checking signals of various input and output pins of a circuit or an IC.

Solder and Soldering Irons

Soldering is the joining of two pieces of metal alloy having a melting point below 800° Fahrenheit. Solder includes a combination of tin and lead. Tin-lead solder ranges from pure tin to pure lead and all proportions in between. In electrical soldering, the alloy mix is usually 60% tin and 40% lead (60/40) or 63% tin and 37% lead (63/37) (**Fig.SP-RB-SRB-4.5**). For soldering surface mounted devices, a solder is available with 62% tin, 28% lead, and 2% silver (62/36/2).



Fig.SP-RB-SRB-4.5: Spool of solder

Characteristics of alloys of tin and lead are plotted against temperature in **Fig.SP-RB-SRB-4.6**. This graph allows one to see that only an alloy of 63/37 has a eutectic point (a single melting point). All other combinations start melting at one temperature, pass through a plastic stage, and then become a liquid at a higher temperature. Any physical movement of the components being soldered while the solder is in the plastic stage will result in a fractured solder connection. Such a connection will appear dull and grainy, and is mechanically weaker and less reliable. Therefore, 63/37 or 60/40 solder is commonly used in electronics because it does not remain in the plastic stage very long.

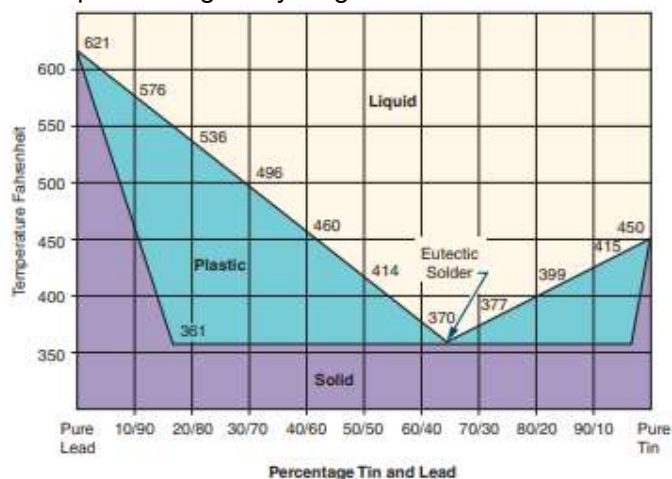


Fig.SP-RB-SRB-4.6: Solder alloy vs. temperature

Flux helps the solder alloy flow around the connections. Flux cleans the component leads of oxide and film, allowing the solder to adhere. Soldering flux is usually included in the solder in the central core. Rosin flux is always used in electrical soldering because it is noncorrosive. The primary tool used for installing, removing, and/or replacing components on a printed circuit board is the soldering iron. An improper choice in selecting a soldering iron or soldering iron tip may result in damage to the component and/or the printed circuit board. Select only a lowwattage pencil soldering iron (**Fig.SP-RB-SRB-4.7**) with an appropriate tip for the soldering being done.



Fig.SP-RB-SRB-4.7: Example of low wattage pencil soldering iron

Solder guns (**Fig.SP-RB-SRB-4.8**) are inappropriate for printed circuit board fabrication. They are designed to supply heat to large electrical connections and will ruin the small detail of a printed circuit board.



Fig.SP-RB-SRB-4.8: Solder gun

The soldering iron is usually specified by wattage. In most situations a 25- to 35-watt soldering iron is adequate. Wattage represents the amount of heat capacity available at the tip. Soldering irons of all wattages usually run at the same temperature. A low-wattage soldering iron will tend to cool faster during the soldering process than a high-wattage soldering iron. The handle of the soldering iron should be comfortable to hold and should be well insulated to prevent heat buildup. Soldering irons can produce static voltage spikes that will destroy many integrated circuit components. A grounded soldering iron tip is a good safety measure. **Fig.SP-RB-SRB-4.9** shows several types of soldering irons suitable for printed circuit boards. The least expensive is the pencil soldering iron, and with proper care it will last many years. The temperature-controlled soldering station provides precise tip temperature through a closed loop circuit. It maintains a constant tip temperature throughout the soldering process. The cordless soldering iron is similar to the pencil soldering iron except it uses a battery for its power source.



Fig.SP-RB-SRB-4.9: Several types of soldering iron

Iron-clad tips are the best selection for soldering printed circuit boards. An unplated tip will provide better heat transfer but wears rapidly and requires frequent cleaning and tinning. The iron-clad tip requires only an occasional cleaning and tinning.

Soldering iron tips come in a wide variety of shapes and sizes to meet a specific application. Many soldering irons have replaceable tips. The tip selected should provide maximum contact area between the surfaces to be joined and also allow accessibility to the connection. **Table SP-RB-SRB-4.1** shows the most popular shapes of soldering iron tips. Typically, a small bevel tip is preferred for general printed circuit board work.

Table SP-RB-SRB-4.1: Soldering iron tips

SOLDERING IRON TIP			DESCRIPTION
TYPE	END VIEW	SIDE VIEW	
CONICAL			Used for general assembly and repair
PYRAMID			Used for general assembly and repair
BEVEL			Allows rapid transfer of heat. Used for printed circuit boards.
CHISEL			Allows large areas to be heated rapidly. Used for tinning wires, soldering terminals.

Maintaining the soldering iron tip is essential for proper soldering. Use an antiseize compound when installing the tip to prevent it from becoming frozen. Make sure the tip is fully installed in the heating element. Clean the tip with steel wool prior to heating it. Heat the tip and apply solder as it warms up. Applying solder to the tip is called tinning. Rest the soldering iron on a stand to remove excess heat from nontemperature controlled soldering irons, as shown in **Fig.SP-RB-SRB-4.10**. This is referred to as idling. Wipe the tip of the soldering iron across a damp sponge or tip cleaner to clean it of any oxidation prior to soldering.



Fig.SP-RB-SRB-4.10: Idling soldering iron

Soldering a PCB

Use a systematic method for installing components on the printed circuit board. One technique is to install all the resistors first, then the capacitors, ICs, and so on.

With axial components the leads need to be bent at a right angle. Grasp the ends of the component between the thumb and index finger and pull the body outward while maintaining a grip on the ends of the leads (**Fig.SP-RB-SRB-4.11**).



Fig.SP-RB-SRB-4.11: Manually bending component leads

The component should now fit between the pad mounting holes. Position components so the values can be easily read. Arrange resistors so the value can be read top to bottom or left to right. Observe polarity of all polarized components and orientate them properly on the printed circuit board. Pay close attention to installing ICs on the printed circuit board to ensure pin 1 is lined up with its appropriate location on the board. Use 1/4" leg warmers on LEDs and small transistors to avoid overheating (**Fig.SP-RB-SRB-4.12**).



Fig.SP-RB-SRB-4.12: Example of leg warmers

After the component leads have been inserted and prior to soldering, they should be bent to a 45° angle to keep them from falling out.

Once the printed circuit board is assembled, the next area to consider is the off-board components (potentiometers, lamps/LEDs, switches, transformers, etc.). A wiring diagram will make the installation clear and easy to understand. The two types of drawings are the pictorial diagram (**Fig.SP-RB-SRB-4.13**) and the functional diagram (**Fig.SP-RB-SRB-4.14**). There is no standard color code for wiring a printed circuit board. It is best to use different color wires for each discrete component and note the colors on the wiring diagram. There typically aren't enough colors for each signal to have its own color, but different colors can be used to denote different classes of signals.

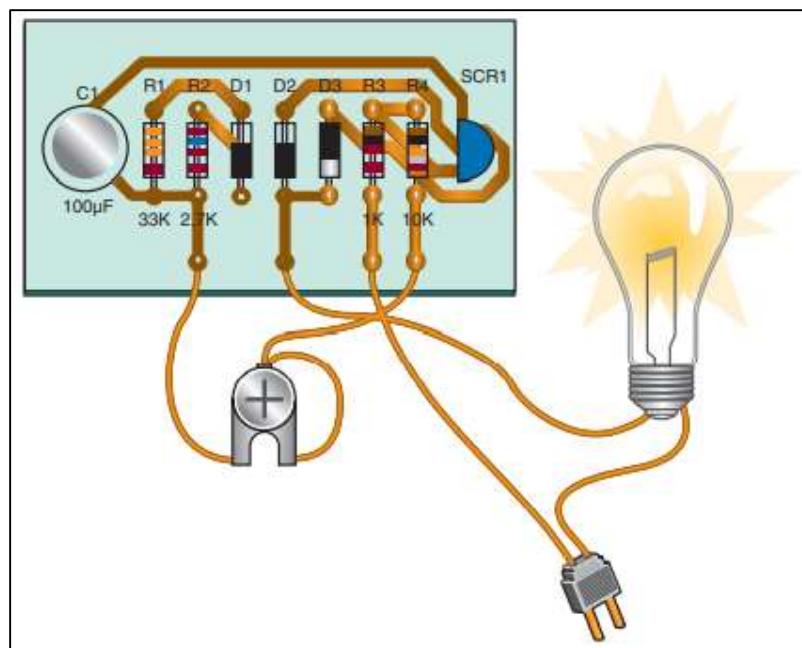


Fig.SP-RB-SRB-4.13: Example of a pictorial diagram

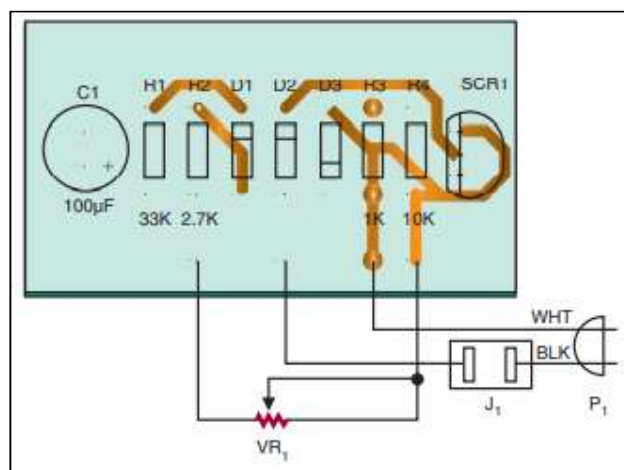


Fig.SP-RB-SRB-4.14: Example of a PCB functional diagram

When constructing a diagram, one set of signals that should be color-coded is power and ground. A different color for the positive voltage, negative voltage, and ground should be used. These colors should not be any of the colors that have been used for other signals. It would be nice to use “standard” colors for these, but unfortunately there are several competing standards. Red is frequently used to indicate the positive supply voltage. Most automotive and electronic wiring diagrams use black to denote ground. Electronics wiring that uses black for ground often uses blue for negative supply voltage. Green is frequently used to denote ground. In electronics, the basic goal of soldering is to electrically and mechanically join two circuit components. For soldering to be successful and reliable, the solder must adhere to the mating surface. The application of solder to a base metal requires the surface to be clean and free of contamination. If the surface to be soldered is contaminated, the solder tends to ball up and not adhere. Sources of contamination include greases, oils, and dirt. Aging also causes surface contamination by the formation of an oxide film.

The best soldering techniques can be outlined simply. First, wipe the tip of the soldering iron on a damp sponge. Then place a small amount of solder on the tip and apply the tip to the connection to be soldered. Solder should not be applied until the connection becomes hot enough to melt the solder. How long this take is quickly learned by a few trials and errors. When the solder has melted and flowed into a con toured fillet, remove the solder. Keep the tip of the soldering iron on the connection for a few seconds longer, and then remove it. Do not disturb the newly made connection until it has had time to solidify. A good solder connection will be bright and shiny (**Fig.SP-RB-SRB-4.15**).

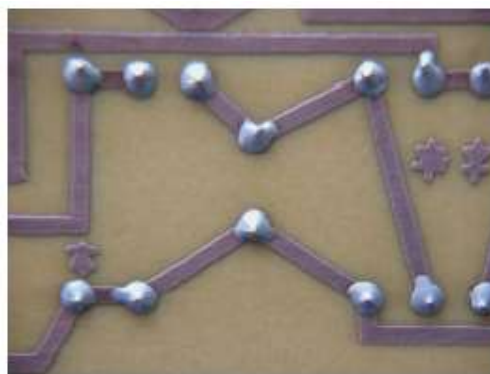


Fig.SP-RB-SRB-4.15: Example of a good solder connection

Handle components and the printed circuit board as little as possible because fingers may be dirty and oily. If there is a question about the cleanliness of a part, clean it using alcohol, steel wool, or a scouring powder. When using scouring powder, ensure that the parts are rinsed thoroughly. If steel wool is used, use a lint-free cloth to remove all pieces of the steel wool from the part or printed circuit board.

Soldering on a printed circuit board requires careful attention because of the heat sensitivity on the pads and traces. Too much heat with too much pressure can cause a pad to lift from the board substrate. Always apply the tip with a light touch—practice will determine approximately how long it needs to be held. The soldering iron should rest on the junction of the lead and pad rather than being pressed to it (**Fig.SP-RB-SRB-4.16**).

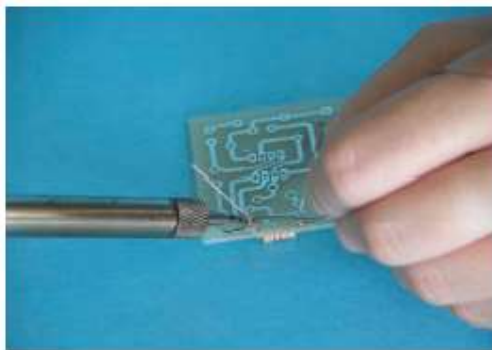


Fig.SP-RB-SRB-4.16: Place the soldering iron tip at the junction of lead and the pad with no pressure

To summarize proper soldering technique:

1. Wipe tip on a damp sponge or tip-cleaning apparatus.
2. Tin the tip by applying solder to it.
3. Apply the tip to both the component lead and pad.
4. Apply solder to the back of the connection; the heat of the tip will draw it to it.
5. Remove the solder.
6. Remove the tip from the connection.
Let the connection cool.
7. Trim the excess leads flush to the printed circuit board.

During the soldering process a flux residue forms on the printed circuit board. The residue is noncorrosive in most cases. However, it should be removed as it is sticky and foreign particles can attach to it. Also, removing the flux makes the inspection of the board easier and gives a more professional appearance to the board. There are several defluxers on the market that make removing easy. Unfortunately, they are not environmentally friendly because they contain trichlorotrifluoroethane as their main ingredient.

To use the flux cleaner, spray the surface to be cleaned in a well-ventilated area. Allow the chemical to saturate the board for ten to fifteen seconds, and then clean it with a medium-bristle brush such as an acid brush. Allow the components to dry completely before powering up the circuit.

Once the skill of soldering is mastered, the skill of desoldering is next. Desoldering requires special implements: desoldering braid, desoldering bulb, desoldering pump, combination bulb and soldering iron, and desoldering station.

In desoldering the first step is to solder. Applying more solder on the connection ensures the connection is heated up prior to removing the solder. Do not heat a connection with very little solder, as it will result in the pad lifting from the substrate.

A desoldering braid consists of a braided copper material with flux imbedded in it that uses capillary action to pull up the hot solder (**Fig.SP-RB-SRB-4.17**). Using the widest braid practical, place it on the pad to be desoldered. Place a hot soldering iron on top of the braid. The solder will melt and be wicked into the braid. When the braid gets saturated, remove it, clip off the saturation end with diagonal cutters, and reapply until the solder is removed. Use a hot soldering iron—the braid acts like a heat sink, cooling the soldering iron tip.



Fig.SP-RB-SRB-4.17: Desoldering braid

The desoldering bulb uses a bulb and a teflon tip. Squeeze the bulb and apply the tip to the melted solder; releasing the bulb draws up the molten solder (**Fig.SP-RB-SRB-4.18(A)**).

A variation of the desoldering bulb is a springloaded desoldering pump (**Fig.SP-RB-SRB-4.18(B)**). It works by depressing a plunger until it locks. To suck up solder, press the release button. The spring forces the plunger up rapidly, creating a vacuum.

The combination rubber bulb and soldering iron uses a hollow tip to suck up the solder (**Fig.SP-RB-SRB-4.18(C)**). It eliminates the rush to get the solder sucker to the pad before the solder rehardens.



(A)



(B)



(C)

Fig.SP-RB-SRB-4.18: (A) Using a desoldering bulb, (B) Solder pump, and (C) Combination desoldering bulb and iron with a hollow tip

A professional desoldering station uses a footoperated vacuum pump and a hollow-tip soldering iron to draw up molten solder. It is the most expensive tool used for desoldering a component from a printed circuit board pad.

Any of the desoldering tools discussed will do the job of removing solder from a printed circuit board connection. Each requires different techniques and fits different budgets.

The mounting or removal and replacement of components for SMT (surface mount technology) printed circuit boards is different than in conventional printed circuit boards.

With SMT printed circuit boards, in addition to the board being smaller, the components are much smaller and must be handled with tweezers. Mount the printed circuit board in a vise. This allows stability when using a magnifying glass to work on the circuit.

The proper method for removing SMT components is with a rework station. To desolder SMT components manually, proceed as described in the next paragraphs.

For an IC remove as much solder as possible from the connection using 0.030"-diameter desoldering braid. The only solder remaining is holding the component leads to the printed circuit board. Wipe the soldering iron clean and heat the lead and pad while using an X-Acto knife to gently pry up the lead. When the lead releases, remove the soldering iron and allow the lead to cool. Repeat on each lead until the IC is removed from the printed circuit board.

Caution

TOO MUCH PRYING PRESSURE CAN LIFT THE PAD.

For SMT resistors and capacitors, add solder to the pad to increase the thermal mass to prevent it from cooling too fast. Alternately heat each end and use tweezers to remove the component while the solder is molten. Use desoldering braid to remove the excess solder.

The solder used should be 63/37 or a silver bearing solder of 62/36/2 with a diameter of 0.020" to 0.015". A noncorrosive liquid flux with a drop dispenser and a light-duty spray defluxer is required for applying and cleaning flux on the SMT leads. Clear tape is useful for taping down components prior to soldering.

After the printed circuit board is completely assembled and soldered, apply power. If the circuit works properly, enjoy. Otherwise, troubleshoot by first removing power from the printed circuit board. Inspect all traces for bridging or breaks and all solder connections for shorts. Check for improper solder connections. Correct all problems and retest.

Analyzing Soldered Connections

The surface of a properly soldered connection should be smooth, well feathered at the edges, and bright and shiny with a slight concave configuration (**Fig.SP-RB-SRB-4.19**).

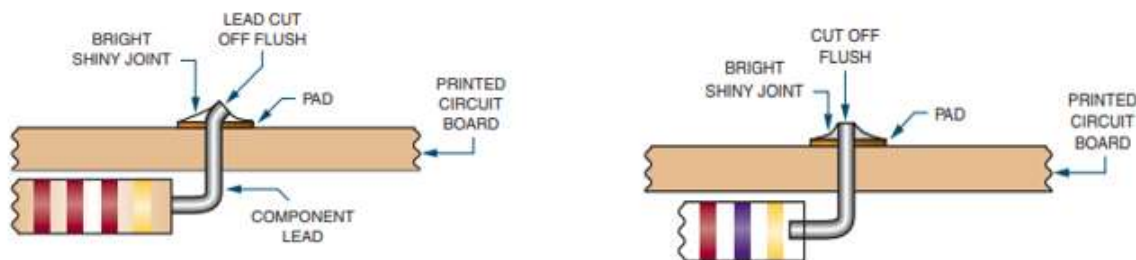


Fig.SP-RB-SRB-4.19: Characteristic of a solder connection

Two of the most common soldering defects include cold solder connections and disturbed or fractured connections. Cold solder connections are the result of withdrawing the soldering iron too soon. The solder does not become liquid and leaves a connection that is dull gray in appearance. A fractured or disturbed connection is the result of movement of the component leads or wire before the solder sets. The fractured joint is evident by a granulated and frosty appearance with many cracks showing.

Other types of defects include:

- Rosin connection is caused by the flux and oxidation particles staying between the solder pad and lead. It is the result of not enough heat.
- Wicking is caused by the capillary action of stranded wire. It is the result of too much heat.
- Voids are caused by impurities in the surface when soldering. They are the result of too much heat.
- Removing the solder too soon results in not enough solder.
- Too much solder is caused by not removing the solder soon enough.
- Bridging (inadvertently connecting two pads) is caused by not removing the solder soon enough.
- Damage to a component is caused by too much heat—the iron is too large or held on too long.

Protective Coating

Conformal coatings are nonconductive materials applied in a thin layer on printed circuit boards. They provide environmental and mechanical protection to significantly extend the life of the components and circuitry. Conformal coatings are traditionally applied by dipping, spraying, or flow coating.

Conformal coatings protect electronic printed circuit boards from abrasion, corrosion, dust, fungus, moisture, and environmental stresses. Common conformal coatings include acrylic, epoxy, parylene, silicone, and urethane.

Most conformal coatings consist of a synthetic resin developed in a volatile solvent. When they are properly applied to a clean surface, the solvents evaporate, leaving a thin film of solid resins. Conformal coatings have one or more of the following characteristics to protect the printed circuit board effectively:

1. Good thermal heat conductivity to carry heat away from the components.
2. Inorganic composition to prevent fungus growth.
3. Low moisture absorption.
4. Low shrinkage factors during application and curing to prevent coating from applying strains or stresses to the components or leads.

5. Resilience, hardness, and strength to support and protect the components.
6. Qualities of electrical insulation.

Safety Precautions

Effective safety habits are necessary to succeed in the electronics field. Practicing good safety habits promotes a safe work environment. Everyone needs to be aware of and support a safe work environment.

A primary concern in the field of electronics is electrical shock. Electrical shock is the passage of current through the body. Current can flow through the body just as it does through an electrical circuit. How much current flows depends on these factors?

- Body resistance is high if the skin is dry and there are no cuts or abrasions at the point of electrical contact. This results in little current flow with only a mild shock.
- Body resistance is low if the skin is moist. This results in a high current flow, and if the current flow is through the chest region the heart can receive a lethal dose of current. The heart may go into fibrillation and breathing may be stopped (see **Table SP-RB-SRB-4.2**).

Table SP-RB-SRB-4.2: Hazards of current flow through the human body

SENSATION FELT	CURRENT IN AMPERES AT 60 HERTZ	
	WOMEN	MEN
Can not feel anything	0.0003	0.0004
Tingling—mild sensation	0.0007	0.0011
Shock, not painful, can let go	0.0012	0.0018
Shock, painful, can let go	0.006	0.009
Shock, painful, barely let go (threshold), may be thrown clear	0.0105	0.016
Shock, painful, severe, can not let go, muscles immobilized, breathing stops	0.015	0.023
Ventricular fibrillation, fatal,		
Length of time—0.03 seconds	1.0	1.0
Length of time—3.0 seconds	0.1	0.1
Heart stops for the length of time current flows, heart may start again if current flows a short time	4	4
Burning of skin, usually not fatal unless heart or other vital organs are burned	5 or more	5 or more
Cardiac arrest, severe burns, and probable death	10,000	10,000

Small amounts of current through the body can be dangerous to one's health. Large amounts of current through the body can be fatal. A shock of 1 to 20 mA can cause a painful sensation. At a shock of approximately 20 to 30 mA, breathing will be stopped. A shock above 100 mA will result in electrocution.

The work environment should be reviewed to identify problem areas. Maintaining a safe work environment is easy when observing the following guidelines:

1. Keep the work area neat and orderly.
2. Be alert and attentive at all times.
3. Know the correct operation and safety procedures of test equipment, tools, and machinery.
4. If an accident occurs, know what action to take.

Note

IF A CHEMICAL IS INVOLVED CHECK THE MSDS. BASIC FIRST-AID AND CPR (CARDIOPULMONARY RESUSCITATION) TRAINING IS ESSENTIAL FOR WORKING IN THE FIELD OF ELECTRONICS.

Personal safety when working with electricity requires the following:

- Work with one hand in back pocket or behind back when working on live circuits. This technique avoids a complete path for current flow that passes through the heart region.
- Use an isolation transformer when working on AC-powered equipment. The transformer isolates the powered equipment from the power source.
- Make sure all capacitors are discharged before beginning troubleshooting. Use a capacitor discharge tool to discharge the capacitor.
- Use grounded line cords and polarized plugs with AC-operated equipment and circuits. They help to reduce the danger from hot chassis.
- Keep hands off of live circuits. Test all circuits prior to working on them with a voltmeter or test lamp.

Prior to handling toxic and hazardous chemicals, read the MSDS to know the best protection and first-aid procedures in case of emergency. When around these types of chemicals, be concerned about inhaling vapors, swallowing liquids, acid burns on the skin, contact with the eyes, and danger of fire and explosions. Follow these guidelines to minimize risk:

- Read the labels of all chemicals being used, paying special attention to the printed warnings.
- Work in well-ventilated areas, especially when working with paint and chemical sprays.
- Wear safety glasses when working with hazardous chemicals.
- Wear rubber gloves when working with acids and acid solutions.
- Use tongs or rubber gloves when placing or removing printed circuit boards from the acid solution.
- Clean all tools that contact hazardous chemicals in case anyone accidentally touches them.

Safety is the number one priority when soldering. Always follow these safety guidelines:

1. Always wear safety glasses when soldering or cutting leads. When clipping leads on the printed circuit boards, flying leads can be an eye hazard. When clipping leads, aim down and away from the face and anyone close.
2. Never touch the soldering iron tip or heating element. Only pick up the iron by the handle. Never change the tip when the iron is hot.
3. Beware of fingers slipping over the iron handle and contacting the heating element.
4. Beware of pressing the soldering iron too hard against the leads when heating. The soldering iron may slip.
5. Beware of molten solder dripping from the connection. Cotton and wool clothing are the safest to wear when soldering. Many synthetics will melt on the skin if molten solder gets splashed on them, causing a greater burn.
6. Use care when handling printed circuit boards. Leads can cut or puncture the skin. Always have cuts or punctures cleaned and treated immediately.
7. Avoid breathing in flux fumes when soldering.
8. Cleaning solvents will dry out the skin. Avoid breathing the vapors.
9. Wash hands before eating or drinking. The lead from the solder remaining on the hands can be ingested with food.

Certain components used on printed circuit boards can be damaged by static electricity referred to as electrostatic discharge (ESD). These static charges are created when nonconductive materials, such as plastic from bags or synthetic materials from clothing, are separated. These charges can be induced on a conductor, such as skin, and are delivered in the form of a spark that passes to another conductor. By touching a printed circuit board or semiconductor device a static discharge can be delivered through it that may damage a sensitive component or device.

Most static charges that are able to damage a printed circuit board component or semiconductor device will be too weak to feel—less than 3,000 volts. Most components of printed circuit board assemblies will have warnings printed on them noting that they are sensitive to damage by static electricity, such as the labels shown in **Fig.SP-RB-SRB-4.20**. Keep in mind that just because a component does not have a symbol or label does not mean it is not ESD sensitive.



Fig.SP-RB-SRB-4.20: Warning labels for components sensitive to static electricity

The following protective guidelines will help to prevent damage to components:

- Keep sensitive components enclosed in antistatic bags, boxes, or wraps when not working on them.
- Remove sensitive components from their protective wrapping only when in the antistatic work area.
- Keep antistatic workstations free of staticgenerating materials such as Styrofoam, plastic desoldering tools, plastic notebook folders, combs, and so on.

Electrostatic Charge

Static electricity is created when two substances are rubbed together or separated. The substances can be solid or fluid. The rubbing or separating causes the transfer of electrons from one substance to the other. This results in one substance being positively charged and the other substance being negatively charged. When either substance comes in contact with a conductor, an electrical current flow until it is at the same electrical potential as ground. This is referred to as electrostatic discharge (ESD).

Static electricity is commonly experienced during the winter months when the environment is dry. Synthetics, especially plastic, are excellent generators of static electricity. When walking across a vinyl or carpeted floor and touching a metal doorknob or other conductor, an electrical arc to ground may result and a slight shock is felt. For a person to feel a shock, the electrostatic potential must be 3,500 to 4,000 volts. It typically takes 5,000 volts to jump 1/4 of an inch. Lesser voltages are not apparent to a person's nervous system even though they are present. Inside an integrated circuit, a static discharge can destroy oxide layers or junctions. These static discharges can cause an ESD latent defect. The device may appear to operate properly, but it is damaged. It has been weakened and failure will result at a later time. There is no method for testing for ESD latent defect. **Table SP-RB-SRB-4.3** shows potential sources for generating an electrostatic discharge.

Table SP-RB-SRB-4.3: Potential electrostatic sources

OBJECT/PROCESS	MATERIAL/ACTIVITY
Clothes	• Synthetic garments • Nonconductive shoes • Virgin cotton at low humidity levels
Chairs	• Finished wood • Vinyl • Fiberglass
Floors	• Sealed concrete • Waxed finished wood • Vinyl tile or sheeting
Work Surface	• Waxed, painted, or varnished surfaces • Vinyl or plastic
Work Area	• Spray cleaners • Plastic solder suckers • Soldering iron with ungrounded tip • Solvent brushes with synthetic bristles • Heat guns
Personal Items	• Styrofoam or plastic drinking cups • Plastic/rubber hair combs or brushes • Cellophane or plastic candy/gum wrappers • Vinyl purses
Packing and Handling	• Regular plastic bags, wraps, envelopes • Bubble wrap or foam packing material • Plastic trays, tote boxes, parts bin

Table SP-RB-SRB-4.3 shows typical measured electrostatic charges in a work environment.

Table SP-RB-SRB-4.4: Electrostatic charges in the work environment

PROCESS	RELATIVE HUMIDITY	
	LOW (10–20%)	HIGH (65–90%)
Walking on carpet	35,000 V	1,500 V
Walking on vinyl floor	12,000 V	250 V
Working at workbench	6,000 V	100 V
Work stool padded with urethane foam	18,000 V	1,500 V
Plastic bag picked up from bench	20,000 V	1,200 V
Opening vinyl envelope used for instructions	7,000 V	600 V

Table SP-RB-SRB-4.5 shows a list of electronic parts that may be damaged by sensitivity to electrostatic discharge.

Table SP-RB-SRB-4.5: Semiconductor devices that can be damaged by electrostatic discharge

DEVICE TYPE	SENSITIVITY (VOLTS)
Bipolar transistors	380–7000
CMOS	250–2000
ECL	500
JFET	140–10,000
MOSFET	100–200
Schottky diodes, TTL	300–2500
SCR	680–1000

Antistatic workstations (**Fig.SP-RB-SRB-4.21**) are designed to provide a ground path for static charges that could damage a component. They have a conductive or antistatic work surface that is connected to both a ground and the worker's skin through a wrist strap. The wrist strap has a minimum of 500 k resistance to prevent shock in case of contact with a live circuit. When direct grounding is impractical, an ionized air blower is required.

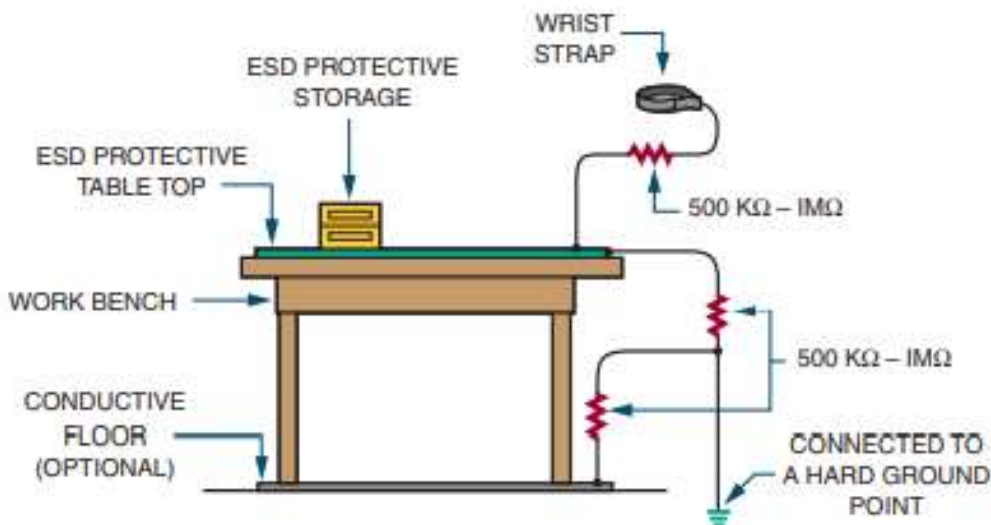


Fig.SP-RB-SRB-4.21: Antistatic work station

The following list identifies precautions required to minimize electrostatic discharge:

1. Prior to starting work on sensitive electronic equipment or circuits, the electronics technician should be grounded using a wrist strap to discharge any static electric charge built up on the body.
2. Always check manuals and package materials for ESD warnings and instructions.
3. Always discharge the package of an ESD sensitive device prior to removing it. Keep the package grounded until the device is placed in the circuit.
4. Minimize the handling of ESD devices. Handle an ESD device only when ready to place it in the circuit.
5. When handling an ESD device, minimize physical movement such as scuffing feet.
6. When removing and replacing an ESD device, avoid touching the component leads.
7. Do not permit an ESD device to come in contact with clothing or other ungrounded materials that could have an electrostatic discharge.
8. Prior to touching an ESD device, always touch the surface on which it rests for a minimum of one second to provide a discharge path.
9. When working on a circuit containing an ESD device, do not touch any material that will create a static charge.
10. Use a soldering iron with the tip grounded. Do not use plastic solder suckers with ESD devices.
11. Ground the leads of test equipment momentarily before energizing the test equipment and before probing an ESD device.

Activity 2: MAJOR GROUP PROJECT

Title: Designing of a PCB

For this project, students are to be divided into 2 groups. Each group is to build a simple PCB and present the product at the end of the term. They are to explain the materials involved in designing the PCB and elaborate on its functions as well as the steps involved in the design process.

Materials needed for making a PCB

- Copper clad laminate
- Ferric chloride solution
- Etch resist chemical or paint
- Painting brush or pen
- Container for etching the PCB
- Drill machine and drill bit
- Etchant remover
- Scouring pad, kitchen paper

Steps involved to design a PCB

1. Cutting copper clad laminate into proper size.
2. Punching indentations for drilling holes for the component leads, as per the schematic.
3. Drawing pads around the indentations with etch resist paint, and linking the pads through tracks using etch resist paint.
4. Immersing the painted board in ferric chloride solution, until the chemical eats away the exposed copper, leaving the painted layout sections intact.
5. Drying the board and scrubbing out the etch resist paint from the tracks and the pads.
6. Drilling holes on the indentations.
7. Polishing the finished board with fine emery paper.
8. Using the ready PCB for assembly and soldering the parts.

Resources

Students can visit this website for aid in designing the PCB: <https://www.homemade-circuits.com/how-to-make-pcb-at-home/>

Appendix

A. Parts of a Material Safety and Data Sheet

Top Portion - Name of Manufacturer - Manufacturer's Address - Emergency Phone Number - Trade name of Product(s) with stock number(s)	Section 4 Reactivity Hazard Data Stability - If known, conditions to avoid preventing hazardous or violent decomposition. Conditions to Avoid - If known, conditions to avoid preventing hazardous reactions. Hazardous Polymerization - If known, conditions to avoid preventing hazardous polymerization resulting in a large release of energy.
Section 1 Hazardous Materials - Exposure recommendations for individual components. - Components are listed if present at or above 1% in the mixture and present a physical or health hazard. - Components identified as carcinogens by NTP, IARC, and OSHA, are listed if present at or above 0.1% in the mixture. - All exposure limits are those that OSHA and ACGIH (American Conference of Governmental Industrial Hygienists) concur unless specifically listed as an OSHA Permissible Exposure Limit (PEL) and/or an ACGIH Threshold Limit Value (TLV).	Section 5 Health Hazard Data Primary Route(s) of Entry - Based on properties and expected use. Effects of Overexposure (Acute) - Potential local and systemic effects due to single or short-term overexposure to the eyes and skin, or through inhalation or ingestion. Signs and Symptoms of Exposure - If known, warning signs which may indicate exposure to the skin or eyes, or through inhalation or ingestion. Medical Conditions Generally Aggravated by Overexposure - If known, preexisting conditions which may contribute to the effects of overexposure to the eyes and skin or through inhalation or ingestion. Emergency and First Aid Procedures - Procedures to be followed when dealing with accidental overexposures.
Section 2 Physical And Chemical Properties Boiling Point - Of product, if known. If unknown, the lowest value of the component is listed for mixtures. Vapor Pressure - Of product, if known. If unknown, the lowest value of the component is listed for mixtures. Vapor Density - Compared to Air = 1. If specific vapor density of product is not known, the value is expressed as lighter or heavier than air. Evaporation Rate - Indicated as faster or slower than Ethyl Ether or Butyl Acetate. Specific Gravity - Compared to water = 1. If specific gravity of product is not known, the value is expressed as less than or greater than water. Appearance and Odor - Describes the physical appearance, color and smell of the product.	Section 6 Control And Protective Measures Protective Equipment - Identifies equipment that may be needed when handling the product. Ventilation - Describes the need for forced mechanical ventilation if required.
Section 3 Fire And Explosion Hazard Data Flash Point - Method Identified Flammability Limits - For product, if known. The lowest and highest value of the components is listed for mixtures. Extinguishing Media - Following National Fire Protection Association criteria. Special Fire Fighting - Minimum equipment to protect firefighters from toxic products of vaporization, combustion, or decomposition in fire situations. Unusual Fire Hazard - Known or expected hazardous products resulting from heating, burning, or other reactions. D.O.T. Category - Describes the classification for shipping by road.	Section 7 Spill Or Leak Procedures - Identifies precautions to be taken and methods of containment, cleanup, and disposal. - RCRA hazardous wastes and CERCLA hazardous substances are listed in this section.
	Section 8 Hazardous Material Identification - Provides hazardous materials identification system rating based on ratings of individual components. - D.O.T. emergency response codes are provided in this section.
	Section 9 Special Precautions And Comments - Covers any relevant points not previously mentioned. - Reinforces pertinent points previously covered.

B. Material Safety and Data Sheet for Ferric Chloride

Company Logo						Section 6: Accidental Release Measures									
Material Safety Data Sheet						Spill Procedure: Provide adequate ventilation. Wear appropriate personal protection. Sprinkle absorbent compound onto spill, then sweep into a plastic container. Wipe up further residue with paper towel and place into container. Wash spill area with soap and water.									
Revision Date		Prepared by		Technical Information		Section 7: Handling and Storage									
Date Sheet Prepared		Who prepared MSDS		Toll Free Phone Number or email address		Handling: Wash thoroughly after handling. Avoid contact with eyes, skin, and clothing. Do not ingest or inhale. Do not expose container to heat or flame.									
Home Office				Emergency		Storage: Keep from freezing. Store in a cool, dry, well ventilated area, away from incompatible substances.									
Company Address				Toll Free Phone Number		Section 8: Exposure Controls									
Section 1: Product Identification						Routes of entry: Eyes, ingestion, inhalation, and skin.									
Name: Ferric Chloride						Ventilation: Use adequate general or local exhaust ventilation to keep airborne concentrations below exposure limits.									
MSDS Code: XXX - Liquid; Related Part Numbers: XXX-500ML; XXX-1L; XXX-4L; XXX-20L						Personal Protection: Wear appropriate protective eyeglasses or chemical safety goggles. Wear appropriate protective clothing to prevent skin contact. Use a NIOSH approved respirator when necessary.									
Use: For etching printed circuits boards.						Section 9: Physical and Chemical Properties									
Section 2: Hazardous Ingredients						Physical State: Liquid									
CAS#		Chemical Name		Percentage by weight		ACGIH TWA		Osha Pel		Osha Stei		Odor: Odorless		Solubility: Miscible	
7705-08-0		Ferric Chloride		40		1mg/m ³		N/E		N/E		Boiling Point: 110°C		Vapor Pressure: 1PSI @20°C	
7647-01-0		Hydrochloric acid		1		5ppm		N/E		N/E		Specific Gravity: 1.43		Evaporation Rate: Slow	
Section 3: Hazards Identification						Section 10: Stability and Reactivity									
Eyes: Corrosive. Contact of liquid will cause severe eye burns, and corneal damage.						Stability: Stable at normal temperatures and pressures.									
Skin: Corrosive. May cause severe skin irritation with possible burns.						Conditions to Avoid: Incompatible materials.									
Inhalation: May cause irritation of mucous membranes and respiratory tract.						Incompatibilities: Alkalis, oxidizers, corrosive to metals.									
Ingestion: May be corrosive to the gastrointestinal tract.						Polymerization: Will not occur.									
Chronic: May cause chemical burns in the mouth, throat, esophagus, and stomach.						Decomposition: Prolonged contact with metals may produce flammable hydrogen gas.									
Chronic: Repeated exposure may cause an increased body load of iron, with possible chronic systemic effects.						Section 11: Toxicological Information									
Section 4: First Aid Measure						Sensitization: (effects of repeated exposure) No									
Eyes: Remove contact lenses. Flush with water or saline for 15 minutes. Get medical aid.						Carcinogenicity: (risk of cancer) No									
Skin: Wash skin with soap and water for 15 minutes. Get medical aid if symptoms persist.						Teratogenicity: (risk of malformation in an unborn fetus) No									
Inhalation: Immediately remove from exposure to fresh air. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Get medical aid.						Reproductive Toxicity (risk of sterility): TDLo: 29mg/kg (rat)									
Ingestion: Do not induce vomiting. If conscious, give 1-2 glasses of water. Get medical aid.						Mutagenicity: (risk of heritable genetic effects): No									
Section 5: Fire Fighting Measures						Lethal Exposure Ingestion									
Auto Ignition Temperature: N/A		Flash Point: N/A		LEL/UEL: N/A		Concentrations: (LD50): 4500 mg/kg (rat)		Inhalation		Skin					
Extinguishing Media: N/A								(LC50): N/E		(LD50): N/E					
General Information: Will not burn.															
NFPA Ratings: Health 2		Flammability 0		Reactivity 2											
HMIS Ratings: Health 2		Flammability 0		Reactivity 2		WHMIS Codes E									

C. Tools used by an electronic technician for PCB fabrication and repairing

ITEM	PICTURE	DESCRIPTION	ITEM	PICTURE	DESCRIPTION
Plastic And Orangewood Spudgers		Used for lifting wires and pushing/aligning leads	Tweezers		Aid to lifting small components
Bristle And Acid Brushes		Applying solvent to PCB for cleaning	Flat Nose Pliers (Duck Bill)		Straighten component leads
Wire Strippers		To strip the end of insulated wire leads	Needle Nose		Form leads
Surgical Scissors		General cutting	Round Nose Pliers		Form leads
X-Acto Knife		General cutting, removal of bridges on PCB	Diagonal Cutters (Flush)		Cut leads flush to PCB and general lead trimming
ITEM	PICTURE	DESCRIPTION	ITEM	PICTURE	DESCRIPTION
Screwdrivers (Flat blade, Phillips)		Install/remove screws	Alcohol And Flux Bottle		Use to hold alcohol and flux
Nut Drivers		Install/remove bolts and nuts	Soldering Iron		Used to solder
1/16" And 1/8" Chisel Tip For Soldering Iron		Options for components to be soldered	Solder Wick		Used for solder removal
Heat Sinks		Used to remove excess heat with semiconductors	Manual Desoldering Tool		Used for solder removal

D: Test equipment used by an electronic technician for determining proper operation, alignment or troubleshooting of a circuit.

ITEM	PICTURE	DESCRIPTION
Isolated Power Supply		Used to power a PCB under test.
Analog Multimeter (VOM)		Used to measure and/or align voltage, resistance and current of a PCB under test.
Function Generator		Used to generate a sinusoidal, triangular or square waveform for troubleshooting and aligning an electronic circuit.
Logic Probes		Used to detect a high, low, clock or lack of a signal in digital electronic circuits.
Logic Pulser		Used to inject a digital signal into a digital circuit for troubleshooting.

ITEM	PICTURE	DESCRIPTION
Digital Multimeter (DMM)		Used to measure and/or align voltage, resistance and current of a PCB under test.
Analog Oscilloscope		Used to measure and/or align amplitude, frequency and compare waveforms.
Frequency Counter		Used to measure and/or align frequency and period of recurring signals.
Signal Generator		Used to generate a sinusoidal waveform for troubleshooting and aligning an electronic circuit.

E: Those terms *italics* may be cross-referenced in this glossary

action potential The stereotypical voltage spike that constitutes an active output from a neuron. They are propagated along the *axon* to other neurons.

afferent With respect to a particular neuron, an axon that impinges on (or *innervates*) that neuron.

arbor Usually used in the context of a dendritic arbor—the tree-like structure associated with dendritic branching.

axon The fibre that emanates from the neuron cell body or *soma* and that conducts *action potentials* to other neurons.

axon hillock The junction of the *axon* and cell body or *soma*. The place where *action potentials* are initiated if the *membrane potential* exceeds a threshold.

axon terminal An axon may branch into several *collaterals*, each terminating at an axon terminal, which constitutes the *presynaptic* component of a *synapse*.

chemical binding The process in which a *neurotransmitter* joins chemically with a *receptor site* thereby initiating a *PSP*.

collateral An axon may divide into many collateral branches allowing contact with many other neurons or many contacts with one neuron.

dendrite One of the branching fibres of a neuron, which convey input information via *PSPs*.

depolarization The *membrane potential* of the neuron has a negative resting or equilibrium value. Making this less negative leads to a depolarization. Sufficient depolarization at the *axon hillock* will give rise to an action potential.

efferent A neuron sends efferent axon *collaterals* to other neurons.

EPSP Excitatory Postsynaptic Potential. A *PSP* that acts to *depolarize* the neural membrane.

hyperpolarization The *membrane potential* of the neuron has a negative resting or equilibrium value. Making this more negative leads to a hyperpolarization and inhibits the action of *EPSPs*, which are trying to *depolarize* the membrane.

innervate Neuron *A* sending signals to neuron *B* is said to innervate neuron *B*.

IPSP Inhibitory Postsynaptic Potential. A *PSP* that acts to *hyperpolarize* the neural membrane.

membrane potential The voltage difference at any point across the neural membrane.

neurotransmitter The chemical substance that mediates synaptic activity by propagation across the *synaptic cleft*.

organelle Subcellular components that partake in metabolism, etc.

postsynaptic membrane That part of a *synapse* which is located on the *dendrite* and consists of the dendritic membrane together with *receptor sites*.

potential difference The voltage difference across the cell membrane.

presynaptic membrane That part of a *synapse* which is located on the *axon terminal*.

PSP Postsynaptic Potential. The change in *membrane potential* brought about by activity at a *synapse*.

receptor sites The sites on the *postsynaptic membrane* to which molecules of *neurotransmitter* bind. This binding initiates the generation of a *PSP*.

refractory period The shortest time interval between two *action potentials*.

soma The cell body.

synapse The site of physical and signal contact between neurons. On receipt of an *action potential* at the *axon terminal* of a *synapse*, *neurotransmitter* is released into the *synaptic cleft* and propagates to the *postsynaptic membrane*. There it undergoes *chemical binding* with *receptors*, which, in turn, initiates the production of a postsynaptic potential (*PSP*).

synaptic cleft The gap between the pre- and postsynaptic membranes across which chemical *neurotransmitter* is propagated during synaptic action. vesicles The spherical containers in the *axon terminal* that contain *neurotransmitter*. On receipt of an action potential at the axon terminal, the vesicles release their neurotransmitter into the *synaptic cleft*.

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